Calculus I

(*following* "Thomas' Calculus, 15th Ed.")

Lectures on YouTube:
https://www.youtube.com/channel/UCmRbK4vlGDht-joOQ5g0J2Q

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Prologue

This lecture note is ...

Currently the lecture note is not fully grown up; other useful techniques and interesting examples would be soon incorporated. Any questions, suggestions, comments will be deeply appreciated.

Seongjai Kim
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CHAPTER 1

Functions

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Lecture-Note: https://skim.math.msstate.edu/LectureNotes/Calculus-1-Lecture.pdf

1.1. Functions and Their Graphs

Introduction. Functions are a tool for describing the real world in mathematical terms. A **function** can be represented by **an equation, a graph, a numerical table, or a verbal description**; we will use all four representations throughout the lecture note. This section reviews these ideas.

1.1.1. Functions; Domain and Range

Definition 1.1. A **function** f from a set D to a set Y is a rule that **assigns a unique value** $f(x) \in Y$ to each $x \in D$. That is,

$$\begin{aligned} f: \ & D \to Y \\ & x \mapsto f(x), \quad \text{where } f(x) \text{ is a unique value for } x \end{aligned} \tag{1.1}$$

- A function produces the same output for the same input.
- Let y be the output value of f at x. Then,

$$y = f(x),$$

where x is the **independent variable**, representing the input value to f, and y is the **dependent variable** or **output value of** f at x.

Definition 1.2. The set D of all possible input values is called the **domain** of the function. The set of all output values $f(x)$, as x varies throughout D, is called the **range** of the function. The range might not include every element in the set Y.

- For a function $y = f(x)$, if the domain is not stated explicitly or restricted by context, the domain is assumed to be the largest set of real x-values for which the function gives real y-values. This is called the **natural domain** of f.

Example 1.3. Find the natural domain and associated range of each functions. The domain in each case is the values of x for which **the formula makes sense.**

(a) $y = x^2$ (b) $y = 1/x$ (c) $y = \sqrt{1 - x^2}$

Solution.

1.1.2. Graphs of Functions

Definition 1.4. If f is a function with domain D, its **graph** consists of the points in the Cartesian plane whose coordinates are the input-output pairs for f. In set notation, the graph is

$$\boldsymbol{G}(f) := \{(x, f(x)) \mid x \in D\}. \tag{1.2}$$

Example 1.5. Graph the functions

(a) $y = x^2$ over $[-2, 2]$ (b) $y = 1/x$ over $[-2, 2]$ (c) $y = \sqrt{1 - x^2}$

Solution.

Representing a Function Numerically

Note: We have seen how a function may be represented **algebraically by a formula** and **visually by a graph**.

- Another way to represent a function is **numerically**, **through a table of values**.
- From an appropriate table of values, a graph of the function can be obtained, possibly with the aid of a computer. The graph consisting of only the points in the table is called a **scatterplot**.

Example 1.6.

A Matlab code

```
a=0; b=pi;
n=8; h=(b-a)/n; angle = (a:h:b)';

% y = sqrt(1-x^2)
X = cos(angle); Y = sin(angle);
Data = [X Y]

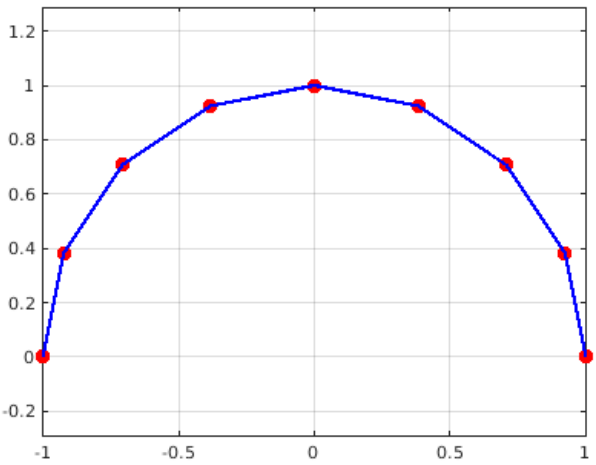
plot(X,Y,'r.','MarkerSize',30);
hold on; axis equal; grid on
plot(X,Y,'b-','LineWidth',2);
hold off

print -dpng 'plot-Data.png';
```

Data set

x	y
1.0000	0
0.9239	0.3827
0.7071	0.7071
0.3827	0.9239
0.0000	1.0000
-0.3827	0.9239
-0.7071	0.7071
-0.9239	0.3827
-1.0000	0.0000

A **Matlab** code generated a data set in a table form. The data can be visualized using a computer program.



1.1.3. The Vertical Line Test for a Function

Claim 1.7. Vertical Line Test

- Not every curve in the coordinate plane can be the graph of a function.
- A function f can have only one value $f(x)$ for each x in its domain, so **no vertical line** can intersect the graph of a function more than once.
- If a is in the domain of the function f, then the vertical line $x = a$ will intersect the graph of f at the single point $(a, f(a))$.

Example 1.8. Are they functions?

(a) Circle $x^2 + y^2 = 1$ (b) Upper semicircle (c) Lower semicircle

Solution.

Ans: (a) no. (b) yes. (c) yes.

Example 1.9. Derive an equation of the circle centered at (h, k) and having radius r.

Solution.

1.1.4. Various Function Types

Piecewise-Defined Functions

Example 1.10. Graph the piecewise-defined functions.

(a) $f(x) = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$

(b) $g(x) = \begin{cases} -x, & x < 0 \\ x^2, & 0 \leq x \leq 1 \\ 1, & x > 0 \end{cases}$

Solution.

Example 1.11.

- The function whose value at any number x is *the greatest integer less than or equal to* x is called the **greatest integer function** or the **integer floor function**. e.g., $\lfloor 2.3 \rfloor = 2, \ \lfloor 1.0 \rfloor = 1, \ \lfloor -1.5 \rfloor = -2.$
- The function whose value at any number x is *the smallest integer greater than or equal to* x is called the **least integer function** or the **integer ceiling function**. e.g., $\lceil 2.3 \rceil = 3, \ \lceil 1.0 \rceil = 1, \ \lceil -1.5 \rceil = -1.$

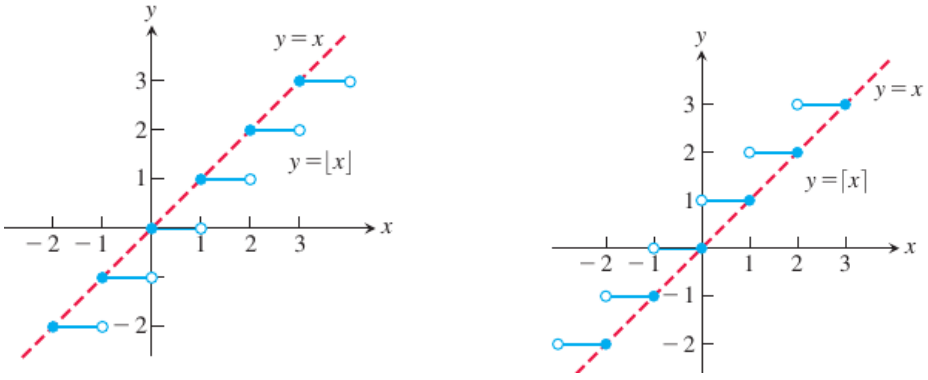


Figure 1.1: The greatest integer function (left) and the least integer function (right).

Increasing and Decreasing Functions

Definition 1.12. Let f be a function defined on an interval I and let x_1 and x_2 be two distinct points in I.

1. If $f(x_1) < f(x_2)$ whenever $x_1 < x_2$, then f is said to be **increasing** on I.
2. If $f(x_1) > f(x_2)$ whenever $x_1 < x_2$, then f is said to be **decreasing** on I.

Note: It is important to realize that the definitions of increasing and decreasing functions must be satisfied for every pair of points x_1 and x_2 in I with $x_1 < x_2$. Because we use the inequality $<$ to compare the function values, instead of $\leq$, it is sometimes said that f is **strictly increasing** or **strictly decreasing** on I.

Example 1.13. For each of given functions, find regions on which it is either increasing or decreasing.

(a) $y = x^2$ (b) $y = x^3$

Solution.

Ans: (a) $(-\infty, 0]$: decreasing; $[0, \infty)$: increasing

Even Functions and Odd Functions: Symmetry

Definition 1.14. A function $y = f(x)$ is an

$$\begin{array}{ll} \textbf{even function} \text{ of } x & \text{if } f(-x) = f(x) \\ \textbf{odd function} \text{ of } x & \text{if } f(-x) = -f(x) \end{array} \tag{1.3}$$

for every x in the function's domain.

Note: The names **'even'** and **'odd'** come from **powers of x**.

Example 1.15. Find if the given functions are even, odd, or neither.

(a) $y = 2x^2 - 2$ (b) $y = x^3 - 4x$ (c) $y = x^2 + x$

Solution.

Remark 1.16. Symmetry

- The graph of an even function is **symmetric about the y-axis**. A **reflection** across the y-axis leaves the graph unchanged.
- The graph of an odd function is **symmetric about the origin**. A **rotation** of 180° about the origin leaves the graph unchanged.

1.1.5. Common Functions

Linear Functions

- A function of the form $f(x) = mx + b$ is called a **linear function**. The function $f(x) = x$, where $m = 1$ and $b = 0$, is called the **identity function**.

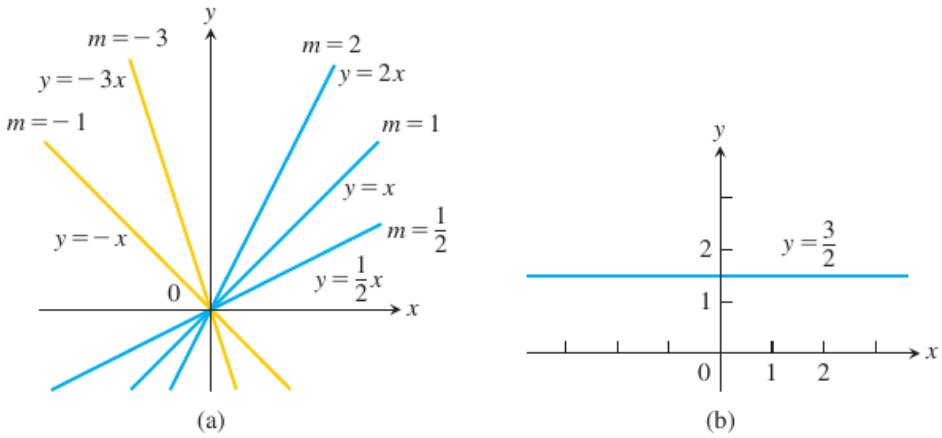


Figure 1.2: (a) Lines through the origin with slope m. (b) A constant function with slope $m = 0$.

- Two variables y and x are **proportional** (to one another), if one is always a constant multiple of the other – that is, if $y = kx$ for some nonzero constant k.
  - If the variable y is proportional to the reciprocal $1/x$, then sometimes it is said that y is **inversely proportional** to x.

Power Functions

A function $f(x) = x^a$, where a is a constant, is called a **power function**. There are several important cases to consider.

(a) $f(x) = x^a$ with $a = n$, a positive integer.

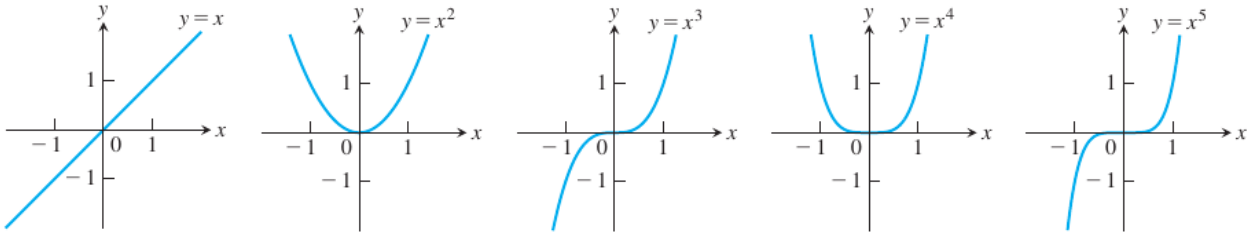


Figure 1.3: Graphs of $f(x) = x^n$, $n = 1, 2, 3, 4, 5$.

Each curve passes through the origin and the point $(1, 1)$.
Are they symmetric? Where are they increasing? What happens when x approaches $-\infty$ or ∞?

(b) $f(x) = x^a$ with $a = -1$ or $a = -2$.

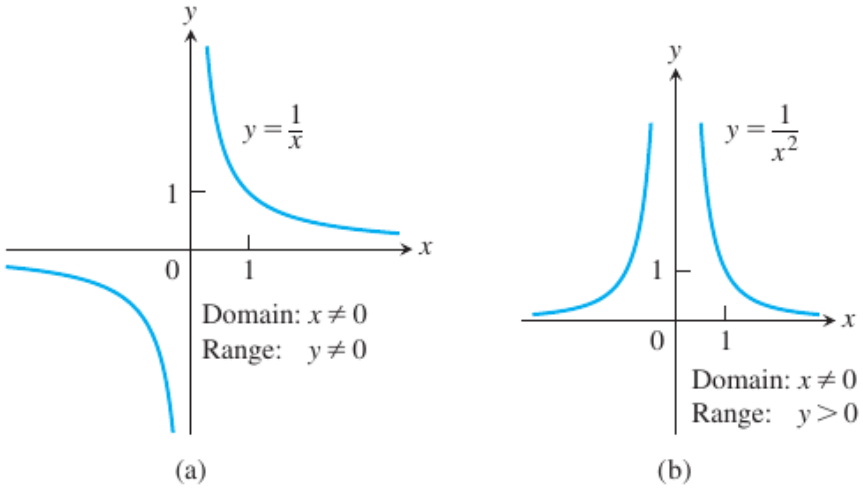


Figure 1.4: Graphs of the power functions $f(x) = x^a$. (a) $a = -1$. (b) $a = -2$.

(c) $a = 1/2,\ 1/3,\ 3/2$, and $2/3$.

- $f(x) = x^{1/2} = \sqrt{x}$ and $g(x) = x^{1/3} = \sqrt[3]{x}$ are the **square root** and **cube root** functions, respectively.
- Note that $x^{3/2} = (x^{1/2})^3$ and $x^{2/3} = (x^{1/3})^2$.

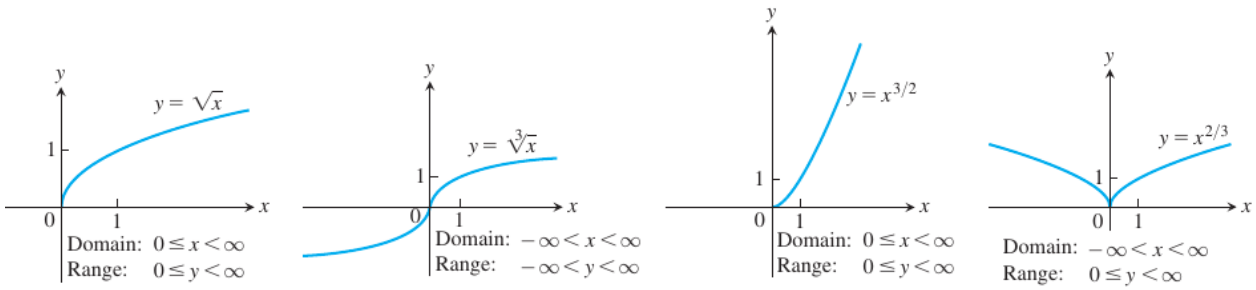


Figure 1.5: Graphs of the power functions $f(x) = x^a$ for $a = 1/2,\ 1/3,\ 3/2$, and $2/3$.

Polynomials

A function p is a **polynomial** if

$$p(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0, \tag{1.4}$$

where n is a nonnegative integer, $a_0, a_1, \cdots, a_n \in \mathbb{R}$, and $a_n \neq 0$.

- n is called the **degree** of the polynomial
- $a_0, a_1, \cdots, a_n$ are called the **coefficients** of the polynomial
- p is called a **quadratic polynomial** when $n = 2$, while it is a **cubic polynomial** if $n = 3$.
- All polynomials have domain (∞, ∞).

Example 1.17. Sketch the graph of each polynomial function.

(a) $y = x^3 - 3x^2 + 2x$

(b) $y = (x+1)(x-1)^3(x-2)^2$

Rational Functions

A **rational function** is a quotient or ratio

$$f(x) = \frac{p(x)}{q(x)}, \tag{1.5}$$

where p and q are polynomials. The **domain** of a rational function is the set of all real x for which $q(x) \neq 0$.

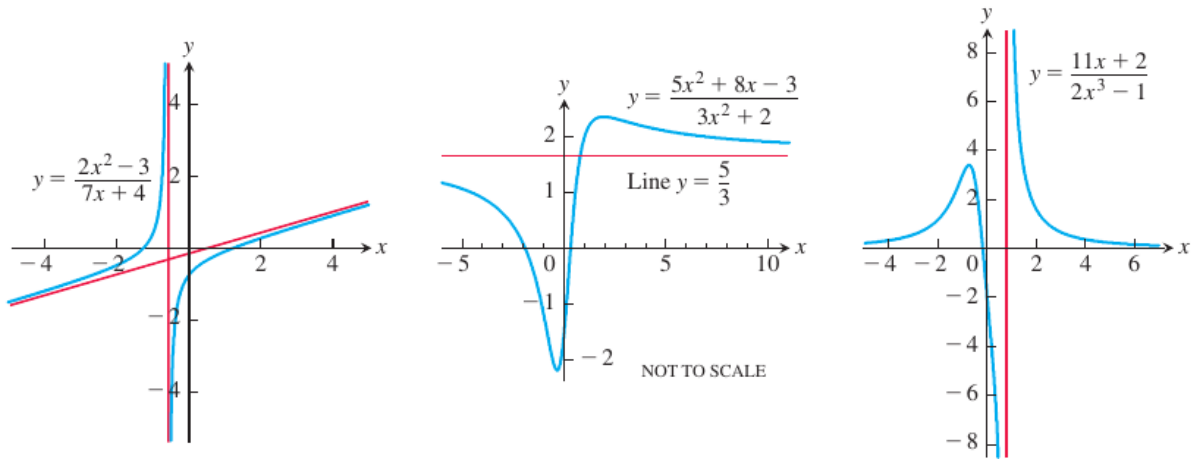


Figure 1.6: Graphs of three rational functions. The straight red lines approached by the graphs are called **asymptotes** and are not part of the graphs.

Self-study **1.18.** Sketch the graph of $f(x) = \dfrac{x^2 - 2x}{x - 1}$

Note: Other common functions are trigonometric functions, exponential functions, and logarithmic functions, which we will deal with in detail later in this chapter.

Exercises 1.1

1. Sketch the graph and find the domain and range of each function.

   (a) $f(x) = 1 - 2\sqrt{x}$

   (b) $g(t) = \dfrac{3}{t-4}$

   (c) $h(x) = \sqrt{4x - x^2}$

   Ans: (b) D: $\{t \neq 4\}$. R: $\{y \neq 0\}$.

2. Consider the point (x, y) lying on the graph of the line $2x + y = 1$. Let L be the distance from the point (x, y) to the origin $(0, 0)$. Write L as a function of x.
   Clue: Start with the distance: $L = (x^2 + y^2)^{1/2}$, which comes from the **Pythagorean theorem**.

   Ans: $L = \sqrt{5x^2 - 4x + 1}$

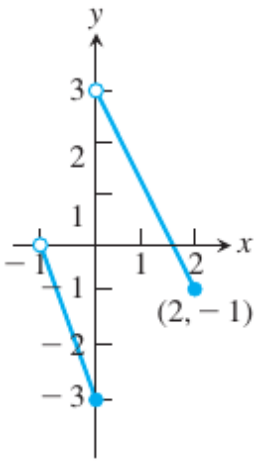
3. Graph the following equations and explain why they are not graphs of functions of x.

   (a) $|y| = 2x$

   (b) $y^2 = x^2$

   Hint: (a) Consider cases that $|y|$ is nonnegative or negative. (b) It can be written as $|y| = |x|$.

4. Find a formula for the piecewise-defined function.



5. Say whether the function is even, odd, or neither. Give reasons for your answer.

   (a) $f(x) = x^3 - 5x$

   (b) $f(x) = x^{-5}$

   (c) $g(t) = |t^3| + t$

   (d) $g(x) = \dfrac{x^2 - 1}{x^2 + 1}$

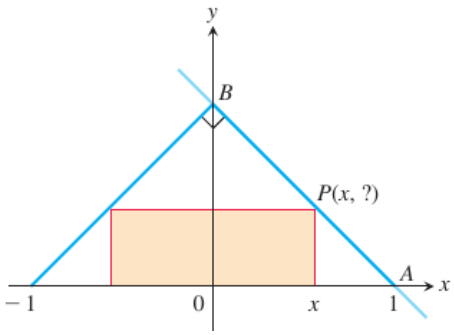
   Ans: (c) neither, $\because\ g(-t) \neq g(t)$ and $g(-t) \neq -g(t)$. (d) even, $\because\ g(-x) = g(x)$.

6. The accompanying figure shows a rectangle inscribed in an *isosceles right triangle* whose hypotenuse is 2 units long.

   (a) Express the y-coordinate of P in terms of x.

   (b) Express the area of the rectangle in terms of x.

   Clue: (a) You might start by writing an equation for the line AB.



Ans: (b) $A = x(1 - x)$.

1.2. Combining Functions; Shifting and Scaling Graphs

Sums, Differences, Products, and Quotients

- Like numbers, functions can be added, subtracted, multiplied, and divided to produce new functions.
- If f and g are functions, we define functions $f+g$, $f-g$, fg, f/g by the formulas

$$\begin{aligned}(f+g)(x) &= f(x)+g(x)\\ (f-g)(x) &= f(x)-g(x)\\ (fg)(x) &= f(x)\,g(x)\\ (f/g)(x) &= f(x)/g(x)\end{aligned} \tag{1.6}$$

  - **Domain of $f+g$, $f-g$, and fg**: every x that belongs to the domains of both f and g; that is, $x \in D(f) \cap D(g)$.
  - **Domain of f/g**: $x \in D(f) \cap D(g)$, except $g(x) = 0$.

Example 1.19. The functions defined by the formulas

$$f(x) = \sqrt{x} \text{ and } g(x) = \sqrt{1-x}$$

have domains $D(f) = [0, \infty)$ and $D(g) = (-\infty, 1]$. The points common to these domains are the points in

$$[0, \infty) \cap (-\infty, 1] = [0, 1].$$

Find the formulas and domains for the various algebraic combinations of the two functions, where $f \cdot g = fg$.

Solution. *Let's complete filling the following table.*

Function	Formula	Domain
$f+g$		
$f-g$		
$f \cdot g$	$(f \cdot g)(x) = f(x)g(x) = \sqrt{x(1-x)}$	
f/g		
g/f	$\left(\frac{g}{f}\right)(x) = \frac{g(x)}{f(x)} = \sqrt{\frac{1-x}{x}}$	$(0, 1]$ ($x = 0$ excluded)

Composite Functions

Definition 1.20. If f and g are functions, the **composite** function $f \circ g$ ("f composed with g") is defined by

$$(f \circ g)(x) = f(g(x)). \tag{1.7}$$

The domain of $f \circ g$ consists of the numbers x in the domain of g for which g(x) lies in the domain of f:

$$D(f \circ g) = \{x \in G(g) \mid g(x) \in D(f)\}. \tag{1.8}$$

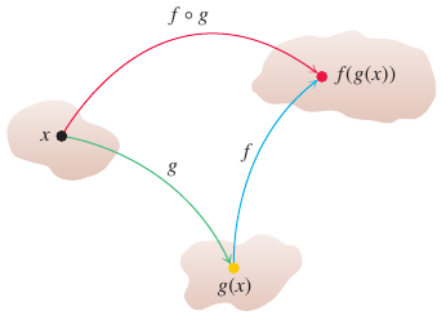


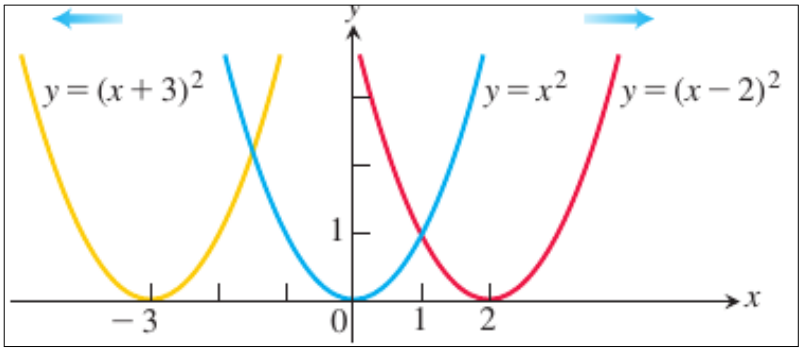
Figure 1.7: Arrow diagram for $f \circ g$.

Example 1.21. Let $f(x) = \sqrt{x}$ and $g(x) = x + 1$. Find $f \circ g$, $g \circ f$, $f \circ f$, and $g \circ g$, and their domains.

Solution.

Composition		Formula	Domain
$(f \circ g)(x)$	=	$f(g(x)) = \sqrt{g(x)} = \sqrt{x+1}$	$[-1, \infty)$
$(g \circ f)(x)$	=		
$(f \circ f)(x)$	=	$f(f(x)) = \sqrt{f(x)} = \sqrt{\sqrt{x}} = x^{1/4}$	$[0, \infty)$
$(g \circ g)(x)$	=		

Shifting a Graph of a Function



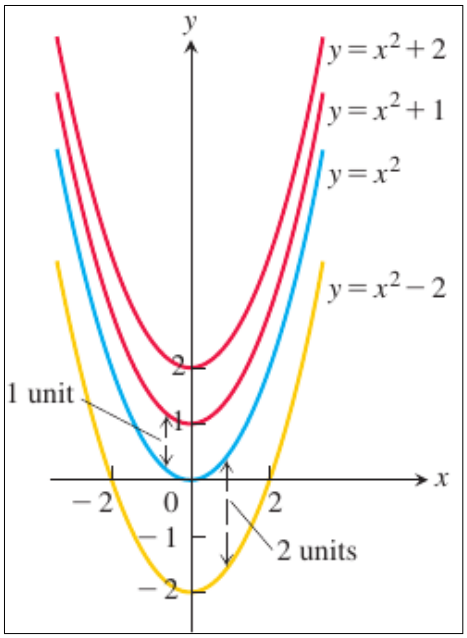


Figure 1.8: Shift from the graph of $y = x^2$.

Observation 1.22. Note that $\rightarrow$ and $\uparrow$ are positive directions of x and y, respectively.

shift the graph	$\rightarrow$	2	$\Longrightarrow$	$x \rightarrow (x-2)$
shift the graph	$\rightarrow$	-3	$\Longrightarrow$	$x \rightarrow (x+3)$
shift the graph	$\uparrow$	2	$\Longrightarrow$	$y \rightarrow (y-2)$
shift the graph	$\uparrow$	-2	$\Longrightarrow$	$y \rightarrow (y+2)$

Shift Formula

If the graph of $y = f(x)$ **is shifted** (c, d), then the resulting graph is formulated as

$$y - d = f(x - c). \tag{1.9}$$

Example 1.23. Sketch the graph of the functions.

(a) $y = -(x-3)^2$ (b) $y - 2 = -x^2$ (c) $y = -(x+1)^2 + 4$

Solution.

Ans: (a) vertex$= (3, 0)$. (c) vertex$= (-1, 4)$.

Summary **1.24.** If the graph of $y = f(x)$ **is shifted** (c, d), then the resulting graph is formulated with

$$\begin{array}{ll} x & \text{being replaced by } x - c \\ y & \text{being replaced by } y - d \end{array} \tag{1.10}$$

Why does this formula make sense algebraically?

- The equation

$$y = f(x) \tag{1.11}$$

  expresses the relation between x and y of each point (x, y) on the graph of f.
- Let the graph be shifted (c, d). Let (x, y) **be shifted to** (X, Y).
- Then, we should find the **relation between X and Y**, to formulate the shifted graph.
- Since $(X, Y) = (x + c, y + d)$, we may get

$$(x, y) = (X - c, Y - d). \tag{1.12}$$

- Using (1.11) and (1.12), we obtain

$$Y - d = f(X - c). \tag{1.13}$$

  So (1.10) follows.

Scaling a Graph of a Function

Scaling Formula

- If the graph of $y = f(x)$ is scaled **horizontally** by a factor of $b > 0$, then the resulting graph is formulated as

$$y = f(x/b). \tag{1.14}$$

- If the graph of $y = f(x)$ is scaled **vertically** by a factor of $a > 0$, then the resulting graph is formulated as

$$y/a = f(x). \tag{1.15}$$

- If the graph of $\boldsymbol{y = f(x)}$ **is scaled by factors of** (b, a), then the resulting graph is formulated as

$$\boldsymbol{y/a = f(x/b)} \quad \text{or} \quad \boldsymbol{y = a\, f(x/b)} \tag{1.16}$$

Note: The reference book (Thomas' Calculus) explains scaling in terms of "stretch" and "compress".

$$\begin{aligned} &\text{stretched by a factor of } b &&\Leftrightarrow \text{ scaling factor} = b \\ &\text{compressed by a factor of } b &&\Leftrightarrow \text{ scaling factor} = 1/b \end{aligned}$$

Reflecting a Graph of a Function

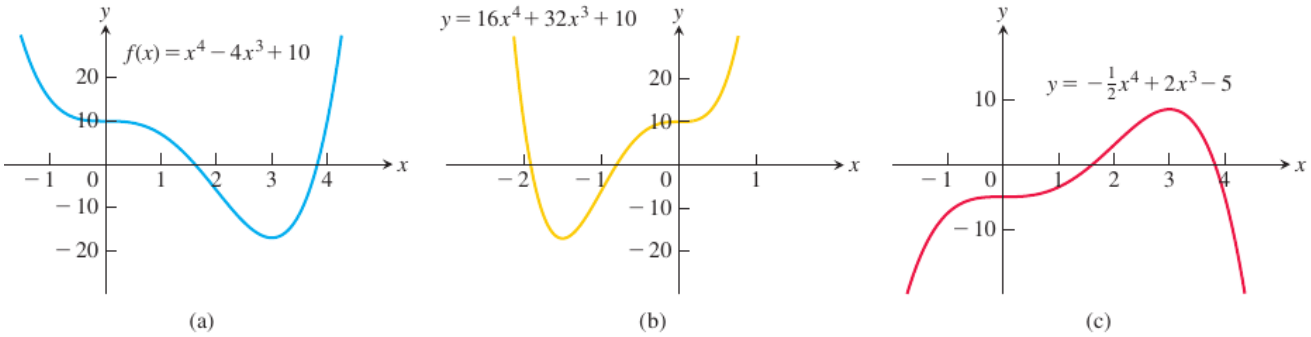
Reflecting Formula

Reflection is the same as scaling by a factor of -1. That is,

$$\begin{aligned} &\text{reflect } G(f) \text{ across the } x\text{-axis} &&\Leftrightarrow\ y \text{ is replaced by } -y \\ &\text{reflect } G(f) \text{ across the } y\text{-axis} &&\Leftrightarrow\ x \text{ is replaced by } -x \end{aligned} \tag{1.17}$$

Example 1.25. The graph in (a) depicts $y = f(x)$. Find the scaling factor and the reflection for graphs in (b) and (c) to be expressed in the form

$$y = pf(qx).$$



Solution.

(b) $c = 1/2$ & y-axis reflection

(c) $d = 1/2$ & x-axis reflection

Ans: (b) $y = f(-2x)$. (c) $\dfrac{y}{-1/2} = f(x) \Rightarrow y = -\dfrac{1}{2}f(x)$.

Exercises 1.2

1. If $f(x) = x - 1$ and $g(x) = 1/(x+1)$, find the following.

   (a) $f(g(x))$
   (b) $f(g(1/2))$
   (c) $g(f(x))$
   (d) $g(f(1/2))$

   Ans: (c) $g(f(x)) = 1/(f(x)+1) = 1/x$. (d) 2.

2. Write a formula for $f \circ g \circ h$.

   (a) $f(x) = 3x + 4,\ g(x) = 2x - 1,\ h(x) = x^2$
   (b) $f(x) = \sqrt{2x+1},\ g(x) = \dfrac{1}{x+4},\ \ h(x) = \dfrac{1}{x}$

   Ans: (b) $\sqrt{\dfrac{5x+1}{4x+1}}$.

3. Copy and complete the following table.

	$g(x)$	$f(x)$	$(f \circ g)(x)$
a.	$x - 7$	$\sqrt{x}$	?
b.	$x + 2$	$3x$	?
c.	?	$\sqrt{x-5}$	$\sqrt{x^2-5}$
d.	$\dfrac{x}{x-1}$	$\dfrac{x}{x-1}$	?
e.	?	$1 + \dfrac{1}{x}$	x
f.	$\dfrac{1}{x}$	?	x

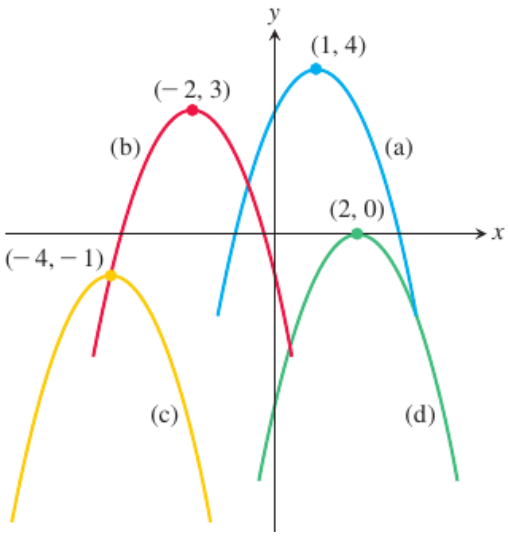
   Ans: (c) x^2. (e) $1/(x-1)$.

4. Find a function $g(x)$ so that

   (a) $f(x) = \dfrac{x}{x-2}$ and $(f \circ g)(x) = x$.
   (b) $f(x) = 2x^3 - 4$ and $(f \circ g)(x) = x + 2$.

   Ans: (a) $g(x) = \dfrac{2x}{x-1}$.

5. The accompanying figure shows the graph of $y = -x^2$ shifted to four new positions. Write an equation for each new graph.



6. **Challenge**[1] Assume that f is an **even** function, g is an **odd** function, and both f and g are defined on the entire real line $(-\infty, \infty)$. Which of the following (where defined) are even? odd?

(a) fg
(b) f/g
(c) g/f
(d) $f^2 = f\,f$
(e) $g^2 = g\,g$
(f) $f \circ g$
(g) $g \circ f$
(h) $f \circ f$
(i) $g \circ g$

Ans: (a) Odd. (c) Odd. (e) Even. (i) Odd.

7. **Challenge** Can a function be both even and odd? Give reasons for your answer.

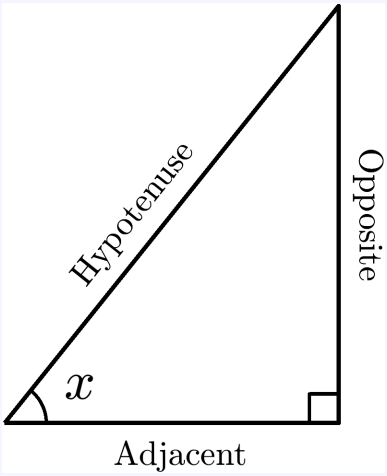
[1] **Challenge** indicates the problem may be harder than usual. Challenge it!

1.3. Trigonometric Functions

You might learn the following in your High school.

Sides of a right triangle:

- The **hypotenuse** of a right triangle is the side opposite the right angle. It must be the longest side of the right triangle.
- The **adjacent side** is the non-hypotenuse side that is adjacent to *the given angle*.
- The **opposite side** is the side across from a given angle.



SOH–CAH–TOA:

$\sin x$	$= \dfrac{\textit{opposite}}{\textit{hypotenuse}}$	$\Rightarrow$	SOH
$\cos x$	$= \dfrac{\textit{adjacent}}{\textit{hypotenuse}}$	$\Rightarrow$	CAH
$\tan x$	$= \dfrac{\textit{opposite}}{\textit{adjacent}}$	$\Rightarrow$	TOA

Cosecant, Secant, Cotangent:

$$\csc x = \frac{1}{\sin x}, \quad \sec x = \frac{1}{\cos x}, \quad \cot x = \frac{1}{\tan x} \tag{1.18}$$

1.3.1. Angles and Sectors

Definition 1.26. An **angle** is the figure formed by two rays sharing a common endpoint, called the **vertex of the angle**.

The **angle** can be defined with the *unit circle*, the circle of radius 1.

"The angle is θ (radian), when the corresponding arc length is θ."

- The angle of the whole circle is 2π (radian).
- $2\pi = 360^\circ$
- $\pi = 180^\circ \quad \Rightarrow \quad \boxed{^\circ = \frac{\pi}{180}}$

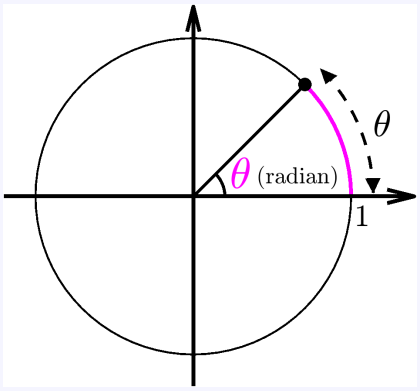


Figure 1.9: Geometric definition of the angle.

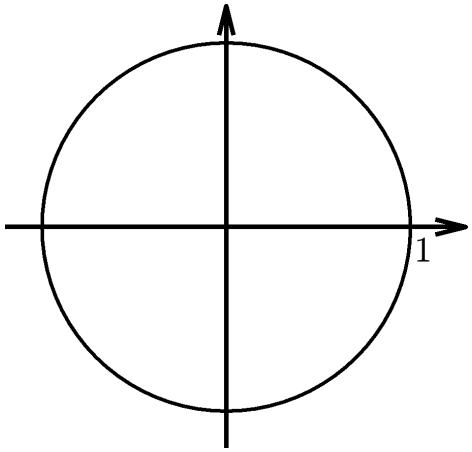
Example 1.27. Figure the following angles, using the unit circle.

(a) 30°

(b) -60°

(c) 420°

(d) $-\frac{3}{2}\pi$



Sectors

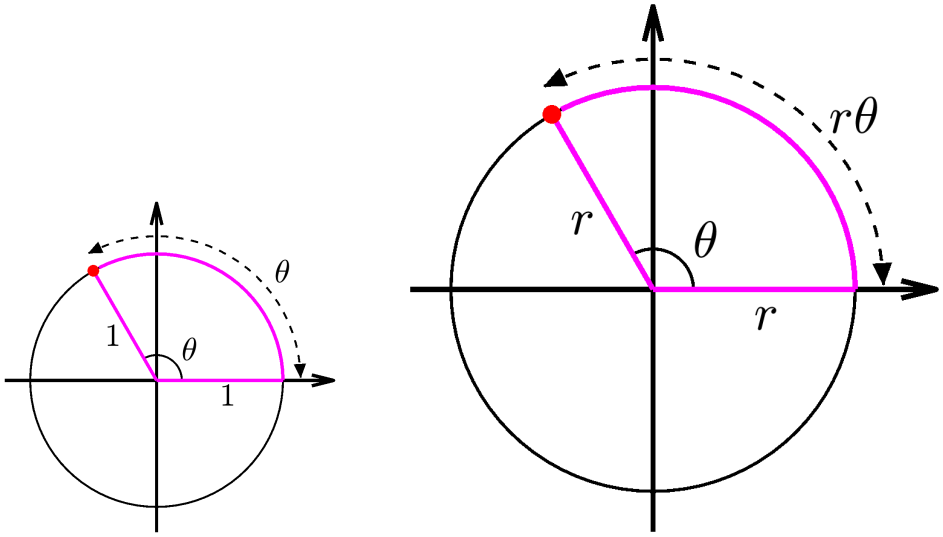


Figure 1.10: The angle and the **arc length** of the **sector**.

Definition 1.28. A **sector** (a.k.a. **circular sector**) is the the portion of a disc enclosed by two radii and a circular arc.

Formula 1.29. A sector is given as in the right of Figure 1.10. Then,

$$\text{Arc length } \ell = r\theta \qquad \text{Area } A = \frac{1}{2}r\ell = \frac{1}{2}r^2\theta$$

** A sector is like a triangle!*

Example 1.30. Find the arc length of the sectors.

(a) $r = 12$, angle = $\frac{5}{4}\pi$

(b) $r = 4$, angle = 240°

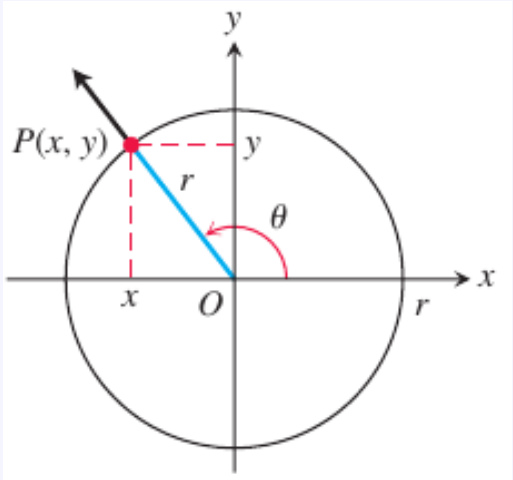
Solution.

Ans: (a) 15π. (b) $16\pi/3$.

1.3.2. The Six Basic Trigonometric Functions

Definition 1.31. The trigonometric functions of an **acute angle** are given in terms of the sides of a right triangle (SOH–CAH–TOA).

- We extend this definition to **obtuse and negative angles** by first placing the angle in standard position in a circle of radius r.
- We then define the trigonometric functions in terms of the coordinates of the point $P(x, y)$ where the angle's terminal ray intersects the circle.



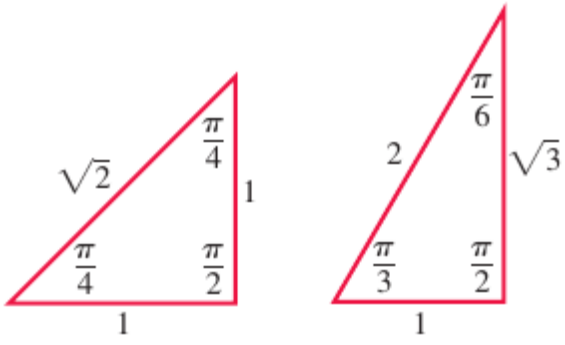
$$\begin{aligned}
\textbf{sine}: \quad \sin\theta &= \frac{y}{r} & \qquad \textbf{cosecant}: \quad \csc\theta &= \frac{1}{\sin\theta} = \frac{r}{y} \\
\textbf{cosine}: \quad \cos\theta &= \frac{x}{r} & \qquad \textbf{secant}: \quad \sec\theta &= \frac{1}{\cos\theta} = \frac{r}{x} \\
\textbf{tangent}: \quad \tan\theta &= \frac{y}{x} & \qquad \textbf{cotangent}: \quad \cot\theta &= \frac{1}{\tan\theta} = \frac{x}{y}
\end{aligned} \tag{1.19}$$

Example 1.32. Evaluate the exact values of these trigonometric ratios for common angles.

(a) $\{\sin, \cos, \tan\}\dfrac{\pi}{4} = \dfrac{1}{\sqrt{2}}, \dfrac{1}{\sqrt{2}}, 1$

(b) $\{\sin, \cos, \tan\}\dfrac{\pi}{6} = \dfrac{1}{2}, \dfrac{\sqrt{3}}{2}, \dfrac{1}{\sqrt{3}}$

(c) $\{\sin, \cos, \tan\}\dfrac{\pi}{3} = \dfrac{\sqrt{3}}{2}, \dfrac{1}{2}, \sqrt{3}$



Geometric interpretation of trigonometric functions

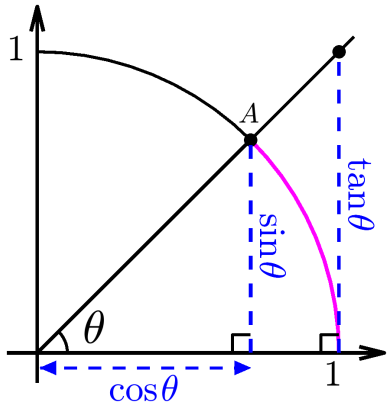


Figure 1.11: Geometric interpretation of trigonometric functions.

Definition 1.33. The **trigonometric functions** are defined with the **unit circle**, the circle of radius 1.

Point $A(x, y)$ is chosen on the unit circle as in the figure.

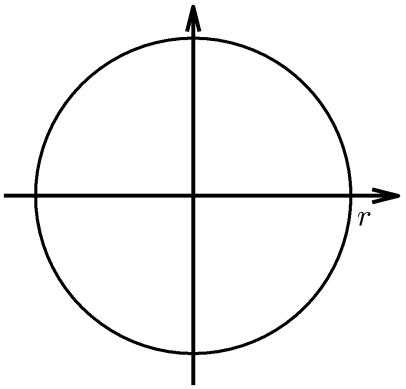
* $A(x, y) = (\cos\theta, \sin\theta)$
* "sin" is the vertical component of the right triangle.
* $\tan\theta = \dfrac{y}{x} = \text{slope} \quad \Rightarrow \quad \tan\theta = \dfrac{\sin\theta}{\cos\theta} = \dfrac{\tan\theta}{1}$
* These hold for all angles θ, i.e., all choices of A on the unit circle.

Example 1.34. Let θ be the angle made by the line segment from $O(0, 0)$ to $P(-4, 3)$. Evaluate the following.

(a) $\sin\theta$

(b) $\cos\theta$

(c) $\tan\theta$



Ans: (a) $3/5$. (b) $-4/5$. (c) $-3/4$.

Formula 1.35. Frequently Used Trigonometric Formulas:
For all angle x,

$$\sin^2 x + \cos^2 x = 1 \qquad \tan x = \frac{\sin x}{\cos x} = \text{slope}$$

Graphs of Trigonometric Functions

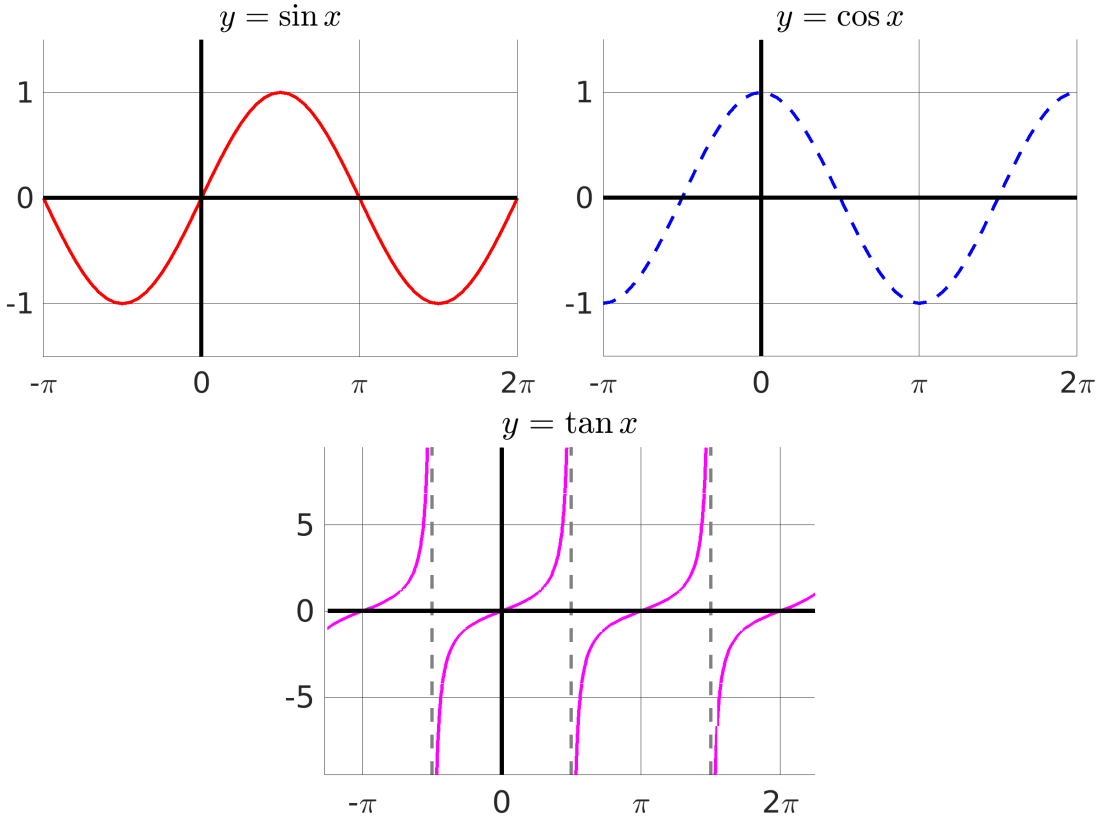


Figure 1.12: Graphs of $y = \sin x$, $y = \cos x$, and $y = \tan x$.

Periodicity of the Trigonometric Functions

Definition 1.36. A function $f(x)$ is **periodic** if there is a number $p > 0$ such that $f(x + p) = f(x)$ for every value of x. The smallest such value of p is the **period** of f.
For example,

Period of "sin" and "cos" $= 2\pi$
Period of "tan" $= \pi$

Observation 1.37. $\sin(-x) = -\sin x$ **and** $\cos(-x) = \cos x$**.**

The following holds also when $\sin$ and $\cos$ are interchanged.

- When the graph of $y = \sin x$ is moved by $\pm\pi$, the result becomes $y = -\sin x$.
- When the graph of $y = \sin(\pm x)$ is moved by $\pm\dfrac{\pi}{2}$ (or, $\pm 3\pi/2$), the result should match with either $y = \cos x$ or $y = -\cos x$.

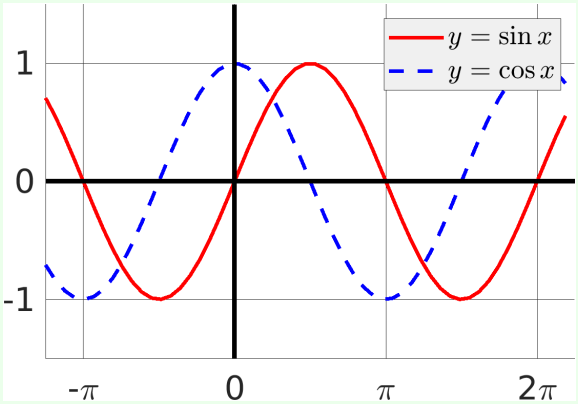


Figure 1.13: Graphs of $y = \sin x$ and $y = \cos x$, superposed.

Example 1.38. Which one is the same as $\cos\left(\dfrac{3\pi}{2} - x\right)$?

A. $\cos x$

B. $-\cos x$

C. $\sin x$

D. $-\sin x$

E. None of these

Transformations of Trigonometric Graphs

The rules for shifting, scaling, and reflecting the graph of a function, studied in §1.2 and summarized in the following, apply to the trigonometric functions:

$$\frac{y-d}{a} = f\Big(\frac{x-c}{b}\Big) \quad \text{or} \quad y = a\, f\Big(\frac{x-c}{b}\Big) + d, \tag{1.20}$$

where (b, a) are scaling factors and (c, d) are shifting units.

Definition 1.39. The transformation rules applied to the sine function give the **general sine function** or **sinusoid** formula

$$f(x) = A \sin\Big(\frac{2\pi}{B}(x-C)\Big) + D, \tag{1.21}$$

where where $|A|$ is the **amplitude**, $|B|$ is the **period**, C is the **horizontal shift**, and D is the **vertical shift**.

Example 1.40. Identify A, B, C, and D for the sine function and sketch its graph.

$$y = 2\sin\Big(\frac{x}{2} + \pi\Big) - 1$$

Solution.

Ans: $y = 2\sin\Big(\frac{2\pi}{4\pi}(x-(-2\pi))\Big) - 1$

1.3.3. Trigonometric Formulas

Identity

$$\cos^2\theta + \sin^2\theta = 1$$
$$1 + \tan^2\theta = \sec^2\theta$$
$$1 + \cot^2\theta = \csc^2\theta$$

Addition Formulas

$$\sin(A+B) = \sin A\cos B + \cos A\sin B$$
$$\cos(A+B) = \cos A\cos B - \sin A\sin B$$

Double-Angle Formulas

$$\sin 2\theta = 2\sin\theta\cos\theta$$
$$\cos 2\theta = \cos^2\theta - \sin^2\theta$$

Half-Angle Formulas

$$\sin^2\theta = \frac{1-\cos 2\theta}{2}$$
$$\cos^2\theta = \frac{1+\cos 2\theta}{2}$$

(1.22)

The Law of Cosines

If a, b, and c are sides of a triangle ABC, then

$$c^2 = a^2 + b^2 - 2ab\cos C. \tag{1.23}$$

Exercises 1.3

1. On a circle of radius 10 m, how long is an arc that subtends a central angle of

   (a) $7\pi/5$
   (b) 120°

2. Graph the functions. What is the period of each function?

   (a) $-\cos(2\pi x)$ (b) $2\sin\left(2x - \dfrac{\pi}{4}\right) + 1$

   Ans: (b) Period: π.

3. Use the Half-Angle Formulas to find the exact function values.

   (a) $\cos^2 \dfrac{\pi}{8}$ (b) $\sin^2 \dfrac{3\pi}{8}$

   Ans: (b) $1/2 + \sqrt{2}/4$.

4. Solve for the angle θ, where $0 \le \theta \le 2\pi$

   (a) $\cos^2\theta = \dfrac{3}{4}$ (b) $\cos^2\theta - \dfrac{3}{2}\sin\theta = 0$.

   Hint: (b). $\cos^2\theta = 1 - \sin^2\theta$. Thus the equation reads $\sin^2\theta + \dfrac{3}{2}\sin\theta - 1 = 0$, which is a quadratic equation of $x = \sin\theta$.

   Ans: (b). $\theta = \pi/6,\ 5\pi/6$.

5. A triangle has sides $a = 2$ and $b = 3$ and angle $C = 60^\circ$. Find the length of side c.

   Ans: $\sqrt{7}$.

1.4. Graphing with Software

- Through the course, you will learn how to
  - visualize functions, solve equations, and analyze data.
- You may use
  - Maple, Mathematica, Matlab, or **Python**

  which are available on computers campus-wide.
- However, for a bit simpler work in Calculus, you may try online graphing software. For example,
  - **GeoGebra** https://www.geogebra.org/

  for which you may find **Tutorials** https://www.geogebra.org/a/14. Learn about "CAS Calculator".

Note: In modern society, becoming a good programmer would be quite advantageous. Thus, first, try to be familiar with computational environments for computer software. I recommend you to start with one of Maple, Mathematica, and Matlab. The simplest one is **Matlab**.

Mathematica

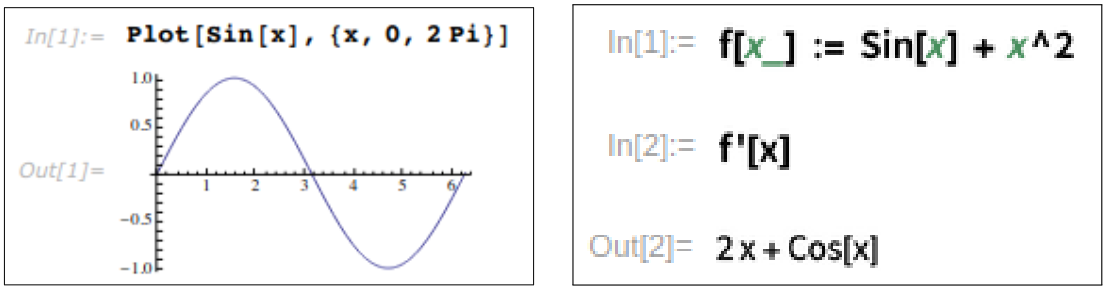


Figure 1.14: Examples in Mathematica.

Maple

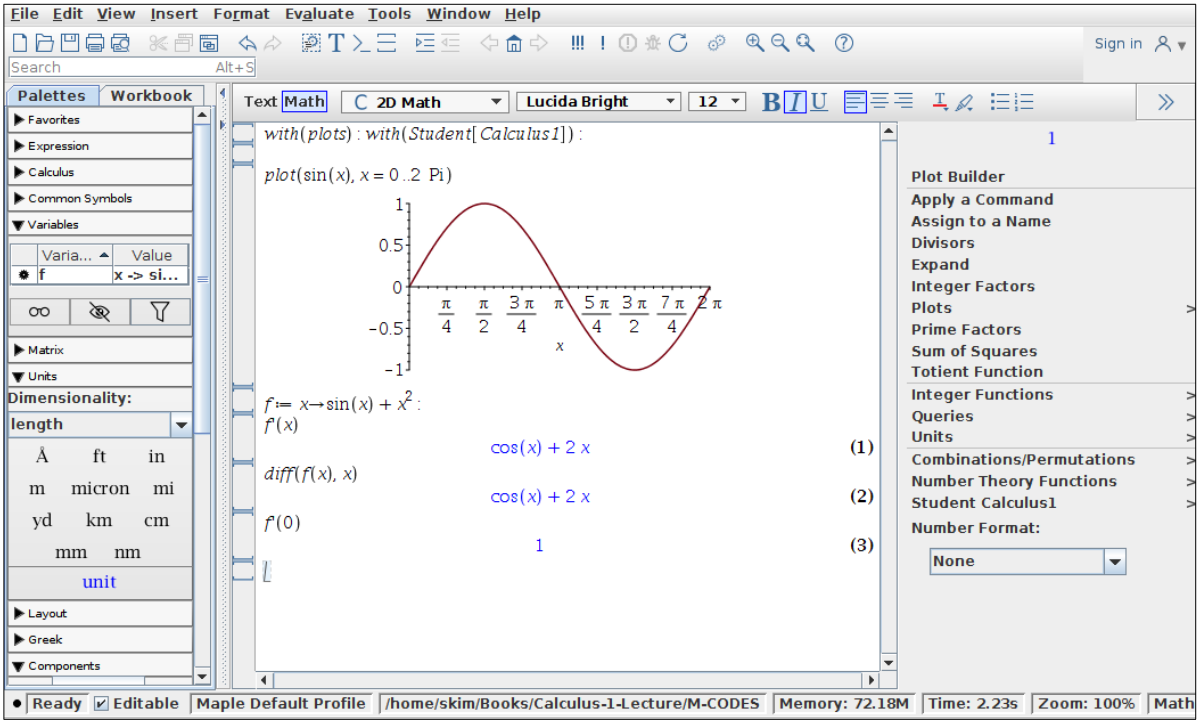


Figure 1.15: An example code in Maple.

Example 1.41. Let's play with Maple a little more.

maple-plot-derivative.mw

```
with(plots, implicitplot):
implicitplot(x^2+2 x = -y^2+4 y-1, x=-3..1, y=0..4)
implicitplot(r=1-cos(theta), r=0..2, theta=0..2 Pi, coords=polar,
     size=[250, 250], axis=[color=red], background=green)
g := x -> if x<3 then x+1 else -x^2+13 end if:
plot(g, 0..5, size=[300,200])
```

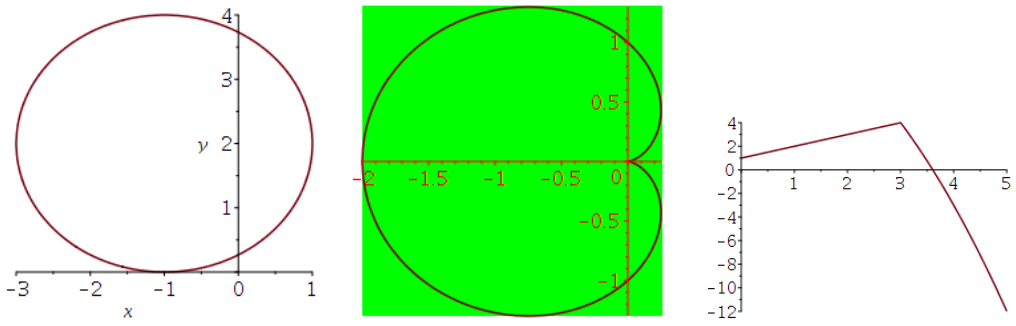


Figure 1.16: Maple plots: A circle, a cardioid, and a piecewise-defined function.

Matlab

y_sin_x.m

```
x = linspace(-pi, 2*pi, 100);
y = sin(x);

plot(x,y,'r',linewidth=2); grid on
xticks([-pi,0,pi,2*pi]); xticklabels({'-\pi','0','\pi','2\pi'})
yticks([-1,-0.5,0,0.5,1])
title('Matlab: $y=\sin x $','fontsize',18,'interpreter','latex')
axis tight
    if exist('__octave_config_info__')  %octave
        print('y=sin-x-matlab.png')
    else
        exportgraphics(gcf,'y=sin-x-matlab.png','Resolution',100)
    end
```

Python

y_sin_x.py

```
import numpy as np
from matplotlib import pyplot as plt

x = np.linspace(-np.pi, 2*np.pi, 100)
y = np.sin(x)

plt.plot(x,y,'r',linewidth=2)
plt.grid(color = 'green', linestyle = '--', linewidth = 0.5)
plt.xticks([-np.pi,0,np.pi,2*np.pi],['$-\pi$',0,'$\pi$','$2\pi$'])
plt.yticks([-1,-0.5,0,0.5,1])
plt.title('Python: $y=\sin x $',fontsize=18)
#plt.show()
plt.savefig('y=sin-x-python.png',bbox_inches='tight')
```

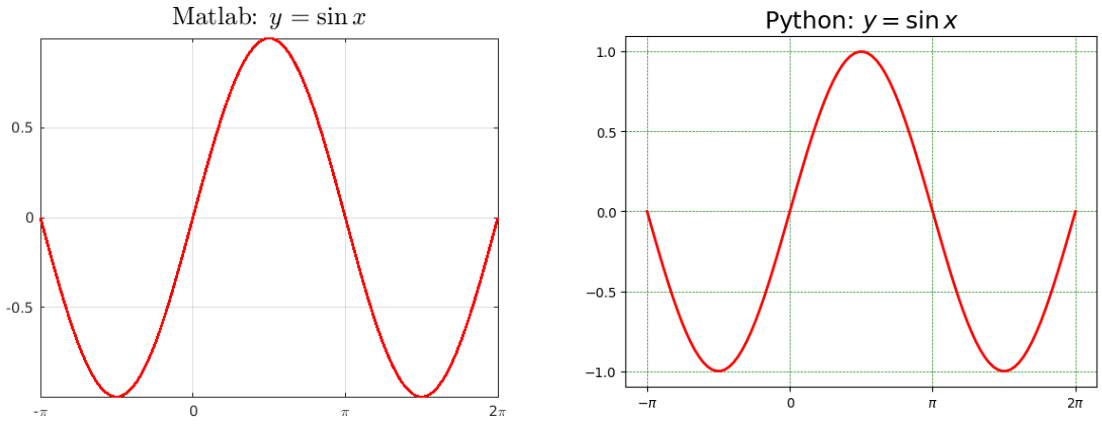


Figure 1.17: `y=sin-x-matlab.png` and `y=sin-x-python.png`

Exercises 1.4

1. CAS[2] Find an appropriate graphing software viewing window for the given function and use it to display its graph. The window should give a picture of the overall behavior of the function. There is more than one choice, but incorrect choices can miss important aspects of the function.

   (a) $f(x) = x^4 - 4x^3 + 15$

   (b) $f(x) = x\sqrt{9 - x^2}$

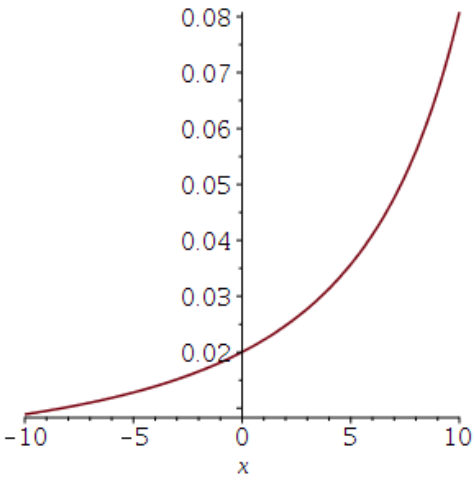
   (c) $y = x^{1/3}(x^2 - 8)$

   (d) $y = \dfrac{8}{(x-20)^2 - 1}$

   Note: **Finding a Viewing Window**:

   If you do not specify the viewing window, then the CAS will use a default window.

   - For example, for (d), if you implement $plot\left(\dfrac{8}{(x-20)^2-1}\right)$ in Maple, then you will see the figure right.
   - You should specify a viewing window, as in Example 1.41, not to miss important aspects of the function.
   - A good strategy for the determination of a viewing window is to (1) ***start with a large window*** and then (2) ***specify a right one to see important features in detail***.

2. CAS Graph the function $y = \cos x + \dfrac{1}{100}\sin 100x$

3. CAS Graph four periods of the function $f(x) = -\tan 2x$.

[2] Problems with the sign CAS must be done with a **Computer Algebra System** such as Maple, Mathematica, or Matlab.

1.5. Exponential Functions

Definition 1.42. A function of the form

$$f(x) = a^x, \quad \text{where } a > 0 \text{ and } a \neq 1, \tag{1.24}$$

is called an **exponential function** (with base a).

- All exponential functions have domain $(-\infty, \infty)$ and range $(0, \infty)$, so an exponential function never assumes the value 0.
- All exponential functions are either increasing ($a > 1$) or decreasing ($0 < a < 1$) over the whole domain.

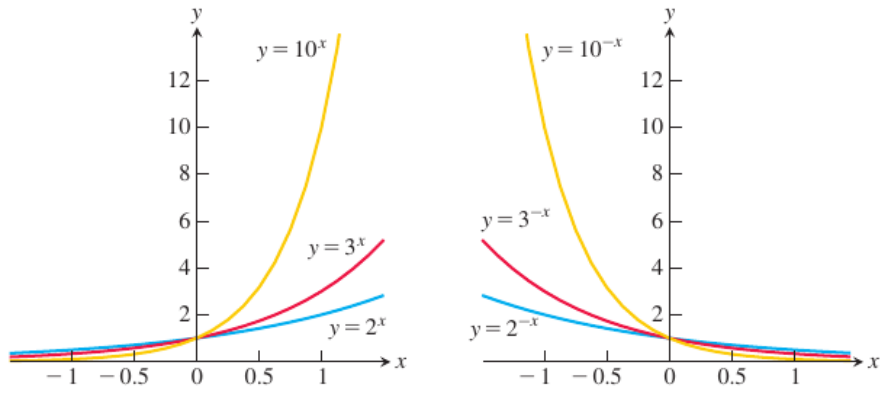


Figure 1.18: Exponential functions.

Example 1.43. Sketch the graph of the function $f(x) = 3 - 2^x$ and determine its domain and range.

Solution.

Example 1.44. Table 1.1 shows data for the population of the world in the 20th century. Figure 1.19 shows the corresponding scatter plot.

- The pattern of the data points in Figure 1.19 suggests an exponential growth.
- Use an **exponential regression** algorithm to find a model of the form

$$P(t) = a \cdot b^t, \tag{1.25}$$

where $t = 0$ corresponds to 1900.

Table 1.1

t (years since 1900)	Population P (millions)
0	1650
10	1750
20	1860
30	2070
40	2300
50	2560
60	3040
70	3710
80	4450
90	5280
100	6080
110	6870

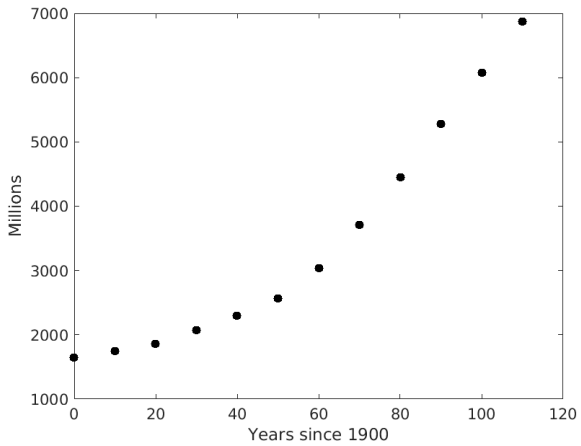


Figure 1.19: Scatter plot for world population growth.

Solution.

population.m

```
Data =[0 1650; 10 1750; 20 1860; 30 2070;
       40 2300; 50 2560; 60 3040; 70 3710;
       80 4450; 90 5280; 100 6080; 110 6870];
m = size(Data,1);

% exponential model, through linearization
A = ones(m,2);
A(:,2) = Data(:,1);
r = log(Data(:,2));
lm = (A'*A)\(A'*r);
a = exp(lm(1)), b = exp(lm(2)),
```

```
12
13  plot(Data(:,1),Data(:,2),'k.','MarkerSize',20)
14     xlabel('Years since 1900');
15     ylabel('Millions'); hold on
16     print -dpng 'population-data.png'
17  t = Data(:,1);
18  plot(t,a*b.^t,'r-','LineWidth',2)
19     print -dpng 'population-regression.png'
20     hold off
```

The program results in

$$a = 1.4365 \times 10^3, \quad b = 1.0140.$$

Thus the exponential model reads

$$P(t) = (1.4365 \times 10^9) \cdot (1.0140)^t. \quad \textbf{(1.26)}$$

Figure 1.20 shows the graph of this exponential function together with the original data points. We see that the exponential curve fits the data reasonably well. □

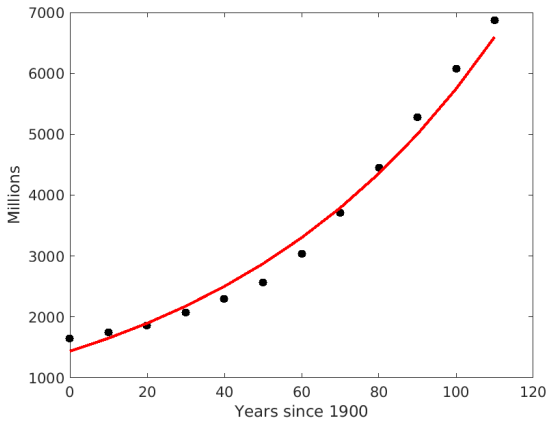


Figure 1.20: Exponential model for world population growth.

Integer and Rational Exponents

- When $x = n$ is a positive integer,
$$a^n = \underbrace{a \cdot a \cdot \ldots \cdot a}_{n \text{ times}}.$$
- When $x = -n$ for some positive integer n,
$$a^{-n} = \frac{1}{a^n} = \left(\frac{1}{a}\right)^n.$$
- When $x = p/q$ is a rational number,
$$a^{p/q} = \sqrt[q]{a^p} = (\sqrt[q]{a})^p$$

Laws of Exponents

If $a > 0$ and $b > 0$, the following rules hold for all real numbers x and y.

1. $a^x \cdot a^y = a^{x+y}$
2. $\dfrac{a^x}{a^y} = a^{x-y}$
3. $(a^x)^y = (a^y)^x = a^{xy}$
4. $a^x \cdot b^x = (ab)^x$
5. $\dfrac{a^x}{b^x} = \left(\dfrac{a}{b}\right)^x$

The Number e

Of all possible bases for an exponential function, there is one that is most convenient for the purposes of calculus. The choice of a base a is influenced by the way the graph of $y = a^x$ crosses the y-axis.

- As we will see later, that some of the formulas of calculus will be greatly simplified, if we choose the base a so that **the slope of the tangent line to $y = a^x$ at $x = 0$ is exactly** 1. See Exercise 5, p. 42.
- In fact, there is such a number and it is denoted by the letter e. (This notation was chosen by the Swiss mathematician **Leonhard Euler** in 1727, probably because it is the first letter of the word exponential.)
- It turns out that the number e lies between 2 and 3. Later, we will see that the value of e, correct to six decimal places, is

$$e \approx 2.718282 \tag{1.27}$$

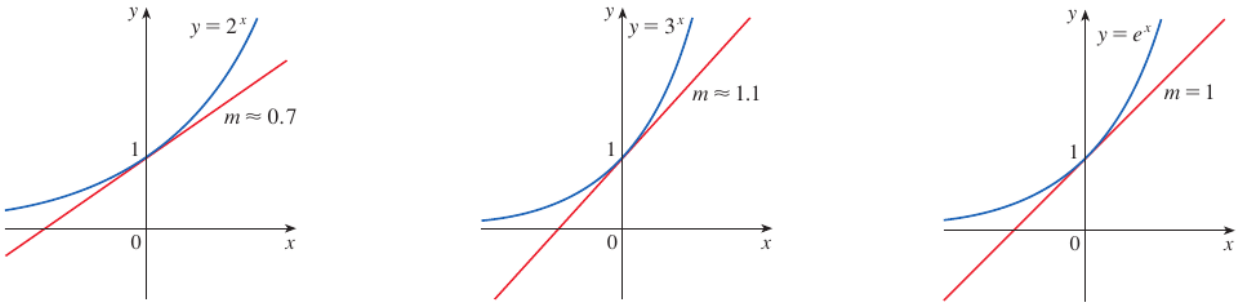


Figure 1.21: The number e.

Remark 1.45. Properties of the Natural Exponential Function
The exponential function $f(x) = e^x$ is an increasing continuous function with domain $\mathbb{R}$ and range $(0, \infty)$. Thus $e^x > 0$ for all x and the x-axis is a **horizontal asymptote** of $f(x) = e^x$.

Example 1.46. Find the domain of the following functions.

(a) $f(x) = \dfrac{1+x}{e^{\cos x}}$

(b) $f(x) = \dfrac{1-e^{x^2}}{1-e^{1-x^2}}$

Solution.

Example 1.47. Graph the function $y = \frac{1}{2}e^{-x} + 1$ and state the domain and range.

Solution.

Exercises 1.5

1. Use the Laws of Exponents to rewrite and simplify each expression.

   (a) $16^2 \cdot 16^{-1.75}$ (b) $\dfrac{3^{5/3}}{3^{2/3}}$ (c) $(\sqrt{3})^{1/2} \cdot (\sqrt{12})^{1/2}$ (d) $(36^{\sqrt{2}})^{\sqrt{2}/4}$

   Ans: (a) 2. (d) 6.

2. Make a rough sketch by hand of the graph of the function.

   (a) $y = e^{|x|}$ (b) $h(x) = 2\Big(\dfrac{1}{2}\Big)^x - 3$

3. The population of **Starkville**, Mississippi, was 2,689 in the year 1900 and 25,495 in 2020. Assume that the population in Starkville grows exponentially with the model

   $$P_n = P_0 \cdot (1 + r)^n,$$

   where n is the elapsed year and r denotes the growth rate per year. Then, we can find $r = 0.018921 (= 1.8921\%)$.

   (a) Estimate the population in 1950 and 2000.
   (b) Approximately when is the population going to reach 50,000?

   Ans: (b) 2056.

4. Let $f(x) = 5^x$. Show that

   $$\frac{f(x+h) - f(x)}{h} = 5^x\Big(\frac{5^h - 1}{h}\Big)$$

5. The number e is determined so that the slope of the graph of $y = e^x$ at $x = 0$ is exactly 1. Let h be a point near 0. Then

   $$S(h) := \frac{e^h - e^0}{h - 0} = \frac{e^h - 1}{h}$$

   represents the **average slope** of the graph between the two points $(0, 1)$ and (h, e^h). Use your calculator to evaluate $S(h)$, for $h = 0.1,\ 0.01,\ 0.001,\ 0.0001$. What can you say about the results?

   Ans: For example, $S(0.01) = 1.0050$.

1.6. Inverse Functions and Logarithms

1.6.1. Inverse Functions

Key Idea **1.48.** Let $f : X \to Y$ be a function. For simplicity, consider

$$y = f(x) = 2x + 1. \tag{1.28}$$

- Then, f is a rule that performs two actions: $\times 2$ and followed by $+1$.
- The **reverse** of f must be: -1 followed by $\div 2$.
  - Let $y \in Y$. Then the reverse of f can be written as

$$x = (y-1)/2 =: g(y) \tag{1.29}$$

    The function g is the **inverse function** of f.
  - However, it is conventional to choose x for the independent variable. Thus it can be formulated as

$$y = g(x) = (x-1)/2. \tag{1.30}$$

- Let's summarize the above:

  (a) **Solve $y = f(x)$ for x:** $x = (y-1)/2 =: g(y)$.
  (b) **Exchange x and y:** $y = g(x) = (x-1)/2$.

Note: The first step for finding the inverse function of f is to solve $y = f(x)$ for x, to get $x = g(y)$. Here the required is **for g to be a function**.

Definition **1.49.** A function f is called a **one-to-one function** if it never takes on the same value twice; that is,

$$f(x_1) \neq f(x_2) \quad \text{whenever } x_1 \neq x_2. \tag{1.31}$$

Claim 1.50. Horizontal Line Test.
A function is **one-to-one** if and only if no horizontal line intersects its graph more than once.

- $f(x) = x^3$ is one-to-one.
- The function in Figure 1.23 is not one-to-one because $f(x_1) = f(x_2)$ for $x_1 \neq x_2$.

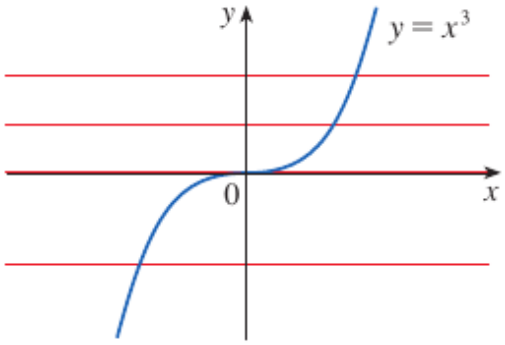


Figure 1.22: $f(x) = x^3$ is one-to-one.

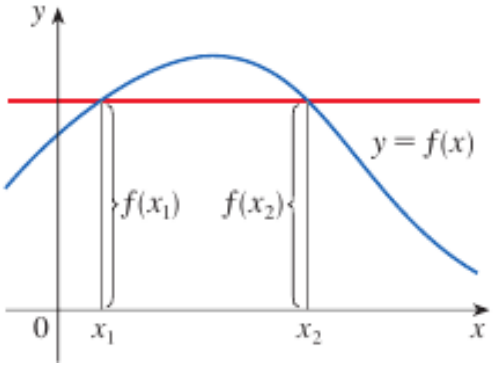


Figure 1.23: This function is not one-to-one because $f(x_1) = f(x_2)$ for $x_1 \neq x_2$.

Example 1.51. Check if the function is one-to-one.

1. $f(x) = x^2$
2. $g(x) = x^2$, $x \geq 0$
3. $h(x) = x^3$

Solution.

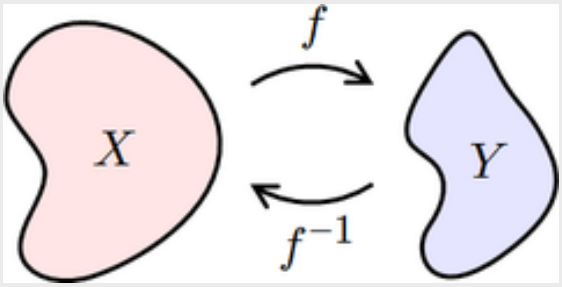
Definition 1.52. Let f be a one-to-one function with domain X and range Y. Then its **inverse function** f^{-1} has domain Y and range X and is defined by

$$f^{-1}(y) = x \iff f(x) = y, \tag{1.32}$$

for any $y \in Y$.

Remark 1.53.
The definition says that if f maps x into y, then f^{-1} maps y back into x. From (1.32), we can obtain the **cancellation equations**

$$\begin{aligned} f^{-1}(f(x)) &= x \quad \text{for all } x \in X \\ f(f^{-1}(y)) &= y \quad \text{for all } y \in Y \end{aligned} \tag{1.33}$$



Example 1.54. For example, if $f(x) = x^3$, then $f^{-1}(x) = x^{1/3}$ and so that the cancellation equations read

$$\begin{aligned} f^{-1}(f(x)) &= f^{-1}(x^3) = (x^3)^{1/3} = x \\ f(f^{-1}(y)) &= f(y^{1/3}) = (y^{1/3})^3 = y \end{aligned}$$

Example 1.55. Assume f is a one-to-one function.

(a) If $f(1) = 5$, what is $f^{-1}(5)$?

(b) If $f^{-1}(8) = -10$, what is $f(-10)$?

Solution.

Caution **1.56.**

- Do not mistake the -1 in f^{-1} for an exponent.

$$f^{-1}(x) \text{ does not mean } \frac{1}{f(x)}. \tag{1.34}$$

- If f were not one-to-one, then its inverse would not be uniquely defined and cannot be a function. $\Rightarrow$ An inverse function does not exist.

Strategy **1.57. How to Find the Inverse Function of a One-to-One Function** f: Write $y = f(x)$.

Step 1: Solve this equation for x in terms of y (if possible).

Step 2: Interchange x and y; the resulting equation is $y = f^{-1}(x)$.

Example **1.58.** Find the inverse of the function $h(x) = \dfrac{6 - 3x}{5x + 7}$.

Solution.

Example 1.59. Find the inverse of the function $f(x) = x^3 - 2$, expressed as a function of x.

Solution. Write $y = x^3 - 2$.

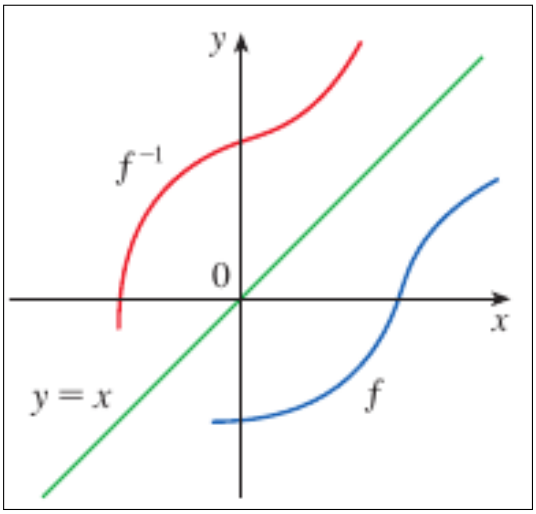
Step 1: Solve it for x:

$$x^3 = y + 2 \;\Rightarrow\; \boldsymbol{x = \sqrt[3]{y+2}}.$$

Step 2: Exchange x and y:

$$\boldsymbol{y = \sqrt[3]{x+2}}.$$

Therefore the inverse function is
$f^{-1}(x) = \sqrt[3]{x+2}$.



Observation 1.60. The graph of f^{-1} is obtained by reflecting the graph of f about the line $y = x$.

Example 1.61. Let $f(x) = x^2 - 2x + 1$, $x \geq 1$. Then the range of f is $[0, \infty)$.

(a) Find the inverse of f.

(b) Find its domain and range.

(c) Plot the graphs of f and f^{-1} in the same coordinates.

Solution.

1.6.2. Logarithmic Functions

Recall: If $a > 0$ and $a \neq 1$, the exponential function $f(x) = a^x$ is either increasing or decreasing and so it is one-to-one by the Horizontal Line Test. It therefore has an inverse function.

Definition 1.62. The **logarithmic function with base a**, written $y = \log_a x$, is the inverse of $y = a^x$ ($a > 0$, $a \neq 1$). That is,

$$\log_a x = y \iff a^y = x. \tag{1.35}$$

Example 1.63. Find the inverse of $y = 2^x$.

Solution.

1. **Solve $y = 2^x$ for x**:

$$x = \log_2 y$$

2. **Exchange x and y**:

$$y = \log_2 x$$

Thus the graph of $y = \log_2 x$ must the reflection of the graph of $y = 2^x$ about $y = x$.

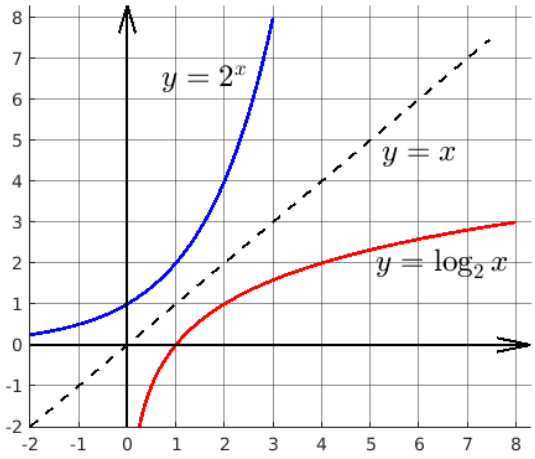


Figure 1.24: Graphs of $y = 2^x$ and $y = \log_2 x$.

Note:

- Equation (1.35) represents the action of "**solving for x**"
- The domain of $y = \log_a x$ must be the range of $y = a^x$, which is $(0, \infty)$.

The Natural Logarithm and the Common Logarithm

Of all possible bases a for logarithms, we will see later that the most convenient choice of a base is the number e.

Definition 1.64.

- The logarithm with base e is called the **natural logarithm** and has a special notation:
$$\log_e x = \ln x \tag{1.36}$$
- The logarithm with base 10 is called the **common logarithm** and has a special notation:
$$\log_{10} x = \log x \tag{1.37}$$

Remark 1.65.

- From your calculator, you can see buttons of LN and LOG, which represent $\ln = \log_e$ and $\log = \log_{10}$, respectively.
- When you implement **a code on computers**, the functions $\ln$ and $\log$ can be called by "`log`" and "`log10`", respectively.

Properties of Logarithms

- **Algebraic Properties**: for $(a > 0,\ a \neq 1)$

Product Rule:	$\log_a xy$	=	$\log_a x + \log_a y$
Quotient Rule:	$\log_a \dfrac{x}{y}$	=	$\log_a x - \log_a y$
Power Rule:	$\log_a x^\alpha$	=	$\alpha \log_a x$
Reciprocal Rule:	$\log_a \dfrac{1}{x}$	=	$-\log_a x$

(1.38)

- **Inverse Properties**

$a^{\log_a x} = x,\ \ x > 0;$	$\log_a a^x = x,\ \ x \in \mathbb{R}$
$e^{\ln x} = x,\ \ x > 0;$	$\ln e^x = x,\ \ x \in \mathbb{R}$

(1.39)

Example 1.66. Use the laws of logs to expand $\ln\left(\dfrac{x^2\sqrt{x^2+3}}{3x+1}\right)$.

Solution.

Example 1.67. Simplify the following.

(a) $\log_3 75 - 2\log_3 5$

(b) $2\log_5 100 - 4\log_5 50$

Solution.

Example 1.68. Solve for x.

(a) $e^{5-3x} = 3$.

(b) $\log_3 x + \log_3(x-2) = 1$

(c) $\ln(\ln x) = 0$

Solution.

Ans: (a) $x = \frac{1}{3}(5 - \ln 3)$. (b) $x = 3$. **(Caution: $x = -1$ cannot be a solution.)**

Claim 1.69.

(a) Every exponential function is a power of the natural exponential function.

$$a^x = e^{x \ln a}. \tag{1.40}$$

(b) Every logarithmic function is a constant multiple of the natural logarithm.

$$\log_a x = \frac{\ln x}{\ln a}, \quad (a > 0,\ a \neq 1) \tag{1.41}$$

which is called the **Change-of-Base Formula**.

Proof. (a). $a^x = e^{\ln(a^x)} = e^{x \ln a}$.

(b). $\ln x = \ln(a^{\log_a x}) = (\log_a x)(\ln a)$, from which one can get (1.41). □

1.6.3. Inverse Trigonometric Functions

Note: Trigonometric functions are periodic.

- When we try to find the inverse trigonometric functions, we have a difficulty: because the trigonometric functions are not one-to-one.
- The difficulty is overcome by restricting the domains of these functions so that they become one-to-one.

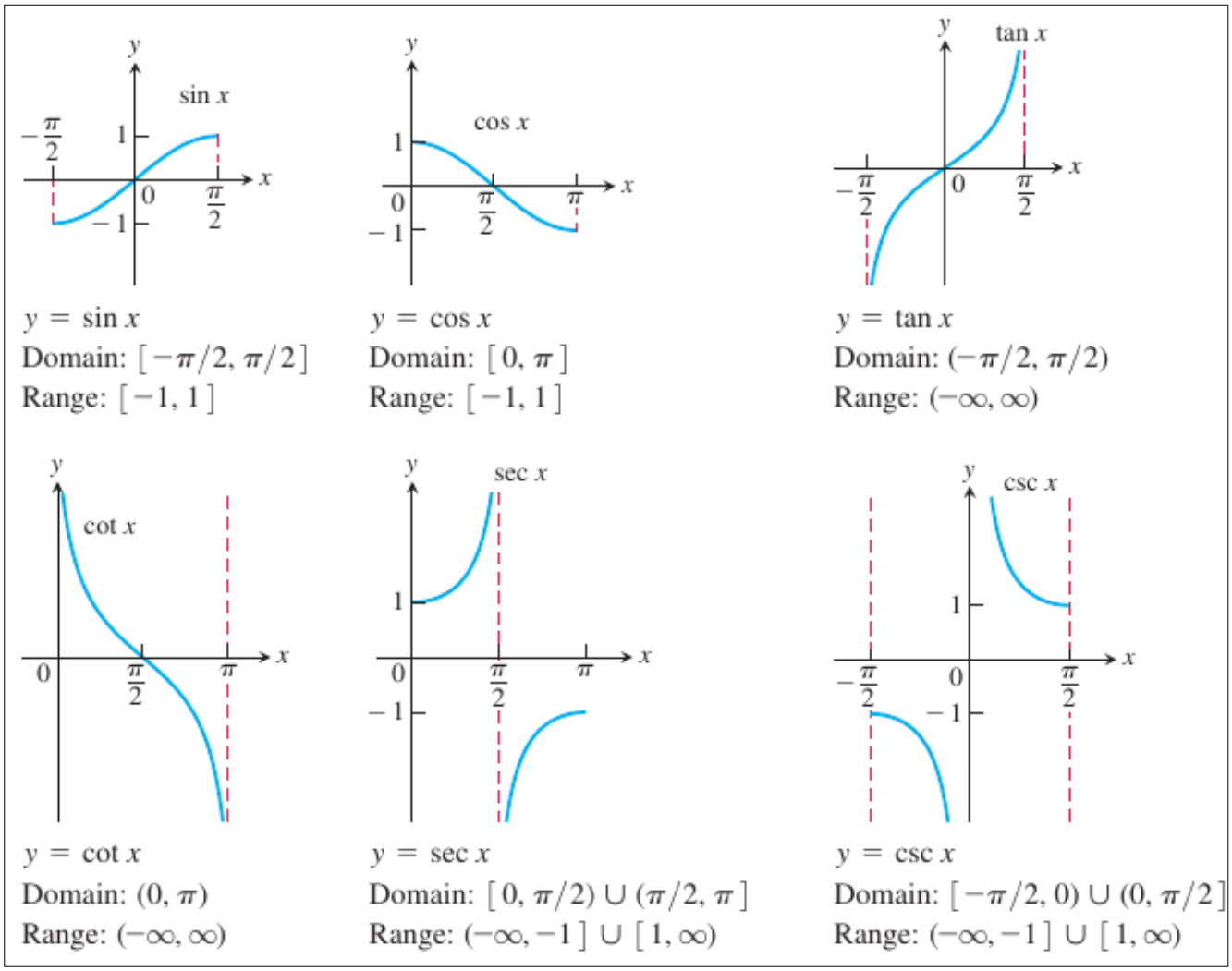


Figure 1.25: Domain restrictions that make the trigonometric functions one-to-one.

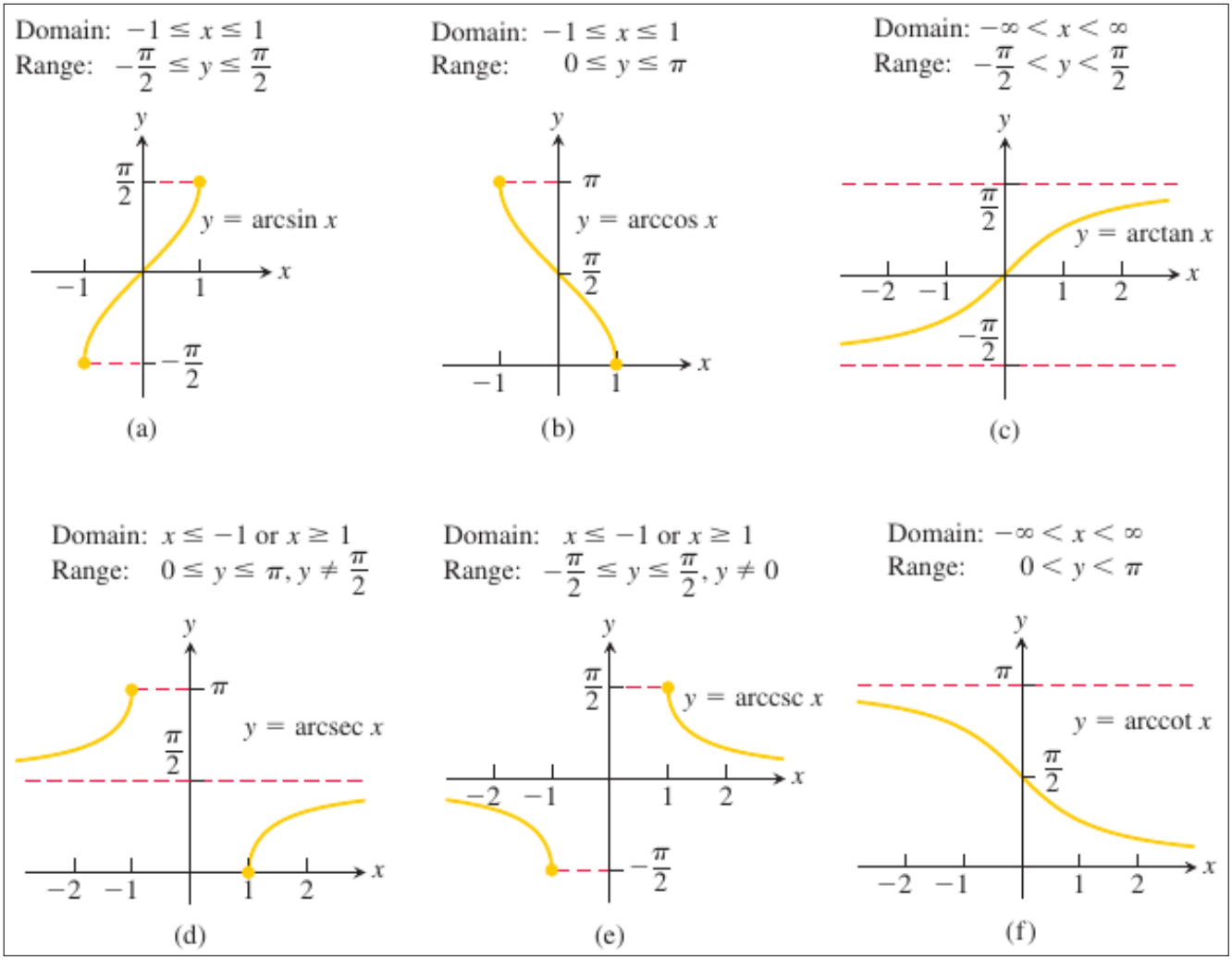


Figure 1.26: Graphs of the six basic inverse trigonometric functions.

Definition 1.70. (The Arcsine and Arccosine Functions)

$y =$ **arcsin** x is the number in $[-\pi/2, \pi/2]$ for which $\sin y = x$
$y =$ **arccos** x is the number in $[0, \pi]$ for which $\cos y = x$

Example 1.71. Evaluate (a) $\arcsin\left(\frac{\sqrt{3}}{2}\right)$ and (b) $\arccos\left(-\frac{1}{2}\right)$.

Solution.

Ans: (b). $2\pi/3$.

Example 1.72. Is it correct? If not, why?

(a) $\arcsin\left(\sin\dfrac{9\pi}{4}\right) = \dfrac{9\pi}{4}$ (b) $\cos(\arccos 2) = 2$

Solution.

Definition 1.73. (Inverses of $\tan x$, $\cot x$, $\sec x$, and $\csc x$)

$y = \textbf{arctan}\, x$ is the number in $(-\pi/2, \pi/2)$ for which $\tan y = x$
$y = \textbf{arccot}\, x$ is the number in $(0, \pi)$ for which $\cot y = x$
$y = \textbf{arcsec}\, x$ is the number in $[0, \pi]\backslash\{\pi/2\}$ for which $\sec y = x$
$y = \textbf{arccsc}\, x$ is the number in $[-\pi/2, \pi/2]\backslash\{0\}$ for which $\csc y = x$

Example 1.74. Find the exact value for the expression.

(a) $\cos^{-1}(-1)$

(b) $\arctan(-1)$

(c) $\sin\left(\arcsin\left(-\dfrac{1}{\sqrt{2}}\right)\right)$

Solution.

Example **1.75.** Simplify the expression $\cos(\tan^{-1} x)$.

Solution. Let $y = \tan^{-1} x$. Then $y \in (-\pi/2, \pi/2)$ and $\tan y = x$. Here

$$\begin{cases} \text{What we must find :} & \cos(y) \\ \text{What we know :} & \tan(y) = x \end{cases} \tag{1.42}$$

Now, $\sec^2 y = 1 + \tan^2 y = 1 + x^2$.

$\Rightarrow \sec y = \sqrt{1 + x^2}$ (because $\sec y > 0$ for $y \in (-\pi/2, \pi/2)$)

$\Rightarrow \cos(y) = \dfrac{1}{\sqrt{x^2 + 1}}$.

Ans: $\cos(\tan^{-1} x) = \dfrac{1}{\sqrt{x^2 + 1}}$.

Note: You can reach the final outcome from (1.42), with *a geometric manipulation*.

Self-study **1.76.** Simplify the expression $\sin(\tan^{-1} x)$.

Solution.

Ans: $\sin(\tan^{-1} x) = \dfrac{x}{\sqrt{x^2 + 1}}$.

Exercises 1.6

1. Let $f(x) = \sqrt{1-x^2}$, $0 \le x \le 1$.

   (a) What symmetry does the graph have?
   (b) Show that f is its own inverse.

   Hint: (b) Use Strategy 1.57.

   Ans: (a). It is symmetric about $y = x$.

2. Find a formula for the inverse of the function; identify its domain and range.

   (a) $f(x) = x^2 - 2x$, $x \le 1$

   (b) $f(x) = \dfrac{x+3}{x-2}$

   Ans: (b). $f^{-1}(x) = (2x+3)/(x-1)$; D: $\{x \ne 1\}$; R: $\{y \ne 2\}$

3. Simplify to find the exact value of each expression.

   (a) $\log_5(1/125)$
   (b) $\ln(\ln e^{e^{30}})$
   (c) $e^{-2\ln 5}$
   (d) $e^{\ln(\ln e^2)}$

   Ans: (b) 30. (d) 2.

4. Solve each equation for x.

   (a) $\ln(x+3) = 2$

   (b) $\ln(\ln x) = 1$

   Ans: (b) $x = e^e$.

5. For $\log_2 x + \log_2(x-3) = 2$.

   (a) Solve it for x.
   (b) How many solutions does it have? If there is only one solution, why?

6. Find the exact value of each expression.

   (a) $\tan^{-1}\sqrt{3}$
   (b) $\arcsin(\sin(5\pi/4))$
   (c) $\sec^{-1} 2$
   (d) $\cos(\sin^{-1}(\frac{5}{13}))$

   Ans: (d) $12/13$.

CHAPTER 2

Limits and Continuity

In This Chapter:

Topics	Applications/Properties
Rates of Change Average Rates of Change Tangent Lines to Curves	Secant Lines
Limit of a Function Limit Laws Sandwich Theorem	Basics of Computer Programming
The Precise Definition of a Limit	
One-Sided Limits	
Continuity Limits of Continuous Functions Intermediate Value Theorem	
Limits Involving Infinity Horizontal Asymptotes Oblique Asymptotes Vertical Asymptotes	

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Lecture-Note: https://skim.math.msstate.edu/LectureNotes/Calculus-1-Lecture.pdf

2.1. Rates of Change and Tangent Lines to Curves

Note: In the late 16th century, **Galileo** discovered that a solid object dropped from rest (initially not moving) near the surface of the earth and allowed to fall freely will fall a distance proportional to the square of the time it has been falling.

- This type of motion is called **free fall**.
- It assumes negligible air resistance to slow the object down, and that gravity is the only force acting on the falling object.
- If y denotes the distance fallen in feet after t seconds, then the **Galileo's law of free-fall** is

$$y = 16t^2 \text{ ft.} \tag{2.1}$$

The Galileo's law of free-fall states that, in the absence of air resistance, all bodies fall with the same acceleration, independent of their mass.

Average and Instantaneous Speed

Average Speed. When $f(t)$ measures the distance traveled at time t,

$$\text{Average speed over } [t_0, t_1] = \frac{\text{distance traveled}}{\text{elapsed time}} = \frac{f(t_1) - f(t_0)}{t_1 - t_0} \tag{2.2}$$

Example 2.1. A rock breaks loose from the top of a tall cliff. What is its average speed

(a) during the first 2 sec of fall?

(b) during the 1-sec interval between second 1 and second 2?

Solution.

Ans: (a) 32 ft/sec. (b) 48 ft/sec.

Example 2.2. Find **the speed** of the falling rock in Example 2.1 at $t = 1$ and $t = 2$.

Solution. We can calculate the average speed of the rock over a time interval $[t_0, t_0 + h]$, having length $\Delta t = (t_0 + h) - (t_0) = h$, as

$$\frac{\Delta y}{\Delta t} = \frac{16(t_0 + h)^2 - 16t_0^2}{h} \frac{\text{ft}}{\text{sec}}. \tag{2.3}$$

- We cannot use this formula to calculate the **instantaneous speed** at the exact moment t_0 by simply substituting $h = 0$, because we cannot divide by zero.
- But we can use it to calculate average speeds over shorter and shorter time intervals.
- When we do so, by taking smaller and smaller values of h, we see a pattern.

Table 2.1: Average speeds over short time intervals $[t_0, t_0 + h]$.

Length of time interval h	Average speed over interval of length h starting at $t_0 = 1$	Average speed over interval of length h starting at $t_0 = 2$
1	48	80
0.1	33.6	65.6
0.01	32.16	64.16
0.001	32.016	64.016
0.0001	32.0016	64.0016

- The ***average speed*** on intervals starting at $t_0 = 1$ seems to approach a limiting value of 32, as the length of the interval decreases.
- This suggests that the rock is falling at a ***speed*** of 32 ft/sec at $t_0 = 1$.

Let's confirm this algebraically.

- When $t_0 = 1$, Equation (2.3) reads

$$\begin{aligned} \frac{\Delta y}{\Delta t}(t_0 = 1) &= \frac{16(1+h)^2 - 16(1)^2}{h} = \frac{16(1+2h+h^2) - 16(1)^2}{h} \\ &= \frac{32h + 16h^2}{h} = 32 + 16h \to 32, \quad \text{as } h \to 0. \end{aligned} \tag{2.4}$$

- Similarly, when we set $t_0 = 2$, Equation (2.3) reads

$$\begin{aligned} \frac{\Delta y}{\Delta t}(t_0 = 2) &= \frac{16(2+h)^2 - 16(2)^2}{h} = \frac{16(4+4h+h^2) - 16(2)^2}{h} \\ &= \frac{64h + 16h^2}{h} = 64 + 16h \to 64, \quad \text{as } h \to 0. \end{aligned} \tag{2.5}$$

As h gets closer and closer to 0, the average speed has the limiting value 64 ft/sec when $t_0 = 2$ sec, as suggested by Table 2.1. □

Average Rates of Change and Secant Lines

Definition 2.3. The **average rate of change** of $y = f(x)$ with respect to x over $[x_1, x_2]$, $x_2 = x_1 + h$, is

$$\frac{\Delta y}{\Delta x} = \frac{f(x_2) - f(x_1)}{x_2 - x_1} = \frac{f(x_1 + h) - f(x_1)}{h}, \quad h \neq 0. \tag{2.6}$$

Remark 2.4. Geometrically, the rate of change of f over $[x_1, x_2]$ is the slope of the line through the points $P(x_1, f(x_1))$ and $Q(x_2, f(x_2))$ (Figure 2.1). In geometry, a line joining two points of a curve is called a **secant line**. Thus, the **average rate of change** of f over $[x_1, x_2]$ is identical with the slope of secant line PQ.

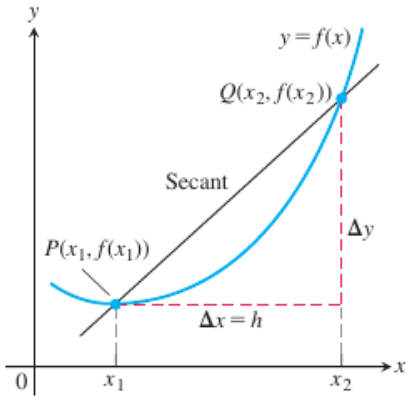


Figure 2.1: A secant to the graph of $y = f(x)$.

Defining the Tangent Line

We use an approach that analyzes the behavior of the secant lines that pass through P and nearby points Q as Q moves toward P along the curve (Figure 2.2).

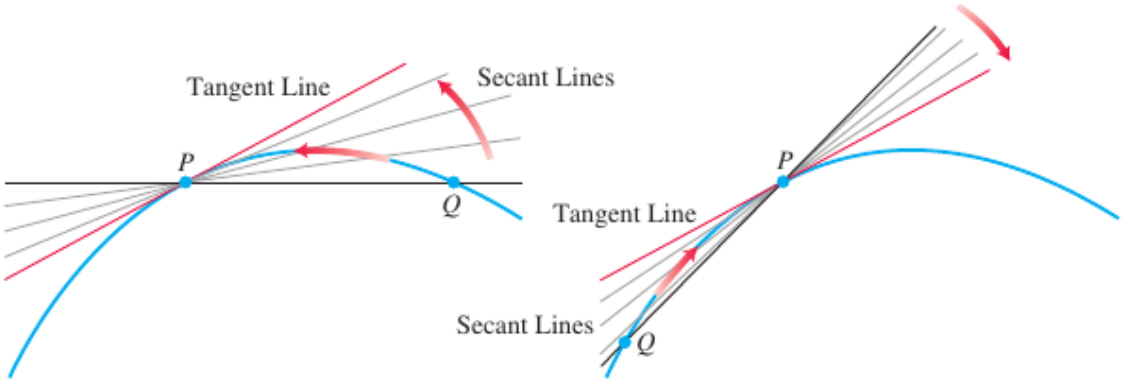


Figure 2.2: Secant lines and the tangent line to the curve at P.

Definition 2.5. (Rough Definition of the Tangent Line)
The **tangent line** to the curve at P is the **limit** of the secant lines, as $Q \to P$ from either side. (We clarify the limit idea in the next section.)

Example 2.6. Find the slope of the tangent line to the parabola $y = x^2$ at the point $(2, 4)$ by analyzing the slopes of secant lines through $(2, 4)$. Write an equation for the tangent line to the parabola at this point.

Solution. The slope of the secant line through $(2, 2^2)$ and $(2 + h, (2 + h)^2)$:

Ans: $y - 4 = 4(x - 2)$.

Remark 2.7. Examples 2.2 and 2.6 show that if the **average rate of change** of $y = f(x)$ converges to a certain number, as $h \to 0$, then the number can be viewed as the **instantaneous rate of change**, i.e., the **slope of the tangent line**.

Exercises 2.1

1. Use the method in Example 2.6 to find (1) the slope of the curve at the given point P, and (2) an equation of the tangent line at P.

   (a) $y = 7 - x^2$, $P(2, 3)$ (b) $y = x^3 - 12x$, $P(1, -11)$

   Ans: (b) slope: -9

2. The deck of a bridge is suspended 275 feet above a river. If a pebble falls off the side of the bridge, the height, in feet, of the pebble above the water surface after t seconds is given by $y = 275 - 16t^2$.

   (a) Find the average velocity of the pebble for the time period beginning when $t = 4$ and lasting

   (i) 0.1 seconds (ii) 0.05 seconds (iii) 0.01 seconds

   (b) Estimate the instantaneous velocity of the pebble when $t = 4$.

   Ans: (a)(ii). -128.8 ft/sec

3. The table shows the position of a motorcyclist after accelerating from rest. Find the average velocity for each time period:

   (a) $[2, 4]$ (b) $[3, 4]$ (c) $[4, 6]$

t (sec)	0	1	2	3	4	5	6
s (feet)	0	4.9	20.6	46.5	79.2	124.8	176.7

4. Let $f(x) = x^2 - x$.

   (a) Use the method in (2.4)-(2.5), p.60, to estimate the instantaneous slopes of f at four different points: $(x, f(x))$ for $x = 0, 1, 2, 3$.

   (b) Plot the results of (a) in the x-and-slope coordinates to find the best-fitting curve.

   Ans: (b) slope$(x) = 2x - 1$.

2.2. Limit of a Function and Limit Laws

Example 2.8. How does the function

$$f(x) = \frac{x^2 - 1}{x - 1}$$

behave near $x = 1$?

Solution. The given formula defines f for all real numbers x except $x = 1$ (since we cannot divide by zero). When $x \neq 1$, it can be simplified as

$$f(x) = \frac{(x+1)(x-1)}{x-1} = x + 1, \text{ for } x \neq 1.$$

- The graph of f is the line $y = x + 1$ with the point $(1, 2)$ removed.
- This removed point is shown as a **hole** in Figure 2.3.
- Even though $f(1)$ is not defined, it is clear that we can make the value of $f(x)$ as close as we want to 2 by choosing x close enough to 1.

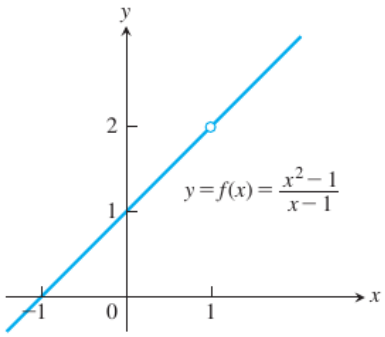


Figure 2.3: The graph of $y = (x^2 - 1)/(x - 1)$.

Intuitive Definition of a Limit

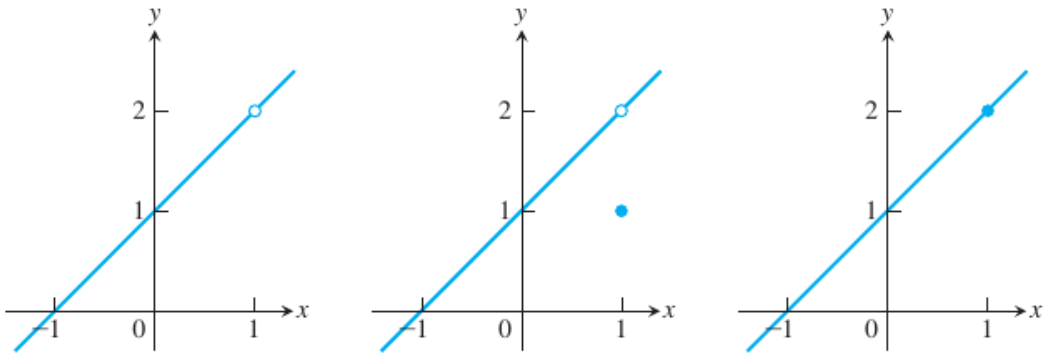
Definition 2.9. Suppose that $f(x)$ is defined on an open interval about a, **except possibly at** a. Then we write

$$\lim_{x \to a} f(x) = L, \tag{2.7}$$

and say **"the limit $f(x)$, as x approaches a, is L"**,
if we can make the values of $f(x)$ arbitrarily close to L (as close to L as we like) by restricting x to be sufficiently close to a (on either side of a) but not equal to a.

Example 2.10.

- The limit of a function does not depend on how the function is defined at the point being approached.
- It does not even matter whether the function is defined at the point.

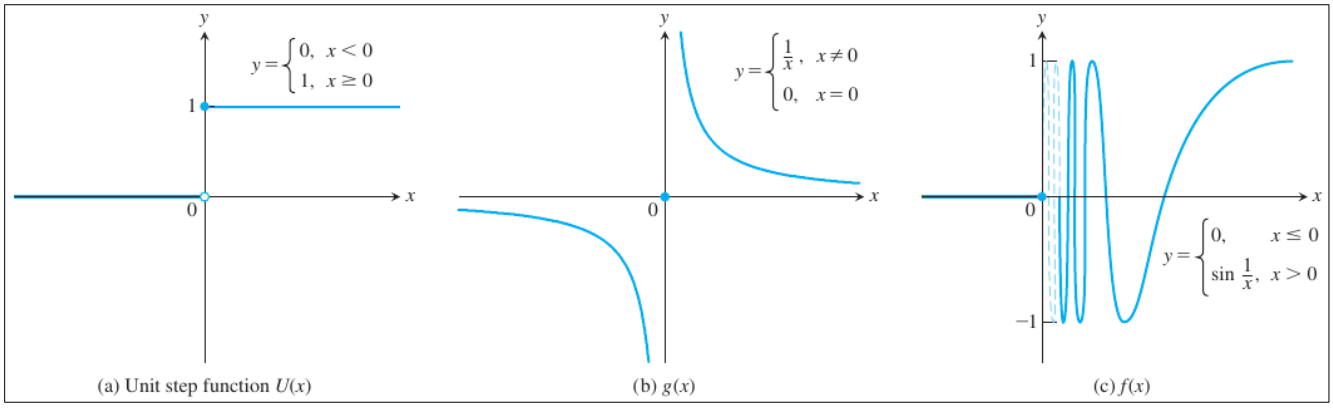


(a) $f(x) = \dfrac{x^2-1}{x-1}$ (b) $g(x) = \begin{cases} \dfrac{x^2-1}{x-1}, & x \neq 1 \\ 1, & x = 1 \end{cases}$ (c) $h(x) = x + 1$

Example 2.11. Discuss the behavior of the following functions, explaining why they have no limit as $x \to 0$.

(a) $U(x) = \begin{cases} 0, & x < 0 \\ 1, & x \geq 0 \end{cases}$ (b) $g(x) = \begin{cases} 1/x, & x \neq 0 \\ 0, & x = 0 \end{cases}$ (c) $f(x) = \begin{cases} 0, & x \leq 0 \\ \sin\dfrac{1}{x}, & x > 0 \end{cases}$

Solution. (a) The **unit step function** $U(x)$ jumps at $x = 0$. (b) *g grows too large to have a limit*. (b) *f oscillates too much to have a limit*.

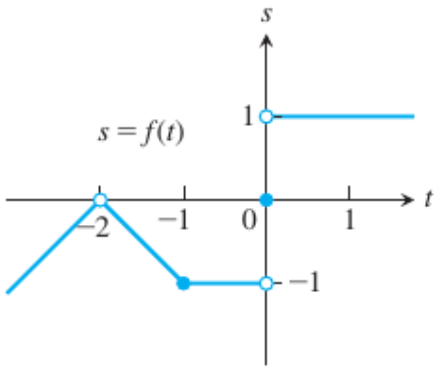


(a) Unit step function $U(x)$ (b) $g(x)$ (c) $f(x)$

Example 2.12. For the function $f(t)$ graphed below, find the following limits or explain why they do not exist.

(a) $\lim_{t\to -2} f(t)$ (b) $\lim_{t\to -1} f(t)$ (c) $\lim_{t\to 0} f(t)$ (d) $\lim_{t\to 1} f(t)$

Solution.



The Limit Laws

Theorem 2.13. ***Limit Laws***

Let L, M, c, and k be real numbers, n a positive integer, and

$$\lim_{x\to a} f(x) = L \quad \textit{and} \quad \lim_{x\to a} g(x) = M. \tag{2.8}$$

Then,

1. *Sum Rule:* $\lim_{x\to a}(f(x)+g(x)) = L+M$
2. *Difference Rule:* $\lim_{x\to a}(f(x)-g(x)) = L-M$
3. *Constant Multiple Rule:* $\lim_{x\to a}(k\cdot f(x)) = k\cdot L$
4. *Product Rule:* $\lim_{x\to a}(f(x)\cdot g(x)) = L\cdot M$
5. *Quotient Rule:* $\lim_{x\to a}\dfrac{f(x)}{g(x)} = \dfrac{L}{M}, \quad M\neq 0$
6. *Power Rule:* $\lim_{x\to a}[f(x)]^n = L^n,$
7. *Root Rule:* $\lim_{x\to a}\sqrt[n]{f(x)} = \sqrt[n]{L} = L^{1/n}$

(If n is even, we assume that $f(x)\geq 0$ for x in an interval containing a.)

The Key: Apply $\lim_{x\to a}$ to each term, if the limit exists.

Example 2.14. Find the limits.

(a) $\lim_{x\to 0}\left(x^4+\dfrac{\cos 2x}{100}\right)$ (b) $\lim_{x\to 1}\dfrac{x^2+x-2}{x^2-x}$

Solution.

Ans: (a) $1/100$. (b) 3.

Example 2.15. Evaluate the limits, by **rationalization**.

(a) $\lim_{x\to 0}\dfrac{\sqrt{x^2+100}-10}{x^2}$ (b) $\lim_{h\to 1}\dfrac{h-1}{\sqrt{3h+1}-2}$

Solution.

Ans: (a) $1/20$. (b) $4/3$.

The Sandwich Theorem

Theorem 2.16. ***Sandwich Theorem***
Suppose that $g(x) \le f(x) \le h(x)$ *for all* x *in some open interval containing* a*, except possibly at* $x = a$ *itself. Suppose also that*

$$\lim_{x\to a} g(x) = \lim_{x\to a} h(x) = L.$$

Then $\lim_{x\to a} f(x) = L$.

Example 2.17. Prove that for all θ,

$$\begin{array}{ll} \text{(a)} & -|\theta| \le \sin\theta \le |\theta| \\ \text{(b)} & 0 \le 1 - \cos\theta \le |\theta| \end{array} \tag{2.9}$$

Solution. ***Hint***: Use the geometric definition of the angle in Figure 1.9, p. 24, and the geometric interpretation of trigonometric functions in Figure 1.11, p. 27.

Example 2.18. Use the Sandwich Theorem to verify the following

(a) $\lim_{\theta\to 0} \sin\theta = 0$

(b) $\lim_{\theta\to 0} \cos\theta = 1$

(c) For any function f, $\lim_{x\to a} |f(x)| = 0$ implies $\lim_{x\to a} f(x) = 0$.

Solution. ***Clue***: (b) Use (2.9) with $\cos\theta = 1 - (1 - \cos\theta)$

Basics of Computer Programming

The **very basics of computer programming** is to understand how to deal with

- recursive iterations (loops)
- conditional statements
- calling functions
- data input/output

Example 2.19. Estimate

$$\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n, \tag{2.10}$$

by choosing $n = 10^i,\ i = 1, 2, \cdots, 7$.

Solution.

estimate_e.m

```
for i=1:7
    n = 10^i;
    f = (1+1/n)^n;
    % print out the quantities
    fprintf('n= %6g; f(n)= %.8g\n',n,f)
end
```

Result

```
n=     10; f(n)= 2.5937425
n=    100; f(n)= 2.7048138
n=   1000; f(n)= 2.7169239
n=  10000; f(n)= 2.7181459
n= 100000; f(n)= 2.7182682
n=  1e+06; f(n)= 2.7182805
n=  1e+07; f(n)= 2.7182817
```

Using Maple:

maple-estimate-e.mw

```
for i to 7 do
    n := 10^i;
    f := (1 + 1/n)^n;
    # print out the quantities
    printf("n= %6g; f= %.8g\n",n,f);
end do:
```

The result must be the same as that of the Matlab program.

As you can see from the example, **computer programming** is not about **computational languages** but about **mathematical logic**.

Note: The number e can be defined as the limit in (2.10).

Exercises 2.2

1. Explain why the limits do not exist.

   (a) $\lim\limits_{x\to 0} \dfrac{x}{|x|}$ (b) $\lim\limits_{x\to 1} \dfrac{1}{x-1}$

   Hint: (a) Consider two cases: $x < 0$ and $x > 0$.

2. Find the limits.

   (a) $\lim\limits_{x\to 6} 8(x-5)(x-7)$ (b) $\lim\limits_{x\to 2} \dfrac{2x+5}{11-x^3}$

   Ans: (b) 3.

3. Find the limits.

   (a) $\lim\limits_{t\to 1} \dfrac{t^2+t-2}{t^2-1}$ (b) $\lim\limits_{x\to 4} \dfrac{4-x}{5-\sqrt{x^2+9}}$

   Ans: (b) $5/4$.

4. Evaluate the limit

   $$\lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$$

   for the given function f and the value of x.

   (a) $f(x) = x^2,\ x = -2$
   (b) $f(x) = 1/x,\ x = 2$

   Ans: (b) $-1/4$.

5. Solve the following.

   (a) If $\lim\limits_{x\to 4} \dfrac{f(x)-5}{x-2} = 1$, find $\lim\limits_{x\to 4} f(x)$.
   (b) If $\lim\limits_{x\to 2} \dfrac{f(x)-5}{x-2} = 1$, find $\lim\limits_{x\to 2} f(x)$.
   (c) If $\lim\limits_{x\to 2} \dfrac{f(x)-5}{x-2} = 2$, find $\lim\limits_{x\to 2} f(x)$.

   Ans: (b) 5.

6. **Challenge** If $\lim\limits_{x\to 2} \dfrac{x^2+ax+b}{x-2} = 1$, find the constants a and b.

2.3. The Precise Definition of a Limit

Example 2.20. Consider the function $f(x) = 2x-1$ near $x = 4$. **Intuitively it seems clear** that $f(x)$ is close to 7 when x is close to 4, so we may write

$$\lim_{x\to 4} f(x) = 7. \tag{2.11}$$

However, **how close to $x = 4$ does x have to be** so that **$f(x)$ differs from 7 by, say, less than 1 unit**?

Solution. **Asked**: For what values of x is $|f(x) - 7| < 1$?

- To find the answer, express $f(x)$ explicitly in terms of x:

$$|f(x) - 7| = |(2x - 1) - 7| = |2x - 8| < 1.$$

- We solve the inequality:

$$\begin{aligned} &-1 < 2x - 8 < 1 \\ \Rightarrow\ &7 < 2x < 9 \\ \Rightarrow\ &3.5 < x < 4.5 \\ \Rightarrow\ &-0.5 < x - 4 < 0.5 \end{aligned}$$

- Thus, keeping x within 0.5 units of $x = 4$ will keep $f(x)$ within 1 unit of $y = 7$. That is,

$$|f(x) - 7| < 1, \quad \text{whenever } |x - 4| < 0.5. \ \square \tag{2.12}$$

Remark 2.21. In general: For the function $f(x) = 2x - 1$, for arbitrary $\varepsilon > 0$, there exists a number $\delta = \delta(\varepsilon) > 0$ such that

$$|f(x) - 7| < \varepsilon, \quad \text{whenever } |x - 4| < \delta. \tag{2.13}$$

- In the above example, if $\varepsilon = 1$, then we have found $\delta = 0.5$.
- Another example: if $\varepsilon = 0.01$, then we can set $\delta = 0.005$.

The Precise Definition of a Limit

Definition 2.22. Let $f(x)$ be defined on an open interval about a, except possibly at a itself. We say that

$$\textbf{the limit of } f(x) \textbf{ as } x \textbf{ approaches } a \textbf{ is } L,$$

and write

$$\lim_{x\to a} f(x) = L, \tag{2.14}$$

if, for every number $\varepsilon > 0$, there exists a corresponding number $\delta > 0$ such that

$$|f(x) - L| < \varepsilon \text{ whenever } 0 < |x - a| < \delta. \tag{2.15}$$

Note: The value of f at $x = a$ does not influence the existence of a limit; $x = a$ is excluded. The above definition is sometimes called the ε-δ **definition of the limit**.

Example 2.23. Use Definition 2.22 to show

$$\lim_{x\to 1}(5x - 3) = 2.$$

Solution. Let $f(x) = 5x - 3$ and $\varepsilon > 0$ be set arbitrarily. Then, we should find $\delta > 0$ such that

$$|f(x) - 2| < \varepsilon \text{ whenever } 0 < |x - 1| < \delta. \tag{2.16}$$

- As in Example 2.20, start with $|f(x) - 2| < \varepsilon$:

$$\begin{aligned} |f(x) - 2| = |(5x - 3) - 2| = |5x - 5| &< \varepsilon \\ 5|x - 1| &< \varepsilon \\ |x - 1| &< \varepsilon/5 \end{aligned}$$

- Thus, we can take $\delta = \varepsilon/5$. Such a δ would satisfy (2.16). □

Remark 2.24. In the last example, the value of $\delta = \varepsilon/5$ is not the only value that satisfies (2.16). Any smaller positive δ will do as well. The definition does not ask for the "best" positive δ, just one that will work.

Example **2.25.** For the limit $\lim_{x\to 5}\sqrt{x-1} = 2$, find a δ that works for $\varepsilon = 1$. That is, find a δ such that

$$|\sqrt{x-1}-2| < 1 \text{ whenever } 0 < |x-5| < \delta.$$

Solution.

Ans: $\delta = 3$.

Strategy **2.26.** The process of finding a $\delta > 0$ such that

$$|f(x) - L| < \varepsilon \text{ whenever } 0 < |x-a| < \delta$$

can be accomplished in two steps.

1. Solve the inequality $|f(x) - L| < \varepsilon$ to find an open interval (c, d) containing a.
2. Find a value of a $\delta > 0$ that places the open interval $(a-\delta, a+\delta)$ inside the interval (c, d).

Exercises 2.3

1. Each of the following gives a function $f(x)$ and numbers L, a, and $\varepsilon > 0$. In each case, an open interval about a on which the inequality $|f(x) - L| < \varepsilon$ holds. Then give a value for $\delta > 0$ such that

$$|f(x) - L| < \varepsilon \text{ whenever } 0 < |x - a| < \delta.$$

   (a) $f(x) = 1/x,\ L = 1/4,\ a = 4,\ \varepsilon = 0.05$

   (b) $f(x) = \sqrt{x-7},\ L = 4,\ a = 23,\ \varepsilon = 1$

   Ans: (a) $(10/3, 5)$, $\delta = 2/3$.

2. Each of the following gives a function $f(x)$ and numbers a and $\varepsilon > 0$. Find $L = \lim_{x\to a} f(x)$. Then find a number $\delta > 0$ such that

$$|f(x) - L| < \varepsilon \text{ whenever } 0 < |x - a| < \delta.$$

   (a) $f(x) = \dfrac{x^2 - 4}{x - 2},\ a = 2,\ \varepsilon = 0.05$

   (b) $f(x) = \sqrt{1 - 5x},\ a = -3,\ \varepsilon = 0.5$

   Ans: (b) $L = 4$, $\delta = 0.75$.

3. Use the ε-δ definition of the limit, Definition 2.22, to prove the limit statements.

   (a) $\lim\limits_{x\to 9} \sqrt{x-5} = 2.$

   (b) $\lim\limits_{x\to -3} \dfrac{x^2 - 9}{x + 3} = -6.$

   Hint: (b) Start with $\left|\dfrac{x^2 - 9}{x + 3} + 6\right| < \varepsilon$. When $x \neq -3$, it can be written as $|(x - 3) + 6| < \varepsilon$. Now, solve the inequality.

4. **Challenge** Use Definition 2.22 to prove that $\lim_{x\to a} \sqrt{x} = \sqrt{a}$.

   Hint: Use $|\sqrt{x} - \sqrt{a}| = \dfrac{|x - a|}{\sqrt{x} + \sqrt{a}}$.

2.4. One-Sided Limits

Definition 2.27. A **one-sided limit** is the value the function $f(x)$ approaches, as x approaches a specified point from **one side only**, either from the left or from the right.

- If x approaches a from the right, the limit L^+ is a **right-hand limit** or *limit from the right*; we write

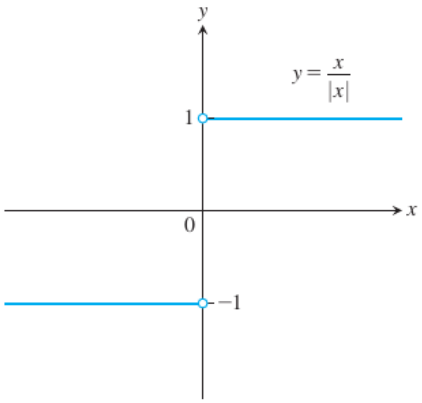
$$\lim_{x \to a^+} f(x) = L^+. \tag{2.17}$$

- If x approaches from the left, the limit L^- is a **left-hand limit** or *limit from the left*; we write

$$\lim_{x \to a^-} f(x) = L^-. \tag{2.18}$$

Example 2.28. Let $f(x) = x/|x|$. Then $\lim_{x \to 0} f(x)$ does not exist; see Exercise 1 of Section 2.2. Find one-sided limits for $f(x)$ at $x = 0$.

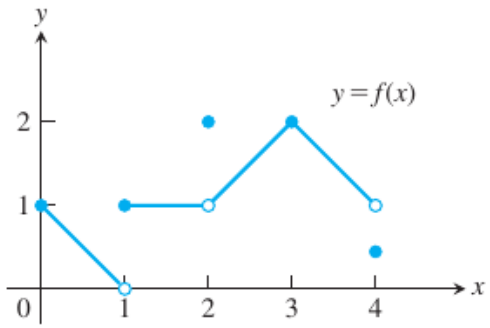
Solution.



Theorem 2.29. *Suppose that a function f is defined on an **open interval** containing a, except perhaps at a itself. Then $f(x)$ has a limit as x approaches a if and only if it has left-hand and right-hand limits there and these one-sided limits are equal:*

$$\lim_{x \to a} f(x) = L \iff \lim_{x \to a^-} f(x) = L \quad \text{and} \quad \lim_{x \to a^+} f(x) = L \tag{2.19}$$

Example 2.30. For the function graphed, complete the table.



a	$\lim_{x\to a^-} f(x)$	$\lim_{x\to a^+} f(x)$	$\lim_{x\to a} f(x)$
0	does not exist	1	1
1			
2			
3			
4			1

Note: Equation (2.19) is exceptionally applied at domain endpoints.

Definition 2.31. Precise Definitions of One-Sided Limits

(a) Assume the domain of f contains an interval (a, c) to the right of a. We say that $f(x)$ has **right-hand limit** L at a, and write

$$\lim_{x\to a^+} f(x) = L, \tag{2.20}$$

if for every number $\varepsilon > 0$ there exists a corresponding number $\delta > 0$ such that

$$|f(x) - L| < \varepsilon \text{ whenever } a < x < a + \delta. \tag{2.21}$$

(b) Assume the domain of f contains an interval (b, a) to the left of a. We say that $f(x)$ has **left-hand limit** L at a, and write

$$\lim_{x\to a^-} f(x) = L, \tag{2.22}$$

if for every number $\varepsilon > 0$ there exists a corresponding number $\delta > 0$ such that

$$|f(x) - L| < \varepsilon \text{ whenever } a - \delta < x < a. \tag{2.23}$$

Example 2.32. Prove that $\lim_{x\to 0^+}\sqrt{x}=0$.

Solution. Let $\varepsilon>0$ be given. Then we should find $\delta>0$ such that

$$|\sqrt{x}-0|<\varepsilon \text{ whenever } 0<x<0+\delta, \tag{2.24}$$

or

$$\sqrt{x}<\varepsilon \text{ whenever } 0<x<\delta.$$

Squaring both sides of this last inequality gives

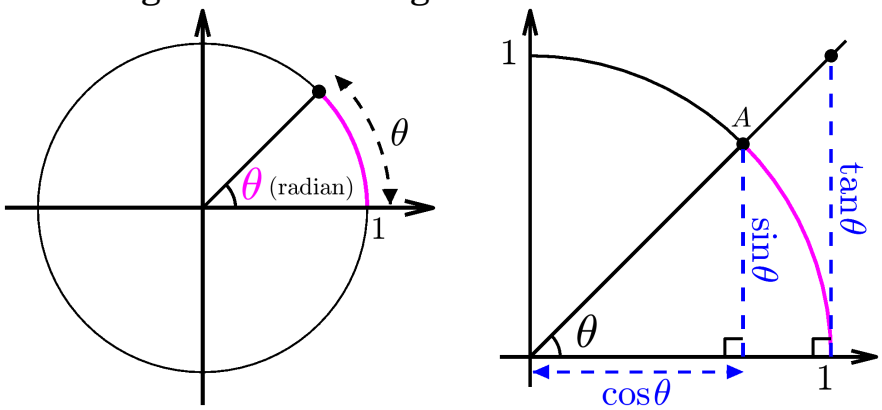
$$x<\varepsilon^2 \text{ whenever } 0<x<\delta.$$

Thus we may choose $\delta=\varepsilon^2$, with which (2.24) follows. □

Theorem 2.33. Limits involving $\sin\theta/\theta$

$$\lim_{\theta\to 0}\frac{\sin\theta}{\theta}=1. \quad (\theta \textit{ in radians}) \tag{2.25}$$

Solution. Collect Figure 1.9 and Figure 1.11.



It follows from the geometric interpretations that for $|\theta|$ small,

$$\begin{aligned}&\sin\theta<\theta<\tan\theta, && \text{when } \theta>0\\ &\sin\theta>\theta>\tan\theta, && \text{when } \theta<0\end{aligned}$$

Dividing the inequalities by $\sin\theta$ reads

$$1<\frac{\theta}{\sin\theta}<\frac{1}{\cos\theta} \quad\Longrightarrow\quad 1>\frac{\sin\theta}{\theta}>\cos\theta.$$

Since $\lim_{\theta\to 0}\cos\theta=1$, (2.25) follows from the Sandwich Theorem. □

Example 2.34. Show that (a) $\lim_{x\to 0} \dfrac{\sin 2x}{5x} = \dfrac{2}{5}$ and (b) $\lim_{x\to 0} \dfrac{\cos x - 1}{x} = 0$.

Solution. ***Hint***: (b) Use the half-angle formula $\cos x = 1 - 2\sin^2(x/2)$.

sinc function

Definition 2.35. The **sinc function**, denoted by $\operatorname{sinc}(x)$, has two forms, normalized and unnormalized.

(a) The **unnormalized sinc function** is defined by

$$\operatorname{sinc} x = \begin{cases} (\sin x)/x, & x \neq 0 \\ 1, & x = 0 \end{cases} \tag{2.26}$$

which is also called the **sampling function**.

(b) The **normalized sinc function** is defined by

$$\operatorname{sinc} x = \begin{cases} (\sin \pi x)/(\pi x), & x \neq 0 \\ 1, & x = 0 \end{cases} \tag{2.27}$$

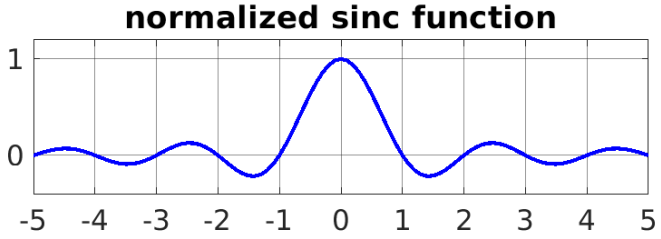
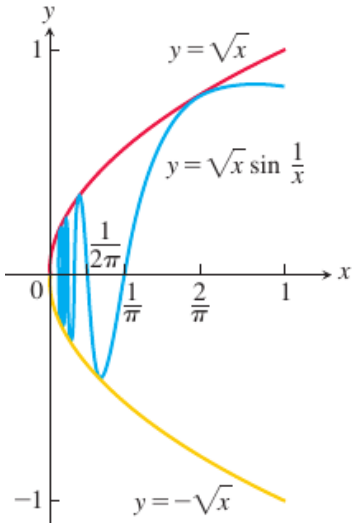


Figure 2.4: The normalized sinc function.

Exercises 2.4

1. Let $g(x) = \sqrt{x}\sin(1/x)$, as graphed in the figure.



(a) Does $\lim_{x\to 0^+} g(x)$ exist? If so, what is it? If not, why not?

(b) Does $\lim_{x\to 0^-} g(x)$ exist? If so, what is it? If not, why not?

(c) Does $\lim_{x\to 0} g(x)$ exist? If so, what is it? If not, why not?

Ans: (c) Yes. 0.

2. Let $f(x) = \begin{cases} \sqrt{1-x^2}, & 0 \le x < 1 \\ 1, & 1 \le x < 2 \\ 2, & x = 2 \end{cases}$ Graph the function and answer these questions.

(a) What are the domain and range of f?

(b) At what points a, if any, does $\lim_{x\to a} f(x)$ exist?

(c) At what points does the left-hand limit exist but not the right-hand limit?

(d) At what points does the right-hand limit exist but not the left-hand limit?

Ans: (b) $[0,1) \cup (1,2]$. (c) $x = 2$.

3. Find the limits.

(a) $\lim_{x\to 1^+} \dfrac{\sqrt{x}\,(x-1)}{|x-1|}$

(b) $\lim_{x\to 1^-} \dfrac{\sqrt{x}\,(x-1)}{|x-1|}$

(c) $\lim_{x\to 0^+} \dfrac{|\sin x|}{x}$

(d) $\lim_{x\to 0^-} \dfrac{|\sin x|}{x}$

Ans: (b) -1.

4. Use (2.25) to find limits.

(a) $\lim_{h\to 0^+} \dfrac{h}{\sin 3h}$

(b) $\lim_{x\to 0} \dfrac{\tan 2x}{x}$

(c) $\lim_{t\to 0} \dfrac{\tan(1-\cos t)}{1-\cos t}$

(d) $\lim_{\theta\to 0} \sin\theta \cot 2\theta$

Ans: (c) 1. (d) $1/2$.

5. Let $f(x) = \begin{cases} x^2 \sin(1/x), & x < 0 \\ \sqrt{x}, & x > 0 \end{cases}$

   (a) Find $\lim_{x \to 0^-} f(x)$.

   (b) Find $\lim_{x \to 0^+} f(x)$.

   (c) Based on your conclusions in parts (a) and (b), can you say anything about $\lim_{x \to 0} f(x)$? Give reasons for your answer.

Ans: (c) The limit exists and is 0. Why?

2.5. Continuity

2.5.1. Definition of Continuity

Note: When the graph of f has **no breaks** at a, we would say that f **is continuous at** a**.**

Definition 2.36. A function f is **continuous** at a if

$$\lim_{x \to a} f(x) = f(a). \tag{2.28}$$

Remark 2.37. Continuity Test
A function $f(x)$ is continuous at a point $x = a$ if and only if it meets the following three conditions.

1. $f(a)$ is defined ($\Leftrightarrow$ a lies in the domain of f)
2. $\lim_{x \to a} f(x)$ exists (f has a limit as $x \to a$)
3. $\lim_{x \to a} f(x) = f(a)$ (the limit equals the function value)

(2.29)

Example 2.38. If any of three conditions in (2.29) fail, then f is **discontinuous** at a. There are four different types of discontinuities:

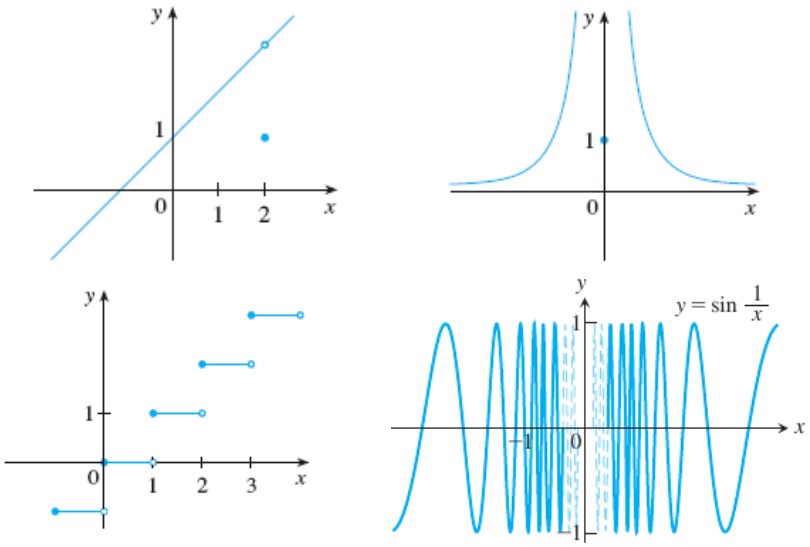


Figure 2.5: **Removable discontinuity**, **infinite discontinuity**, **jump discontinuity**, and **oscillating discontinuity**.

Example 2.39. Where are each of the following functions discontinuous?

(a) $f(x) = \dfrac{x^2 - x - 2}{x - 2}$

(b) $g(x) = \begin{cases} \dfrac{1}{x^2}, & \text{if } x \neq 0 \\ 1, & \text{if } x = 0 \end{cases}$

Solution.

Definition 2.40. One-Sided Continuity

- A function f is **right-continuous** at a (or **continuous from the right**) if
$$\lim_{x \to a^+} f(x) = f(a). \tag{2.30}$$
- A function f is **left-continuous** at a (or **continuous from the left**) if
$$\lim_{x \to a^-} f(x) = f(a). \tag{2.31}$$

Example 2.41. State the numbers at which f is discontinuous. For each number, state whether f is continuous from the right, from the left, or neither.

Solution.

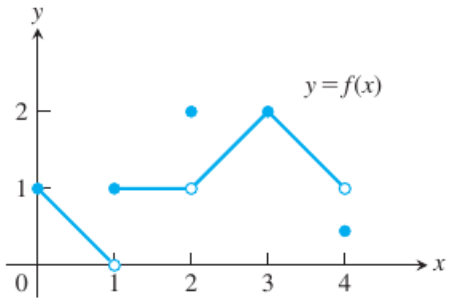


Figure 2.6: The figure used in Example 2.30.

Theorem 2.42. ***Properties of Continuous Functions***
If f and g are continuous at a and k is a constant, then the following functions are also continuous at a:

(a) $f + g$
(b) $f - g$
(c) kf
(d) fg
(e) f/g, *if* $g(a) \neq 0$
(f) f^n, n *a positive integer*

Note: The following types of functions are continuous at every number in their **natural domains**:

- Polynomials
- Rational functions
- Root functions
- Trigonometric functions
- Exponential functions
- Logarithmic functions
- **Inverse functions**: When a continuous function defined on an interval has an inverse, the inverse function is itself a continuous function over its own domain.

Example 2.43. Where is the function $f(x) = \dfrac{\ln x + \tan^{-1} x}{x^2 - 1}$ continuous?

Solution.

Ans: $x > 0$, $x \neq 1$; which is its natural domain.

Example 2.44. Use the definition of continuity and the properties of limits to show that the following function is continuous at the given number a.

$$g(t) = \frac{t^2 + 5t}{2t + 1}, \quad a = 2.$$

Solution.

Example 2.45. Show that f is continuous on $(-\infty, \infty)$.

$$f(x) = \begin{cases} 1 - x^2, & \text{if } x \leq 1 \\ \ln x, & \text{if } x > 1 \end{cases}$$

Solution.

Example 2.46. Explain why the function is discontinuous at $x = 3$. Is it continuous from one side?

$$f(x) = \begin{cases} \dfrac{2x^2 - 5x - 2}{x - 3}, & \text{if } x \neq 3 \\ 6, & \text{if } x = 3 \end{cases}$$

Solution.

Example 2.47. For what value of the constant c is the function f continuous everywhere?

$$f(x) = \begin{cases} cx^2 + 2x, & \text{if } x < 2 \\ x^3 - cx & \text{if } x \geq 2 \end{cases}$$

Solution.

2.5.2. More Properties of Continuous Functions

Continuity of Compositions of Functions

Theorem 2.48. *If f is continuous at a and g is continuous at $f(a)$, then the composition $g \circ f$ is continuous at a.*

Example 2.49. Show that the following functions are continuous on their natural domains.

(a) $y = \left|\dfrac{x-2}{x^2-2}\right|$ (b) $y = \left(\dfrac{\sin^2 x}{x^2+2}\right)^{1/2}$

Solution.

Limits of Continuous Functions

Theorem 2.50. *If $\lim\limits_{x\to a} f(x) = b$ and g is continuous in b, then*

$$\lim_{x\to a} g(f(x)) = g\left(\lim_{x\to a} f(x)\right) = g(b) \tag{2.32}$$

Note: **Continuity and limit are commutative**, when the limit exists.

Example 2.51. Find $\lim\limits_{x\to \pi/2} \cos\left(2x + \sin\left(\dfrac{3\pi}{2} + x\right)\right)$.

Solution.

Ans: -1

Theorem 2.52. ***Intermediate Value Theorem (IVT)***
Suppose that f is **continuous on the closed interval** $[a, b]$ *and let N be any number between $f(a)$ and $f(b)$, where $f(a) \neq f(b)$. Then there exists a number $c \in (a, b)$ such that $f(c) = N$.*

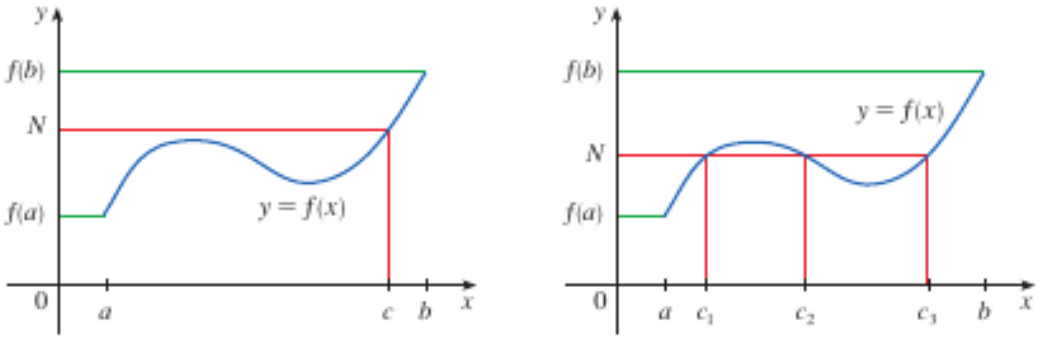


Figure 2.7: There is *at least* one such c that $f(c) = N$.

Remark 2.53. Consequences of the IVT

- **Connectedness of the Graph**: The IVT implies that the graph of a continuous function cannot have any breaks over the interval. It will be **connected** – a single, unbroken curve.
- **Root Finding**: We call a solution of the equation $f(x) = 0$ a **root of the equation** or **zero of the function** f. The IVT tells us that if f is continuous, then any interval on which f changes sign contains a zero of the function.

Example 2.54. Use the Intermediate Value Theorem to show that there is a root of the given equation in the specified interval.

(a) $e^x = 3 - 2x, \quad [0, 1]$ (b) $f(x) = x^3 + x^2 - 4x - 4, \quad [1, 3]$

Solution.

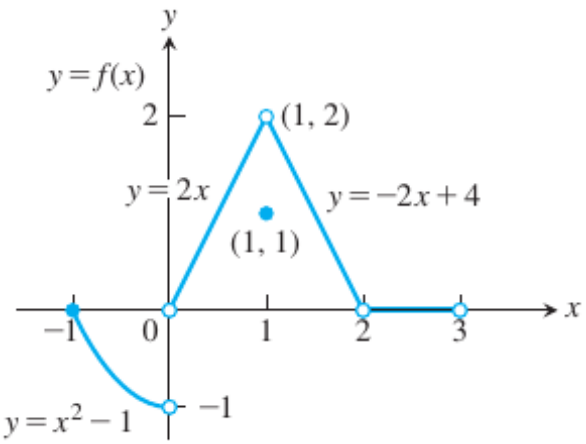
Continuous Extension to a Point

Example 2.55. Define $f(4)$ in a way that extends $f(x) = \dfrac{x^2 - 16}{x^2 - 3x - 4}$ to be continuous at $x = 4$.

Solution.

Exercises 2.5

Exercises 1-4 refer to the function f graphed.



1. (a) Does $f(-1)$ exist?
   (b) Does $\lim_{x\to -1^+} f(x)$ exist?
   (c) Does $\lim_{x\to -1^+} f(x) = f(-1)$?
   (d) Is f continuous at $x = -1$?

   Ans: All: yes

2. (a) Does $f(1)$ exist?
   (b) Does $\lim_{x\to 1} f(x)$ exist?
   (c) Does $\lim_{x\to 1} f(x) = f(1)$?
   (d) Is f continuous at $x = 1$?

   Ans: (c) no.

3. (a) Is f defined at $x = 2$?
   (b) Is f continuous at $x = 2$?

   Ans: no.

4. What value should be assigned to $f(2)$ to make the extended function continuous at $x = 2$?

5. At what points are the functions continuous?

   (a) $y = \dfrac{x+3}{x^2 - 3x - 10}$

   (b) $f(x) = \begin{cases} \dfrac{x^3 - 8}{x^2 - 4}, & x \neq 1,\ x \neq -2 \\ 3, & x = 2 \\ 4, & x = -2 \end{cases}$

   Ans: (b) $x \neq -2$

6. Find the limits. Are the functions continuous at the point being approached?

   (a) $\lim_{t\to 0} \sin\left(\dfrac{\pi}{2}\cos(\tan t)\right)$

   (b) $\lim_{x\to 0} \tan\left(1 - \dfrac{\sin x}{x}\right)$

   Ans: (b) 0, no.

7. For what value of b is

$$g(x) = \begin{cases} \dfrac{x-b}{b+1}, & x \leq 0 \\ x^2 + b, & x > 0 \end{cases}$$

   continuous at every x?

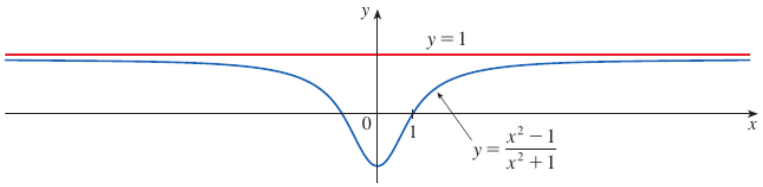
   Ans: $b = 0, -2$

8. Use the Intermediate Value Theorem to show that the equation $x^3 - 15x + 1 = 0$ has three solutions in the interval $[-4, 4]$.

9. (a) Show that the absolute value function $f(x) = |x|$ is continuous everywhere.
   (b) Prove that if f is a continuous function on an interval, then so is $|f|$.
   (c) Is the converse of the statement in part (b) also true?
   In other words, if $|f|$ is continuous, does it follow that f is continuous?
   If so, prove it. If not, find a counterexample.

2.6. Limits Involving Infinity; Asymptotes

Let's begin by investigating the behavior of the function f graphed



You can see

$$\lim_{x\to-\infty}\frac{x^2-1}{x^2+1}=1 \quad \text{and} \quad \lim_{x\to\infty}\frac{x^2-1}{x^2+1}=1$$

Definition 2.56. Intuitive Definition of a Limit at Infinity
Let f be a function defined on some interval (a,∞). Then

$$\lim_{x\to\infty} f(x)=L \tag{2.33}$$

means that the values of $f(x)$ can be made arbitrarily close to L by requiring x **to be sufficiently large**.

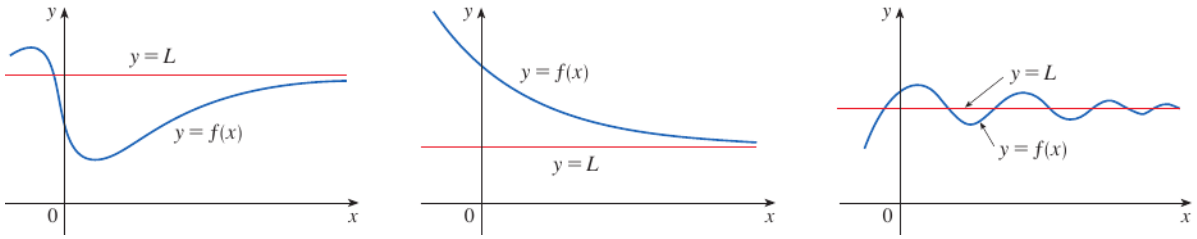


Figure 2.8: Examples illustrating $\lim_{x\to\infty} f(x)=L$.

Remark 2.57. Definition 2.56 can be stated formally:
$f(x)$ has the **limit** L **as** x **approaches infinity** and write

$$\lim_{x\to\infty} f(x)=L, \tag{2.34}$$

if, for every number $\varepsilon>0$, there exists a corresponding number M such that for all x in the domain of f,

$$|f(x)-L|<\varepsilon \text{ whenever } x>M. \tag{2.35}$$

Definition 2.58. Let f be a function defined on some interval $(-\infty, a)$. Then

$$\lim_{x \to -\infty} f(x) = L \tag{2.36}$$

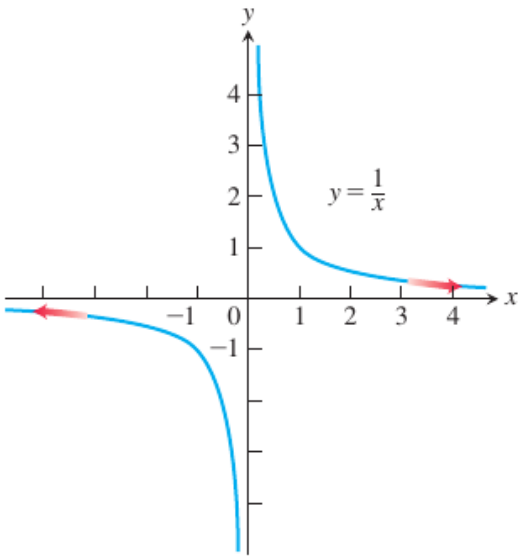
means that the values of $f(x)$ can be made arbitrarily close to L by requiring x **to be sufficiently large negative**.

2.6.1. Horizontal Asymptotes

Definition 2.59. The line $y = L$ is called a **horizontal asymptote** of the curve $y = f(x)$ if either

$$\lim_{x \to \infty} f(x) = L \quad \text{or} \quad \lim_{x \to -\infty} f(x) = L. \tag{2.37}$$

Example 2.60. Find $\lim_{x \to \infty} \frac{1}{x}$ and $\lim_{x \to -\infty} \frac{1}{x}$.



Example 2.61. Evaluate the limit or show that it does not exist.

$$\lim_{x\to\infty} \frac{6x^2+8x-3}{3x^2+2}$$

Solution. ***Hint*: Divide the numerator and the denominator by the highest power of x in the denominator.**

Example 2.62. Find the limits.

(a) $\lim_{x\to\infty} \dfrac{x^{-1}+x^{-4}}{x^{-2}+x^{-3}}$

(b) $\lim_{x\to-\infty} \left(\dfrac{x^2+x-2}{8x^2-3}\right)^{1/3}$

Solution.

Example 2.63. Evaluate the limits or show that they do not exist.

(a) $\lim_{x\to\infty}(\sqrt{x^2+1}-x)$ (b) $\lim_{x\to 2^+}\arctan\left(\frac{1}{x-2}\right)$ (c) $\lim_{x\to 2^-}\arctan\left(\frac{1}{x-2}\right)$

Solution.

Example 2.64. Find

(a) $\lim_{x\to\infty}\sin(1/x)$ (b) $\lim_{x\to\pm\infty} x\sin(1/x)$

Clue: (b) Let $t=1/x$. Then it can be written as $\lim_{t\to 0^\pm}\frac{1}{t}\sin t$.

Solution.

Example 2.65. Find the horizontal asymptotes of the graph of $f(x) = \dfrac{\sqrt{4x^6+1}}{2-x^3}$.

Solution. ***Clue***: When $x<0$, $\sqrt{4x^6+1} = |x^3|\sqrt{4+1/x^6} = -x^3\sqrt{4+1/x^6}$.

Ans: $y = \pm 2$.

Self-study 2.66. Find the horizontal asymptotes of the graph of $f(x) = \dfrac{x^3-2}{|x|^3+1}$.

Solution.

Ans: $y = \pm 1$.

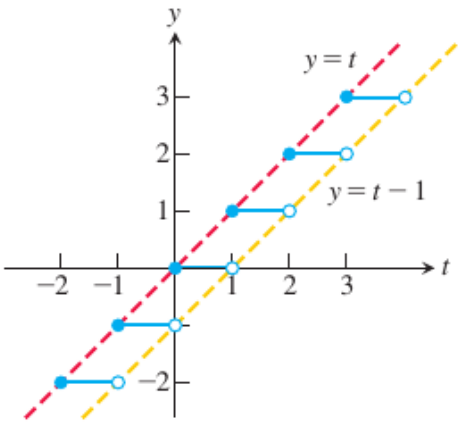
Example 2.67. Evaluate the limits or show that they do not exist.

(a) $\lim_{x\to 0^-} e^{1/x}$ (b) $\lim_{x\to 0^+} e^{1/x}$

Solution.

Example 2.68. Find $\lim_{x\to 0^+} x\left\lfloor \frac{1}{x} \right\rfloor$, where $\lfloor\cdot\rfloor$ denotes the *greatest integer function*.

Solution. ***Clue***: Let $t = 1/x$. For $t > 0$, $t - 1 \le \lfloor t \rfloor \le t$. $\Rightarrow 1 - \frac{1}{t} \le \frac{1}{t}\lfloor t \rfloor \le 1$.



Ans: 1.

2.6.2. Oblique Asymptotes

Example 2.69. Find the **oblique asymptote** (or **slant line asymptote**) of the graph of

$$f(x) = \frac{x^2 - 3}{2x - 4}.$$

Then sketch the graph. **(Note that the numerator has one order higher than the denominator.)**

Solution. ***Hint***: Try to divide $2x - 4$ into $x^2 - 3$, to find a linear part and a remainder for f.

Ans: $y = x/2 + 1$.

2.6.3. Infinite Limits

Example 2.70. Discuss the behavior of

$$f(x) = \frac{1}{1 - x} \quad \text{as } x \to 1^{\pm}, \ -\infty, \ \infty.$$

Solution.

Definition 2.71. Infinite Limits

1. We say that $f(x)$ **approaches infinity as x approaches a**, and write
$$\lim_{x\to a} f(x) = \infty, \tag{2.38}$$
if for every positive real number K there exists a corresponding $\delta > 0$ such that
$$f(x) > K \quad \text{whenever } 0 < |x - a| < \delta. \tag{2.39}$$

2. We say that $f(x)$ **approaches negative infinity as x approaches a**, and write
$$\lim_{x\to a} f(x) = -\infty, \tag{2.40}$$
if for every negative real number K there exists a corresponding $\delta > 0$ such that
$$f(x) < K \quad \text{whenever } 0 < |x - a| < \delta. \tag{2.41}$$

Example 2.72. Prove that $\lim_{x\to 0} \dfrac{1}{x^2} = \infty$.

Solution. ***Clue***: Start with $f(x) = \dfrac{1}{x^2} > K$ for an arbitrary $K > 0$. What to do is to find a corresponding $\delta > 0$.

2.6.4. Vertical Asymptotes

Definition 2.73. A line $x = a$ is a **vertical asymptote** of the graph of a function $y = f(x)$ if either

$$\lim_{x \to a^+} f(x) = \pm\infty \quad \text{or} \quad \lim_{x \to a^-} f(x) = \pm\infty. \tag{2.42}$$

Example 2.74. Find the vertical, horizontal, and oblique asymptotes of each curve.

(a) $f(x) = \dfrac{2}{x^2 - 3x}$ (b) $g(x) = \dfrac{x^2 + 2x}{x + 1}$

Solution.

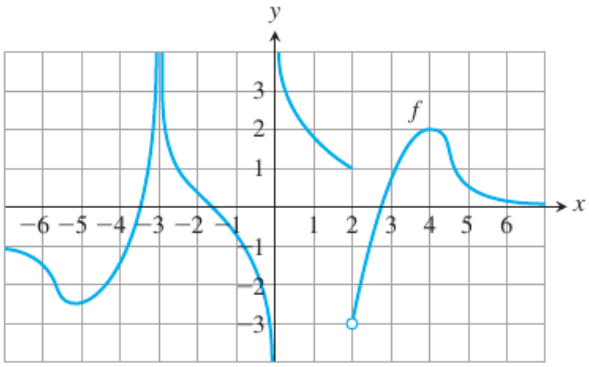
Exercises 2.6

1. For the function f whose graph is given, determine the following limits.

   (a) $\lim_{x\to -3^-} f(x)$
   (b) $\lim_{x\to -3^+} f(x)$
   (c) $\lim_{x\to -3} f(x)$

   (d) $\lim_{x\to 0^-} f(x)$
   (e) $\lim_{x\to 0^+} f(x)$
   (f) $\lim_{x\to 0} f(x)$
   (g) $\lim_{x\to 2^-} f(x)$
   (h) $\lim_{x\to 2^+} f(x)$
   (i) $\lim_{x\to 2} f(x)$

   (j) $\lim_{x\to -\infty} f(x)$
   (k) $\lim_{x\to \infty} f(x)$



2. Find the limits.

   (a) $\lim_{x\to -\infty} \dfrac{\sqrt{x^2+1}}{x+1}$

   (b) $\lim_{x\to \infty} \dfrac{x-3}{\sqrt{4x^2+25}}$

   Ans: (a) -1

3. Find the limits.

   (a) $\lim_{x\to 0^+} \dfrac{2}{3\,x^{1/3}}$

   (b) $\lim_{x\to 0^-} \dfrac{2}{3\,x^{1/3}}$

   Ans: (b) $-\infty$.

4. Find the limits.

   (a) $\lim_{x\to \infty} \left(\sqrt{x^2+25}-\sqrt{x^2-1}\right)$

   (b) $\lim_{x\to -\infty} \left(2x+\sqrt{4x^2+3x-2}\right)$

   Hint: Try multiplying or dividing by the conjugate.

   Ans: (b) $-3/4$.

5. For each, determine the domain and then use various limits to find the asymptotes and the range.

   (a) $y = 4 + \dfrac{3x^2}{x^2+1}$.

   (b) $y = \dfrac{2x}{x^2-1}$.

   Ans: (b) D: $x \neq \pm 1$. R: $(-\infty, \infty)$.
   Specify yourself two vertical asymptotes and a horizontal asymptote.

6. Graph the rational equations, including the graphs and equations of asymptotes.

   (a) $y = \dfrac{x^2}{x-1}$

   (b) $y = \dfrac{x^2-1}{x}$

   Ans: (a) Asymptotes: $y = x+1$ and $x = 1$

CHAPTER 3

Derivatives

In This Chapter:

Topics	Applications/Properties
Derivative & Differentiation Rules Powers, multiples, sums, & differences Derivative of exponential functions Products and quotients Second- and higher-order derivatives	Tangent Lines
Derivatives Trigonometric Function	
Chain Rule Inverse Functions and Logarithms	Implicit Differentiation
Related Rates	
Linearization and Differentials	

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3.9. Inverse Trigonometric Functions . . . 154
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Lecture-Note: **https://skim.math.msstate.edu/LectureNotes/Calculus-1-Lecture.pdf**

3.1. Tangent Lines and the Derivative at a Point

Recall: In Definition 2.5, p. 61, the **tangent line** to the curve at P is *roughly* defined as the **limit** of the secant lines, as $Q \to P$ from either side.

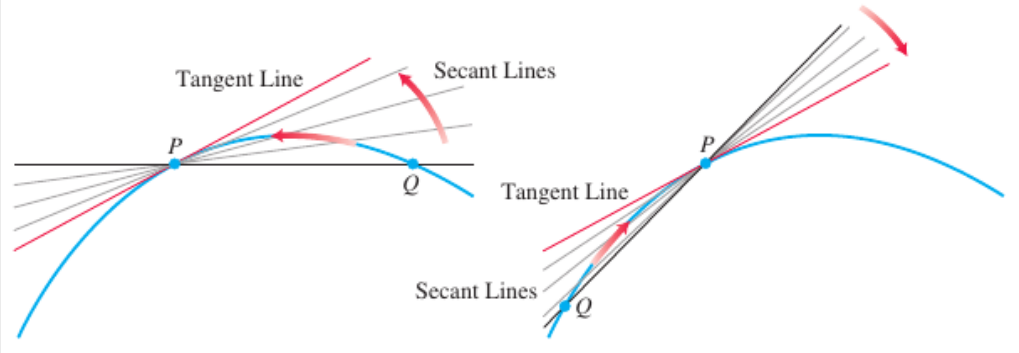


Figure 3.1: Secant lines and the tangent line to the curve at P.

Definition 3.1.
The **slope of the curve** $y = f(x)$ at the point $P(x_0, f(x_0))$ is the number

$$\lim_{h \to 0} \frac{f(x_0 + h) - f(x_0)}{h} \quad \text{(provided the limit exists).} \tag{3.1}$$

The **tangent line** to the curve at P is the line through P with this slope.

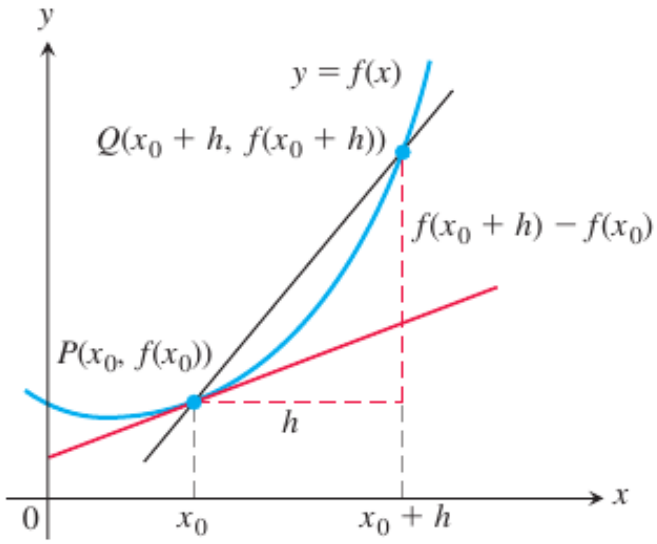


Figure 3.2: The slope at P, as the limit of the average rates of change.

Example 3.2. Find the slope of the parabola $y = x^2$ at the point $P(1, 1)$, and state an equation of the tangent line to the curve at P.

Solution.

Ans: slope $= 2$; $y - 1 = 2(x - 1)$

Example 3.3. In Examples 2.1 and 2.2 in Section 2.1, we studied the speed of a rock falling freely from rest near the surface of the earth. The rock fell

$$y = 16t^2 \text{ (feet)}$$

during the first t sec, and we used a sequence of average rates over increasingly short intervals to estimate the rock's speed at the instant $t = 1$. What was the rock's **exact speed** at this time?

Solution. Let $y = f(t) = 16t^2$. Then

$$f'(1) = \lim_{h \to 0} \frac{f(1 + h) - f(1)}{h} =$$

Definition 3.4. The **derivative** of a function $f(x)$ at a point x_0, denoted $f'(x_0)$, is

$$f'(x_0) = \lim_{h \to 0} \frac{f(x_0 + h) - f(x_0)}{h}, \tag{3.2}$$

provided that the limit exists.

Remark 3.5. If we write $x = x_0 + h$, then (3.2) can be written as

$$f'(x_0) = \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0}. \tag{3.3}$$

Claim 3.6. The tangent line to $y = f(x)$ at $(x_0, f(x_0))$ is the line through point $(x_0, f(x_0))$ whose slope is equal to $f'(x_0)$, the derivative of f at x_0.

Example 3.7. If the tangent line to $y = f(x)$ at $(4, 3)$ passes through the point $(0, 2)$, find $f'(4)$.

Solution.

Summary **3.8.** The following are all interpretations for **the limit of the difference quotient**

$$\lim_{h\to 0}\frac{f(x_0+h)-f(x_0)}{h}. \tag{3.4}$$

1. The **slope of the graph** of $y=f(x)$ at $x=x_0$
2. The **slope of the tangent line** to the curve $y=f(x)$ at $x=x_0$
3. The **rate of change** of $f(x)$ with respect to x at $x=x_0$
4. The **derivative** $f'(x_0)$

Example **3.9.** Find and simplify the difference quotients $\dfrac{f(x+h)-f(x)}{h}$ for the functions, and then apply $\lim_{h\to 0}$.

(a) $f(x)=x^2$ (b) $f(x)=x^3$

Solution.

Exercises 3.1

1. Find an equation for the tangent line to the curve at the given point. Then sketch the curve and tangent line together.

   (a) $y = x^3$, $(2, 8)$

   (b) $y = (x-1)^2 - 1$, $(1, -1)$

   Ans: (b) $y = -1$

2. Find the slope of the curve at the point indicated.

   (a) $y = \sqrt{x}$, $x = 4$

   (b) $y = \dfrac{1}{x-1}$, $x = 3$

   Ans: (b) $f'(3) = -1/4$

3. At what points does the graph of the function have a horizontal tangent line?

   $f(x) = x^3 - 3x$

   Hint: Find $f'(a) = \lim\limits_{h\to 0} \dfrac{f(a+h) - f(a)}{h}$, and solve $f'(a) = 0$ for a.

   Ans: $(1, -2)$ and $(-1, 2)$

4. Find an equation of the straight line having slope $1/4$ that is tangent to the curve $y = \sqrt{x}$.

5. Let $f(x) = \begin{cases} x^2 \sin(1/x), & x \neq 0 \\ 0, & x = 0 \end{cases}$

   (a) Is f continuous at $x = 0$?

   (b) Does $f'(0)$ exist?

   (c) Does f have a tangent line at the origin? If yes, what is it?

   Ans: (c) $y = 0$.

(Continued on the next page)

6. (a) $f(x) = x^3 + 2x, \quad a = 1$ (b) $g(x) = \cos x + 4\sin(2x), \quad a = \pi$

CAS Use a CAS to perform the following steps for the functions.

(i) Plot $y = f(x)$ over the interval $[a-1, a+4]$.

(ii) Define the difference quotient

$$Q(h) = \frac{f(a+h) - f(a)}{h}, \tag{3.5}$$

which is a function of the step size h, when holding a fixed.

(iii) Find the limit of Q as $h \to 0$; define the tangent line

$$y = T(x) := \left(\lim_{h\to 0} Q(h)\right) \cdot (x - a) + f(a) \tag{3.6}$$

(iv) Define the secant lines

$$y = S(x, h) := Q(h) \cdot (x - a) + f(a) \tag{3.7}$$

for $h = 3, 2, 1$.

(v) Graph the secant lines together with f and the tangent line over the interval in part (i).

Note: You may use Maple, Mathematica, or Matlab. The following shows a part of Matlab implementation.

- **Line 6**: In Matlab, to put graphs together, you should say "hold on"
  $\Rightarrow$ You can command "hold off" after adding all graphs.
- **Line 11**: The line is added earlier than Step (v), for your convenience.
- **After Line 11**: You should add `S(x,h)` defined as in (3.7).
- In order to graph e.g. `S(x,3)`, you may add a line
  `fplot(S(x,3),Interval, 'b--','LineWidth',1.5)`
- You result for (a) must look like Figure 3.3 below.

A part of tangent_secant.m

```
1  syms f(x) Q(h)   %also, views x and h as symbols
2
3  f(x) = x^3+2*x; a=1;
4  Interval = [a-1,a+4];
5  fplot(f(x),Interval, 'k-','LineWidth',3) %Step 1
6  hold on
7
8  Q(h) = (f(a+h)-f(a))/h;    % Step 2
9  slope = limit(Q(h),h,0);   % Step 3
10 T(x) = slope*(x-a)+f(a);
11 fplot(T(x),Interval, 'r-','LineWidth',2)
```

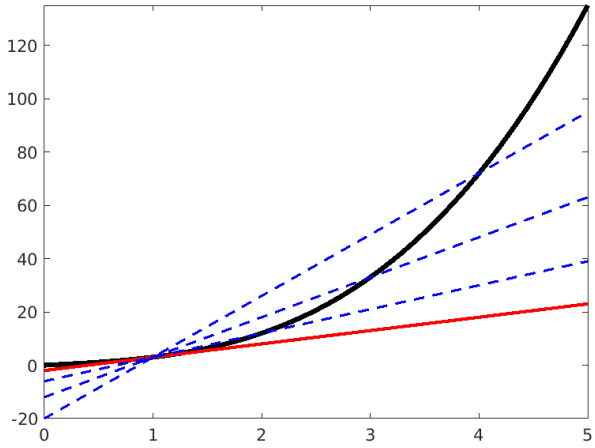


Figure 3.3: The result for Exercise 6 (a).

Note: When you decide to use Maple, you may start with the following. Lines 7, 10, and 13–16 are Maple's outputs respectively for Lines 6, 9, and 12.

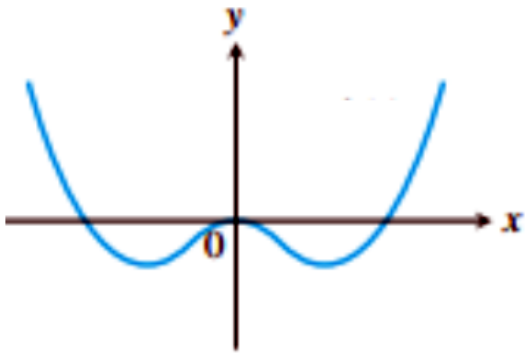
maple-tangent-secant.mw

```
1  with(plots): with(Student[Calculus1]):
2
3  f := x -> x^3 + 2*x:
4  a := 1:
5  Q := h -> (f(a + h) - f(a))/h:
6  slope := limit(Q(h), h = 0);
7                              slope := 5
8  T := x -> slope*(x - 1) + f(a):
9  T(x)
10                              5 x - 2
11 S := (x, h) -> Q(h)*(x - a) + f(a):
12 S(x, h)
13                 /       3          \
14                 \(1 + h)  - 1 + 2 h/ (x - 1)
15                 ---------------------------- + 3
16                              h
```

3.2. The Derivative as a Function

Note: Since derivatives can also be thought of as **the slope of the tangent line** and also as **the rate of change**, we may sketch the graph of $f'(x)$ using the graph of $f(x)$.

Example 3.10. The graph of a function f is given below. Use it to sketch the graph of the derivative $f'(x)$.

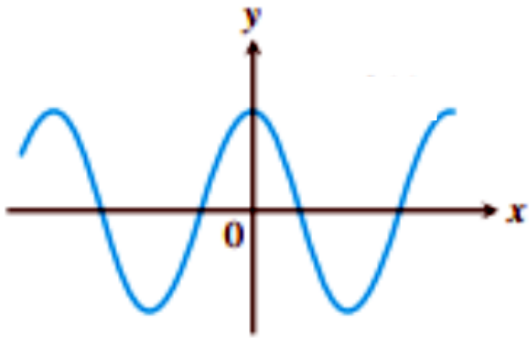


Solution.

Strategy 3.11. Graphing the Derivative

1. See where the rate of change of f is positive, negative, or zero.
2. Estimate the rough size of the growth rate at any x and its size in relation to the size of $f(x)$.
3. See where the rate of change itself is increasing or decreasing.

Example 3.12. The graph of a function f is given below. Use it to sketch the graph of the derivative $f'(x)$.



Solution.

Note: In the previous section, we have focused on finding the derivative of a function f at a particular point $(x_0, f(x_0))$, which is equal to the slope of the tangent line at that point.

- In this section, we investigate the derivative as a function derived from f by considering the limit at each point x in the domain of f.

Definition 3.13. The **derivative** of the function $f(x)$ with respect to the variable x is the function f' whose value at x is

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} \tag{3.8}$$

provided the limit exists.

Remark 3.14. The domain of f' may be either the same or smaller than the domain of f.

Example **3.15.** Using the definition of derivative, find the derivative of $f(x) = 4 - 8x - 5x^2$.

Solution.

Example **3.16.** Let $f(x) = \sqrt{x}$.

(a) Use the definition of derivative to find f'.

(b) State the domain of f'.

(c) Find the tangent line to the curve $y = f(x)$ at $x = 4$.

Solution.

Ans: (a) $f'(x) = 1/(2\sqrt{x})$

Notation 3.17. If we use the traditional notation for $y = f(x)$ to indicate that the independent variable is x and the dependent variable is y, then some common alternative notations for the derivative are:

$$f'(x) = y' = \frac{df}{dx} = \frac{dy}{dx} = \frac{d}{dx}f(x) = D(f)(x) = D_x f(x). \tag{3.9}$$

The symbols D and $\dfrac{d}{dx}$ are called **differentiation operators**, because they indicate the operation of **differentiation** which is the process of calculating a derivative.

3.2.1. Differentiability

Differentiable on an Interval; One-Sided Derivatives

Definition 3.18. Differentiability

1. A function $y = f(x)$ is **differentiable on an open interval** (finite or infinite) if it has a derivative at each point of the interval.
2. It is **differentiable on a closed interval** $[a, b]$ if it is differentiable on the interior (a, b) and if the limits

$$\begin{cases} \lim\limits_{h\to 0^+} \dfrac{f(a+h)-f(a)}{h} & \textbf{Right-hand derivative at } a \\ \lim\limits_{h\to 0^-} \dfrac{f(b+h)-f(b)}{h} & \textbf{Left-hand derivative at } b \end{cases} \tag{3.10}$$

exist at the endpoints.

Example 3.19. In Example 3.16, we found that for $x > 0$,

$$\frac{d}{dx}\sqrt{x} = \frac{1}{2\sqrt{x}}.$$

Apply the definition to examine if the derivative exists at $x = 0$.

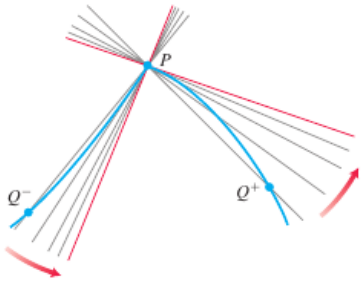
Solution.

Example 3.20. Show that the function $y = |x|$ is differentiable on $(-\infty, 0)$ and on $(0, \infty)$ but has no derivative at $x = 0$.

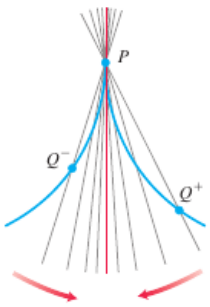
Solution.

When Does a Function *Not* Have a Derivative at a Point?

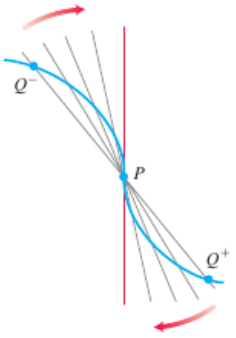
A function fails to have a derivative at a point for many reasons, including the examples below:



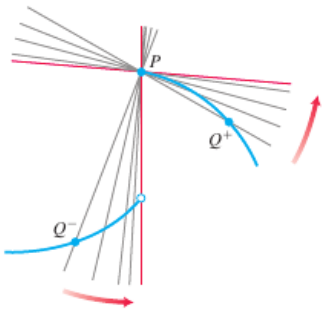
1. a *corner*, where the one-sided derivatives differ

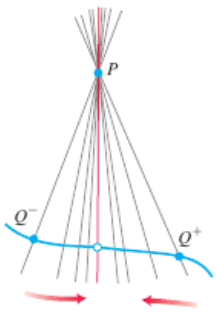


2. a *cusp*, where the slope of PQ approaches ∞ from one side and $-\infty$ from the other

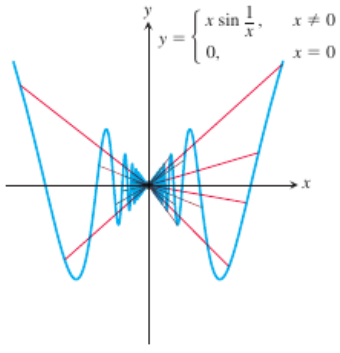


3. a *vertical tangent line*, where the slope of PQ approaches ∞ from both sides or approaches $-\infty$ from both sides (here, $-\infty$)





4. a *discontinuity* (two examples shown)



5. wild oscillation

Theorem 3.21. Differentiability Implies Continuity: *If f has a derivative at $x = c$, then f is continuous at $x = c$.*

Proof. Given that $f'(c)$ exists, we must show that $\lim_{h\to 0} f(c+h) = f(c)$. For $h \neq 0$,

$$f(c+h) = f(c) + f(c+h) - f(c) = f(c) + \frac{f(c+h) - f(c)}{h} \cdot h.$$

Taking limits as $h \to 0$ reads

$$\begin{aligned}
\lim_{h\to 0} f(c+h) &= \lim_{h\to 0} \left[f(c) + \frac{f(c+h) - f(c)}{h} \cdot h \right] \\
&= f(c) + \lim_{h\to 0} \frac{f(c+h) - f(c)}{h} \cdot \lim_{h\to 0} h \\
&= f(c) + f'(c) \cdot 0 \\
&= f(c),
\end{aligned}$$

which completes the proof. □

Note: The converse of Theorem 3.21 is false in general.

Example 3.22. Where are the following functions continuous? Differentiable?

(a) $f(x) = |x|$

(b) $g(x) = x^{1/3}$

Solution.

Exercises 3.2

1. Find the indicated derivatives.

   (a) $\dfrac{dy}{dx}$, if $y = x^3 + x^2$

   (b) $v'(t)$, if $v(t) = t + \dfrac{1}{t}$

   Ans: (b) $v'(t) = 1 - 1/t^2$

2. Differentiate the functions. Then find an equation of the tangent line at the indicated point on the graph of the function.

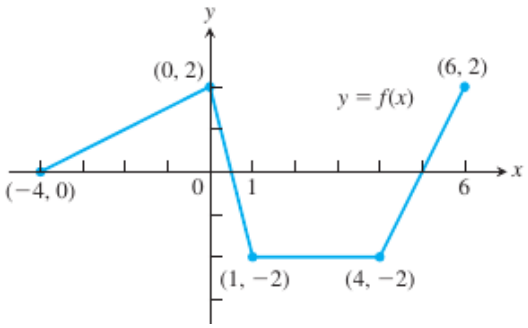
   (a) $y = f(x) = \dfrac{1}{x}, \quad (x, y) = (1, 1).$

   (b) $w = g(z) = 1 + \sqrt{4 - z}, \quad (z, w) = (3, 2).$

   Ans: (b) $w - 2 = -\dfrac{1}{2}(z - 3)$

3. (a) The graph in the accompanying figure is made of line segments joined end to end in the interval $[-4, 6]$. At which points of the interval is f' not defined? Give reasons for your answer.

   (b) Graph the derivative of f. (The graph should show a step function.)

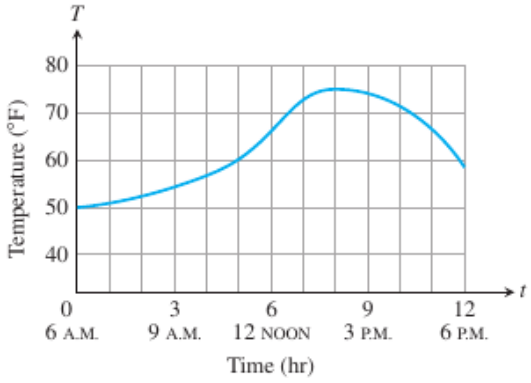


4. The given graph shows the outside temperature T in °F, between 6:00 AM and 6:00 PM.

   (a) Estimate the rate of temperature change at the times 9:00 AM and 2:00 PM.

   (b) At what time does the temperature increase most rapidly? Decrease most rapidly? What is the rate for each of those times?

   (c) Use the graphical technique of Strategy 3.11 to graph the derivative of temperature T versus time t.



   Ans: (a) 2.5°F/hr, 0°F/hr

5. Determine if the piecewise-defined function is differentiable at the origin. If it is differentiable, what is $f'(0)$?

   (a) $f(x) = \begin{cases} 2x + 1 & x \le 0 \\ x^2 + 2x - 1 & x > 0 \end{cases}$

   (b) $f(x) = \begin{cases} x - 1 & x \le 0 \\ \dfrac{1}{2}x + \sqrt{1 + x} - 2 & x > 0 \end{cases}$

   Ans: (a) Not continuous $\Rightarrow$ not differentiable. (b) $f'(0) = 1$.

6. CAS Graph $y = 3x^2$ in a window that has $-2 \le x \le 2$, $0 \le y \le 4$. Then, on the same screen, graph

$$y = \frac{(x+h)^3 - x^3}{h},$$

for $h = -1,\ -0.2,\ 0.2,\ 1$. Explain what is going on.

3.3. Differentiation Rules

3.3.1. Powers, Multiples, Sums, and Differences

- **Derivative of a Constant Function**
  If f has the constant value $f(x) = c$, then
  $$\frac{df}{dx} = \frac{d}{dx}(c) = 0. \tag{3.11}$$
- **Power Rule**: If n is ***any real number***, then
  $$\frac{dx^n}{dx} = nx^{n-1}, \tag{3.12}$$
  for all x where the powers x^n and x^{n-1} are defined.
- **Constant Multiple Rule**
  If f is a differentiable function of x, and c is a constant, then
  $$\frac{d}{dx}[cf(x)] = c\frac{d}{dx}f(x). \tag{3.13}$$
- **Sum and Difference Rules**: If f and g are both differentiable, then
  $$\frac{d}{dx}[f(x) \pm g(x)] = \frac{d}{dx}f(x) \pm \frac{d}{dx}g(x). \tag{3.14}$$

Example 3.23. Find the derivative of the following functions.

(a) $f(x) = 20^5$

(b) $g(t) = 5t + 4t^2$

Example 3.24. Find the derivative of the following functions.

(a) $f(x) = x^{3/2} + x^{-3}$

(b) $r(t) = \dfrac{2}{t^2} + \dfrac{8}{t^4}$

(c) $y = \dfrac{\sqrt{x} + x}{x^2}$

(d) $g(x) = \dfrac{1}{\sqrt{x}} + \sqrt[4]{x}$

3.3.2. Derivative of Exponential Functions

Let $f(x) = a^x$, $a > 0$, $a \neq 1$.

- We first evaluate $f'(0)$:

$$f'(0) = \lim_{h\to 0} \frac{a^h - a^0}{h} = \lim_{h\to 0} \frac{a^h - 1}{h}. \tag{3.15}$$

- When we apply the definition of the derivative to $f(x)$, we get

$$\begin{aligned} \frac{d}{dx}(a^x) &= \lim_{h\to 0} \frac{a^{x+h} - a^x}{h} \\ &= \lim_{h\to 0} \frac{a^x \cdot a^h - a^x}{h} \\ &= \lim_{h\to 0} a^x \cdot \frac{a^h - 1}{h} \\ &= a^x \cdot \lim_{h\to 0} \frac{a^h - 1}{h} \\ &= \boldsymbol{a^x \cdot f'(0)} \end{aligned} \tag{3.16}$$

Remark 3.25. In Section 1.5, the number e is introduced such that the slope of the tangent line to $y = e^x$ at $x = 0$ is exactly 1. That is,

$$\left.\frac{d}{dx}(e^x)\right|_{x=0} = \lim_{h\to 0} \frac{e^h - e^0}{h} = \lim_{h\to 0} \frac{e^h - 1}{h} = 1 \tag{3.17}$$

- **Derivative of the Natural Exponential Function**:

$$\frac{d}{dx}(e^x) = e^x \tag{3.18}$$

Example 3.26. Find the derivative of $f(x) = 3e^x + 4x^2 - \dfrac{1}{x} + 3$

Solution.

3.3.3. Products and Quotients

Note: A common misconception is that the derivative of a product is analogous to the sum and difference rules we saw previously:

$$f \text{ and } g \text{ are differentiable} \Rightarrow (fg)' = f'g'$$

However, this is **not true**.

Let both f and g be differentiable. Then

$$\begin{aligned}
\frac{d}{dx}[f(x)g(x)] &= \lim_{h\to 0}\frac{f(x+h)g(x+h)-f(x)g(x)}{h} \\
&= \lim_{h\to 0}\frac{f(x+h)g(x+h)\boldsymbol{-f(x)g(x+h)\ +f(x)g(x+h)}-f(x)g(x)}{h} \\
&= \lim_{h\to 0}\left[\frac{f(x+h)-f(x)}{h}\cdot g(x+h)+f(x)\cdot\frac{g(x+h)-g(x)}{h}\right] \\
&= f'(x)g(x)+f(x)g'(x).
\end{aligned} \tag{3.19}$$

- **Product Rule**
  If both f and g are differentiable, then so is their product fg, and

$$\frac{d}{dx}[f(x)g(x)] = f'(x)g(x)+f(x)g'(x) \tag{3.20}$$

  Alternative notation:

$$(fg)' = f'g+fg' \quad \text{or} \quad \frac{d}{dx}[uv] = \frac{du}{dx}v+u\frac{dv}{dx}$$

Example 3.27. Find the derivative of the following:

(a) $y = (4x^2+3)(2x+5)$

(b) $y = 4e^x\left(x^3+\dfrac{1}{x}\right)$

Solution.

Similarly to Product Rule, we can prove the following.

- **Quotient Rule**
  If both f and g are differentiable, then so is their product f/g, and

$$\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2} \tag{3.21}$$

  Alternative notation:

$$(f/g)' = \frac{f'g - fg'}{g^2} \quad \textbf{or} \quad \frac{d}{dx}[u/v] = \frac{\frac{du}{dx}v - u\frac{dv}{dx}}{v^2}$$

Example **3.28.** Find the derivative of the following:

(a) $G(x) = \dfrac{6x^4 - 5}{x + 1}$ (b) $y = \dfrac{6e^x}{1 + e^x}$

Solution.

Self-study **3.29.** Find the derivative of $F(x) = \dfrac{x^2 e^x}{x^3 + 5e^x}$

Solution.

3.3.4. Second- and Higher-order Derivatives

Definition 3.30. If $y = f(x)$ is a differentiable function, then

- its derivative $f'(x)$ is also a function,
- so f' may have a derivative of its own, denoted $(f')' = f''$.

This new function is called the **second derivative** of f, because it is the derivative of the first derivative.

Other notations: $f''(x) = \dfrac{d^2f}{dx^2} = \dfrac{d^2y}{dx^2} = \dfrac{d}{dx}\left(\dfrac{dy}{dx}\right) = y''$.

Higher-Order Derivatives: If y'' is differentiable, its derivative $y''' = \dfrac{d}{dx}y'' = \dfrac{d^3y}{dx^3}$ is the **third derivative** of y with respect to x.
The names continue as you imagine, with

$$y^{(n)} = \frac{d}{dx}y^{(n-1)} = \frac{d^ny}{dx^n} = D^ny \tag{3.22}$$

denoting the **n-th derivative** of y w.r.t. x for positive integers n.

Example 3.31. When $f(x) = 6xe^x - 2x^2 + 3$, find f', f'', and f'''.

Example **3.32.** Find the equation of the **tangent line** to the curve at the given point.

(a) $y = 2x^3 - x^2 + 2, \quad (1, 3)$

(b) $y = \dfrac{6e^x}{1 + e^x}, \quad (0, 3)$

Solution.

Example **3.33.** Find the point(s) on the curve where the tangent line is horizontal, for $y = x^3 - 3x - 2$.

Solution.

Exercises 3.3

1. Find the first and second derivatives.

   (a) $y = \dfrac{4x^3}{3} - x + 2e^x$

   (b) $y = \sqrt[3]{x^{9.6}} - x^e$

   (c) $y = \dfrac{x^3}{3} + \dfrac{x^2}{2} + e^{-x}$

   (d) $w = re^{-r} + e^{-3/2}$

   Hint: (b) It can be rewritten as $y = x^{3.2} - x^e$. (c) Note that $e^{-x} = \dfrac{1}{e^x}$. Now you may apply Quotient Rule to get $[e^{-x}]' = -e^{-x}$. Later you will learn how to get derivative of e^{-x} more conveniently.

   Ans: (d) $w' = (1-r)e^{-r}$, $w'' = -(2-r)e^{-r}$.

2. Find the derivatives of all orders.

   (a) $y = \dfrac{x^4}{2} - \dfrac{3}{2}x^2 - x$

   (b) $y = (4x^2 + 3)(2 - x)$

   Ans: (b) $y' = 8x(2-x) - (4x^2+3)$, $y'' = 8[(2-x) - x] - 8x = 16 - 24x$, $y''' = -24$, and $y^{(n)} = 0$, $n \geq 4$

3. Find the second derivatives of the functions.

   (a) $y = \dfrac{x^3 + 7}{x}$

   (b) $w = \left(\dfrac{1+z}{z}\right)(1-z)$

   Ans: (b) $w'' = 2/z^3$

4. Suppose u and v are functions of x that are differentiable at $x = 0$ and that

   $$u(0) = 5, \;\; u'(0) = -3, \quad v(0) = -1, \;\; v'(0) = 2.$$

   Find the values of the following derivatives at $x = 0$.

   (a) $\dfrac{d}{dx}(uv)$ (b) $\dfrac{d}{dx}(u/v)$ (c) $\dfrac{d}{dx}(v/u)$

5. Let $f(x) = x^3 - 4x + 1$.

   (a) **Normal line to a curve**. Find an equation for the line perpendicular to the tangent line to the curve $y = f(x)$ at the point $(2, 1)$.

   (b) **Smallest slope**. What is the smallest slope on the curve? At what point on the curve does the curve have this slope?

   (c) **Tangent lines having specified slope**. Find equations for the tangent lines to the curve at the points where the slope of the curve is 8.

6. **Challenge** Find the values of a and b that make the following function differentiable for all x-values.

(a) $f(x) = \begin{cases} ax, & \text{if } x < 0 \\ x^2 - 3x & \text{if } x \geq 0 \end{cases}$ (b) $g(x) = \begin{cases} ax + b, & \text{if } x > -1 \\ bx^2 - 4, & \text{if } x \leq -1 \end{cases}$

Hint: The limits of values and derivatives, measured from both sides, must be the same. For example, for (b), the limits of values are $-a + b$ and $b - 4$. Thus you can conclude $a = 4$.

7. CAS **Newton's serpentine** is defined as

$$f(x) = \frac{4x}{x^2 + 1}.$$

(a) Plot the graph $y = f(x)$ over the interval $[-4, 4]$.
(b) On the same screen, plot the tangent line to the curve at $(1, 2)$
(c) Again, on the same screen, plot the curve $y = f'(x)$.

You may start with

newtons_serpentine.m

```
syms f(x)

f(x) = 4*x/(x.^2+1);
X = linspace(-4,4,101);
plot(X,f(X),'r-','linewidth',2); hold on
grid on, axis tight

diff(f(x))
```

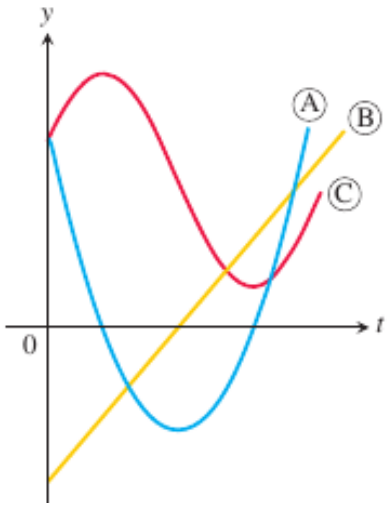
3.4. The Derivative as a Rate of Change

Motion Along a Line: Position, Velocity, Speed, and Acceleration

Definition 3.34. Let $s = f(t)$ be the **position** of a particle moving along a coordinate line. Then,

- Its average rate of change $\dfrac{\Delta s}{\Delta t}$ is the **average velocity** over a time interval Δt.
- Its derivative $\dfrac{ds}{dt} = v(t)$ represents the **instantaneous velocity**.
- **Speed** is the absolute value of velocity: $|v(t)| = \left|\dfrac{ds}{dt}\right|$.
- The instantaneous rate of change of velocity with respect to time is **acceleration**: $a(t) = v'(t) = s''(t)$.

Example 3.35. The graphs in the accompanying figure show the position s, the velocity $v = s'$, and the acceleration $a = s''$ of a moving body along a coordinate line as functions of time t. Which graph is which? Give reasons for your answers.



Example **3.36.** A dynamite blast blows a heavy rock straight up with a launch velocity of $160\,ft/sec$ (about $109\,mph$). It reaches a height of $s(t) = 160t - 16t^2\,ft$ after t seconds.

(a) How high does the rock go?

(b) What are the velocity and speed of the rock when it is $256\,ft$ above the ground on the way up? On the way down?

(c) What is the acceleration of the rock at any time t during its flight (after the blast)?

(d) When does the rock hit the ground again?

Solution.

Example 3.37. At time t, the position of a body moving along the s-axis is $s = t^3 - 6t^2 + 9t\,m$.

(a) Find the body's acceleration each time the velocity is zero.

(b) Find the body's speed each time the acceleration is zero.

(c) Find the **total distance traveled** by the body from $t = 0$ to $t = 2$.

Solution.

Derivatives in Economics and Biology

Example 3.38. The **marginal cost of production** is the rate of change of cost with respect to the level of production, so it is $\frac{dc}{dx}$. Suppose that the dollar cost of producing x washing machines is

$$c(x) = 2000 + 100x - 0.1x^2.$$

(a) Find the average cost per machine of producing the first 100 washing machines.

(b) Find the marginal cost when 100 washing machines are produced.

(c) The marginal cost also represents the approximate cost of the next item to be produced. Show that the marginal cost when 100 washing machines are produced is approximately equal to the cost of producing one more washing machine after the first 100 have been made by calculating this cost directly.

Solution.

Example 3.39. When a bactericide was added to a nutrient broth in which bacteria were growing, the bacterium population continued to grow for a while, but then stopped growing and began to decline. The size of the population at time t hours was

$$b = 10^6 + 10^4 t - 10^3 t^2.$$

Find the growth rates at a) $t = 0$ hours, b) $t = 5$ hours, and c) $t = 10$ hours.

Exercises 3.4

1. **Lunar projectile motion**. A rock thrown vertically upward from the surface of the moon at a velocity of $24\,m/s$ reaches a height of $s = 24t - 0.8t^2\,m$ in t sec.

   (a) Find the rock's velocity and acceleration at time t. (The acceleration in this case is the acceleration of gravity on the moon.)
   (b) How long does it take the rock to reach its highest point?
   (c) How high does the rock go?
   (d) How long does it take the rock to reach half its maximum height?
   (e) How long is the rock aloft?

   Ans: (c) $180\,m$

2. **Vehicular stopping distance**. Based on data from the U.S. Bureau of Public Roads, a model for the total **stopping distance** of a moving car in terms of its speed is

   $$s = 1.1v + 0.054v^2,$$

   where s is measured in *ft* and v in *mph*. The linear term $1.1v$ models the distance the car travels during the time the driver perceives a need to stop until the brakes are applied, and the quadratic term $0.054v^2$ models the additional braking distance once they are applied. Find ds/dv at $v = 35$ and $v = 70\,mph$, and interpret the meaning of the derivative.

3. CAS **Golf Ball Carry**. Using a club of a loft angle θ, a golf player hits a ball with the horizontal head speed v_0. Then, the initial velocity of the ball becomes

   $$\langle v_x, v_z\rangle = v_0\cos\theta\langle\cos\theta,\ \sin\theta\rangle = \langle v_0\cos^2\theta,\ v_0\cos\theta\sin\theta\rangle,$$

   where v_x and v_z are the horizontal and vertical velocities, respectively. For simplicity, we assume that the golf field is complete flat. Then, incorporating the gravity acceleration $\mathbf{a} = \langle 0, -g\rangle$, where $g = 9.8\,m/s^2$, the ball location after t seconds can be expressed as

   $$L(t) = \langle x(t), z(t)\rangle = \langle 0, -g\rangle\frac{t^2}{2} + \langle v_x, v_z\rangle t = \Big\langle (v_0\cos^2\theta)t,\ -\frac{g}{2}t^2 + (v_0\cos\theta\sin\theta)t\Big\rangle. \tag{3.23}$$

   (a) Find the time for the ball to hit the ground, t_g.
   Hint: The ball will hit the ground when $z(t) = 0$.
   (b) Show that the ball carry, as a function of θ, becomes

   $$C(\theta) = \frac{v_0^2\cos^2\theta\sin(2\theta)}{g}. \tag{3.24}$$

   (c) The average club-head speed for male amateur golfers is between 80-90 *mph*. Let the golfer hit the ball with the head speed $90\,mph$, which is the same as $40\,m/s$. Plot $C(\theta)$ over the interval $[0, \pi/2]$.
   (d) At what angle θ does the club make the longest ball carry?

   > Although we have ignored the ball spin and air resistance, the analysis in this problem is mostly realistic. Can you see why beginners hit the longest with the **6-Iron** or **7-Iron** among Irons?

3.5. Derivatives of Trigonometric Functions

- Consider the **angle sum identity** for the sine function:

$$\sin(x+h) = \sin x \cos h + \cos x \sin h. \tag{3.25}$$

If $f(x) = \sin x$, then

$$\begin{aligned} f'(x) &= \lim_{h\to 0} \frac{f(x+h)-f(x)}{h} = \lim_{h\to 0} \frac{\sin(x+h)-\sin x}{h} \\ &= \lim_{h\to 0} \frac{\sin x \cos h + \cos x \sin h - \sin x}{h} \\ &= \lim_{h\to 0} \left[\sin x \cdot \frac{\cos h - 1}{h} + \cos x \cdot \frac{\sin h}{h} \right] \\ &= \sin x \cdot \underbrace{\lim_{h\to 0} \frac{\cos h - 1}{h}}_{=0;\ \text{Example 2.34}} + \cos x \cdot \underbrace{\lim_{h\to 0} \frac{\sin h}{h}}_{=1;\ \text{Theorem 2.33}} \\ &= \cos x \end{aligned} \tag{3.26}$$

- Similarly, we can find $\dfrac{d}{dx}\cos x = -\sin x$, using the **angle sum identity** for the cosine function:

$$\cos(x+h) = \cos x \cos h - \sin x \sin h. \tag{3.27}$$

- Note that $\tan x = \dfrac{\sin x}{\cos x}$. For $\tan x$ and other trigonometric functions, **Quotient Rule** can be used to get their derivatives.

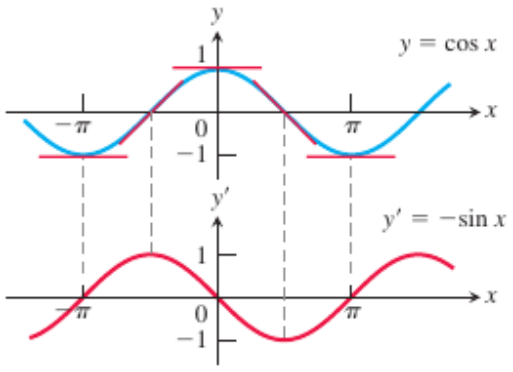


Figure 3.4: Visualization of $\dfrac{d}{dx}\cos x$, by graphing the slopes of the tangent lines.

Derivatives of Trigonometric Functions

$$\begin{aligned} \frac{d}{dx}\sin x &= \cos x & \frac{d}{dx}\csc x &= -\csc x \cot x \\ \frac{d}{dx}\cos x &= -\sin x & \frac{d}{dx}\sec x &= \sec x \tan x \\ \frac{d}{dx}\tan x &= \sec^2 x & \frac{d}{dx}\cot x &= -\csc^2 x \end{aligned} \tag{3.28}$$

Example 3.40. Differentiate the following:

(a) $g(x) = 3e^x + x^2 \cos x$

(b) $f(\theta) = \dfrac{\sin\theta}{1-\cos\theta}$

(c) $y = x \sec x \tan x$

(d) $h(t) = \dfrac{t \sin t}{e^t}$

Example **3.41.** Find an equation of the tangent line to the curve $y = 2x \sin x$ at the point $(\pi/2, \pi)$.

Solution.

Example **3.42.** Differentiate $f(x) = \dfrac{\sec x}{3 + \sec x}$ and find the values of x where the graph of f have a horizontal tangent line.

Solution.

Example **3.43.** By computing the first few derivatives and looking for a pattern, find the following.

$$\frac{d^{2022}}{dx^{2022}}(\cos x).$$

Solution.

Example **3.44.** Assume that a particle's position on the x-axis is given by $x = 3\cos t + 4\sin t$, where x is measured in feet and t is measured in seconds. Find the particle's velocity when a) $t = 0$ and b) $t = \dfrac{\pi}{2}$.

Solution.

Exercises 3.5

1. Differentiate the following:

   (a) $y = x^2 \cos x$

   (b) $f(x) = \cos x \tan x$

   (c) $s = t^2 - \sec t + 5e^{-t}$

   (d) $g(x) = \dfrac{\tan x}{1 + \tan x}$

   Ans: (d) sympy result: `1/((tan(x) + 1)**2*cos(x)**2)`

   sympy_diff.py

```
import sympy as sym

x = sym.symbols('x')
y = sym.tan(x)/(1+sym.tan(x))
print( sym.simplify( y.diff(x) ) )
```

2. *Sketch* the curves over the given intervals, together with their tangent lines at the given values of x. Label each curve and tangent line with its equation.

   (a) $y = \sin x, \quad -3\pi/2 \le x \le 2\pi, \quad x = 0,\ 3\pi/2$

   (b) $y = 1 + \cos x, \quad -3\pi/2 \le x \le 2\pi, \quad x = -\pi/3,\ 3\pi/2$

3. Is there a value of b that will make

$$g(x) = \begin{cases} x + b, & x < 0 \\ \cos x, & x \ge 0 \end{cases}$$

   continuous at $x = 0$? Differentiable at $x = 0$? Give reasons for your answers.

   Ans: No value of b for g to be differentiable.

4. Find values of a and b that will make

$$f(x) = \begin{cases} ax + b, & x < 0 \\ 1 - \tan x, & x \ge 0 \end{cases}$$

   differentiable at $x = 0$?

3.6. The Chain Rule

Let's differentiate the function

$$f(x) = (3x^2 + 1)^2. \tag{3.29}$$

- The expanded formula for f reads

$$f(x) = 9x^4 + 2 \cdot (3x^2) + 1 = 9x^4 + 6x^2 + 1.$$

Thus

$$\frac{df}{dx} = \frac{d}{dx}(9x^4 + 6x^2 + 1) = 36x^3 + 12x. \tag{3.30}$$

- Now, we will try a different way. Let

$$u(x) = 3x^2 + 1$$

Then the function f can be formulated in terms of u:

$$f(x) = (3x^2 + 1)^2 = u^2.$$

- The derivative $\boldsymbol{\frac{df}{dx}}$ **is a rate of change of f with respect to x.** Thus it can be written as

$$\frac{df}{dx} = \frac{df}{du} \cdot \frac{du}{dx} \tag{3.31}$$

so that

$$\frac{df}{dx} = 2u \cdot 6x = 12u \cdot x = 12(3x^2 + 1)x = 36x^3 + 12x,$$

which is the same as in (3.30).

Theorem 3.45. The Chain Rule: *If $f(u)$ is differentiable at the point $u = g(x)$ and $g(x)$ is differentiable at x, then the composite function $(f \circ g)(x) = f(g(x))$ is differentiable at x, and*

$$(f \circ g)'(x) = f'(g(x)) \cdot g'(x) = \frac{d\, f(g(x))}{d\, g(x)} \cdot \frac{dg(x)}{dx}. \tag{3.32}$$

Letting $y = f(u)$ and $u = g(x)$, a simpler form of ***Leibniz's notation*** *reads*

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}, \tag{3.33}$$

where dy/du is evaluated at $u = g(x)$.

Proof. For $y = f(u) = f(g(x))$, define $\Delta u = g(x + h) - g(x)$. Then $h \to 0$ implies $\Delta u \to 0$ and

$$\begin{aligned}
\frac{dy}{dx} &= \lim_{h\to 0} \frac{f(g(x+h)) - f(g(x))}{h} = \lim_{h\to 0} \frac{f(g(x) + \Delta u) - f(g(x))}{h} \\
&= \lim_{h\to 0} \frac{f(g(x) + \Delta u) - f(g(x))}{\Delta u} \cdot \frac{\Delta u}{h} \\
&= \lim_{h\to 0} \frac{f(g(x) + \Delta u) - f(g(x))}{\Delta u} \cdot \lim_{h\to 0} \frac{\Delta u}{h} \\
&= \lim_{\Delta u\to 0} \frac{f(g(x) + \Delta u) - f(g(x))}{\Delta u} \cdot \lim_{h\to 0} \frac{g(x+h) - g(x)}{h} \\
&= f'(g(x)) \cdot g'(x),
\end{aligned}$$

which completes the proof. □

Example 3.46. Differentiate $\sin(x^2 + e^x)$ with respect to x.

Solution.

Example **3.47.** Find dy/dx, by first writing the function in the form $y = f(u)$.

(a) $y = (x^3 - 1)^{100}$

(c) $g(x) = \tan^3 x$

(b) $y = \sin(2x^2)$

(d) $y = e^{(4\sqrt{x}+x^2)}$

Note: We sometimes have to use the Chain Rule two or more times to find the derivative and combine with other rules we've previously learned.

Example **3.48.** Find the derivative of the following:

(a) $y = \sqrt{e^{\cos x} + 1}$

(b) $y = (2x - 5)^{-2}(x^2 - 5x)^6$

Solution.

Example **3.49.** Find the derivative of the following:

(a) $y = e^{\sin^2(\pi t - 1)}$ (b) $y = \cos^4(\sec 3t)$ (c) $y = \left(\dfrac{t^2}{t^3 - 4}\right)^3$

Example **3.50.** Find the value of $(f \circ g)'$ at the given value of x. Use this to find an equation of the tangent line to the curve at the given point.

$$f(u) = u^5 + 1, \quad u = g(x) = \sqrt{x}, \quad x = 1$$

Solution.

Exercises 3.6

1. Given $y = f(u)$ and $u = g(x)$, find $dy/dx = f'(g(x))g'(x)$.

   (a) $y = \sin u, \quad u = 3x + 1$

   (b) $y = -\sec u, \quad u = \dfrac{1}{x} + 7x$

   Ans: (b) $dy/dx = -\sec(1/x + 7x)\tan(1/x + 7x)(-1/x^2 + 7)$

2. Find the derivatives of the functions.

   (a) $y = \left(\dfrac{x^2}{8} + x - \dfrac{1}{x}\right)^4$

   (b) $y = \cos^{-4} x$

   (c) $y = xe^{-x} + e^{x^3}$

   (d) $f(x) = x^2 \sec(1/x)$

   Ans: (b) $-4\cos^{-5} x(-\sin x) = 4\cos^{-5} x \sin x$. (c) $(1 - x)e^{-x} + 3x^2 e^{x^3}$

3. Find y''.

   (a) $y = e^{x^2} + (2x + 1)^4$

   (b) $y = \sin(e^x)$

   Ans: (b) $\big(-e^x \sin(e^x) + \cos(e^x)\big)e^x$

4. Suppose that functions f and g and their derivatives with respect to x have the values in the table at $x = 2$ and $x = 3$.

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
2	8	2	1/3	−3
3	3	−4	2π	5

   Find the derivatives with respect to x of the following combinations at the given value of x.

   (a) $f(x) \cdot g(x), \quad x = 3$

   (b) $f(g(x)), \quad x = 2$

   (c) $\sqrt{f(x)}, \quad x = 2$

   (d) $1/g^2(x), \quad x = 3$

   Ans: (b) $f'(g(x))g'(x)\big|_{x=2} = f'(2)g'(2) = -1$. (c) $\sqrt{2}/24$. (d) $(1/g^2)' = -2g'/g^3\big|_{x=3} = 5/32$

5. Let $f(x) = \tan(\pi x/4)$.

   (a) Find an eauation of the tangent line to the curve $y = f(x)$ at $t = 1$.

   (b) **Slopes on a tangent curve**. What is the smallest value the slope of the curve can ever have on the interval $-2 < x < 2$? Give reasons for your answer.

   Ans: (b) $f'(x) = \sec^2(\pi x/4) \cdot \frac{\pi}{4}$. Since $\min_{-2<x<2} \sec^2(\pi x/4) = 1$, the minimum of $f'(x) = \pi/4$.

6. **Challenge** Verify each of the following statements.

   (a) If f is even, then f' is odd.

   (b) If f is odd, then f' is even.

3.7. Implicit Differentiation

Implicitly Defined Functions

- The functions we have dealt with so far have been defined **explicitly** – for example, $y = \sqrt{x^3 + 1}$ and $y = x^2 \sin x$.
- Some functions are defined **implicitly** by a relation between x and y such as
$$x^2 + y^2 - 25 = 0 \quad \text{and} \quad x^2 + y^3 - 9xy = 0.$$
- **Implicitly Defined Functions** are in the form of $F(x, y) = 0$, in which ***y* is a function of *x***.

Implicit Differentiation: ***An application of Chain Rule.***

- Differentiate the equation (both sides) with respect to x, treating y as a differentiable function of x.
- Collect the terms with dy/dx on one side of the equation and solve for it.

Example 3.51. If $x^2 + y^2 - 25 = 0$, find the derivative dy/dx.

Solution.

Example **3.52.** Find the derivative.

(a) $2xy + y^2 = x + y$ (b) $e^{2x} = \sin(x + 3y)$

Solution.

Example **3.53.** Find the derivative:

$$x^3 = \frac{2x - y}{x + 3y}$$

Solution.

Example 3.54. Use implicit differentiation to find equations of the **tangent line** and the **normal line** to the curve at the given point.

$$x^2 + xy - y^2 = 1, \quad (2, 3)$$

Solution. Start with figuring out $y' = -(2x + y)/(x - 2y)$.

Example 3.55. Use implicit differentiation to find an equation of the tangent line to the curve at the given point.

$$2xy + \pi \sin y = 11\pi, \quad (2, 5\pi/2)$$

Solution. Start with figuring out $y' = -2y/(2x + \pi \cos y)$.

Exercises 3.7

1. Use implicit differentiation to find dy/dx.

   (a) $2xy + y^2 = x + y$
   (b) $e^{2x} = \sin(x + 3y)$
   (c) $x = \sec y$
   (d) $3 + \sin y = y - x^3$

   Ans: (a) $(1 - 2y)/(2x + 2y - 1)$. (b) $2e^{2x}/(3\cos(x + 3y)) - 1/3$.

2. Use implicit differentiation to find dy/dx and then d^2y/dx^2. Write the solutions in terms of x and y only.

   (a) $x^2 + y^2 = 1$
   (b) $xy + y^2 = 1$

   Ans: (b) $y'' = 2y(x + y)/(x^3 + 6x^2y + 12xy^2 + 8y^3)$

3. Verify that the given point is on the curve and find the lines that are (1) tangent and (2) normal to the curve at the given point.

   (a) $x^2 + y^2 = 25, \quad (3, -4)$
   (b) $x \sin 2y = y \cos 2x, \quad (\pi/4, \pi/2)$

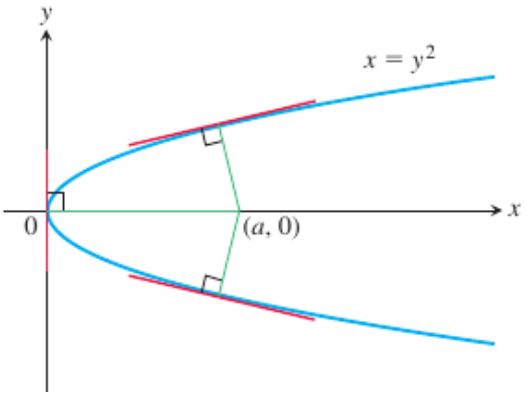
   Hint: (b) $y' = -(2y\sin(2x) + \sin(2y))/(2x\cos(2y) - \cos(2x))$

4. Verify that the following pairs of curves meet orthogonally.

   $$x^2 + y^2 = 4, \qquad x^2 = 3y^2$$

5. Challenge **Normals to a parabola**.

   (a) Show that if it is possible to draw three normals from the point $(a, 0)$ to the parabola $x = y^2$ shown in the accompanying diagram, then a must be greater than $1/2$.
   (b) One of the normals is the x-axis. For what value of a are the other two normals perpendicular?



Hint: $y' = 1/(2y)$. Thus the normal line at (x_0, y_0) reads $y - y_0 = -2y_0(x - x_0)$. (a) Now, the value a is decided when $y = 0$. (b) The two normal lines will be perpendicular when each of them makes an angle of $45°$ with the coordinate axes, i.e., their slope is either 1 or -1.

3.8. Derivatives of Inverse Functions and Logarithms

3.8.1. Derivatives of inverses of differentiable functions

Let f be differentiable and have inverse f^{-1}.

- Then
$$f^{-1}(f(x)) = x.$$
- Applying Chain Rule results in
$$[f^{-1}]'(f(x)) \cdot f'(x) = 1 \iff [f^{-1}]'(f(x)) = \frac{1}{f'(x)}, \tag{3.34}$$
which can be interpreted geometrically as in Figure 3.5.

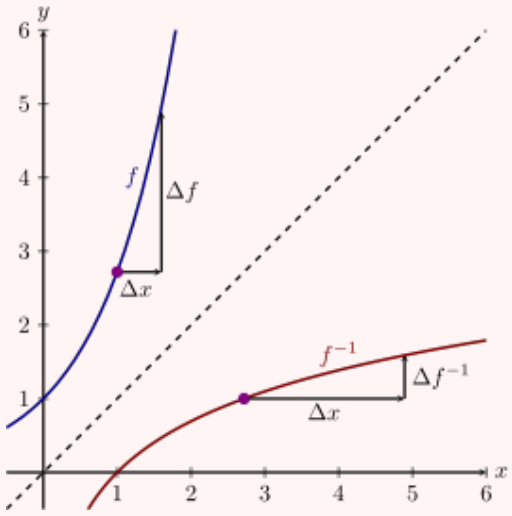


Figure 3.5: The derivative of the inverse function.

- Similarly, starting from $f(f^{-1}(x)) = x$, we obtain
$$[f^{-1}]'(x) = \frac{1}{f'(f^{-1}(x))}. \tag{3.35}$$

We will see exactly what (3.35) means, solving examples.

Example 3.56. Let $f(x) = x^3 - 2$, $x > 0$. Find $\dfrac{df^{-1}}{dx}(6)$.

Solution.

(a) **Explicit construction of f^{-1}**: The inverse of f can be found explicitly as

$$f^{-1}(x) = \sqrt[3]{x+2}.$$

(b) **Using Formula (3.35)**: Note that f^{-1} is defined in the range of f, $f(X)$. The point 6 must be one of them, that is,

$$6 = f(x)$$

for some x. Thus

$$x = f^{-1}(6) = 2.$$

Ans: $1/12$

Example 3.57. Let $f(x) = x^3 - 3x^2 - 1$, $x \geq 2$. Find $\dfrac{df^{-1}}{dx}(-1)$.

Solution.

Example **3.58.** Let $f(x) = x^2 - 4x - 5$, $x > 2$. Fund the value of df^{-1}/dx at $x = 0$.

Solution.

3.8.2. Derivatives of Logarithmic Functions

Let $f(x) = \log_a x$.

- It follows from the definition of $\log$ that

$$y = \log_a x \iff x = a^y = e^{\ln a^y} = e^{y \ln a}. \tag{3.36}$$

- Apply implicit differentiation to the right side of (3.36) to get

$$1 = e^{y \ln a} \cdot y' \ln a \implies y' = \frac{1}{e^{y \ln a} \cdot \ln a} = \frac{1}{x \ln a}. \tag{3.37}$$

Note: If $x > 0$, then $x = e^{\ln x}$.

Summary **3.59.** We may apply above arguments and Chain Rule to get the following formulas.

$$\begin{aligned} \frac{d}{dx}\log_a x &= \frac{1}{x \ln a} & \frac{d}{dx}\ln x &= \frac{1}{x} \\ \frac{d}{dx}a^x &= a^x \ln a & \frac{d}{dx}e^x &= e^x \\ \frac{d}{dx}\ln f(x) &= \frac{f'(x)}{f(x)} & \frac{d}{dx}\ln|x| &= \frac{1}{x} \end{aligned} \tag{3.38}$$

See Example 3.64 below for the formula: $\dfrac{d}{dx}a^x = a^x \ln a$.

Example **3.60.** Differentiate $f = x^x$, $x > 0$.

Solution. Note that $x^x = e^{\ln x^x} = e^{x \ln x}$.

Ans: $f'(x) = x^x(\ln x + 1)$

Example **3.61.** Find the derivative of the following:

(a) $y = \ln(3x^2)$

(b) $g(x) = \sqrt{\ln x}$

(c) $y = \ln(\cos\theta)$

(d) $y = \ln(3te^{-t})$

Solution.

Example **3.62.** Find $\dfrac{d}{dx}\left[\ln\left(\sqrt{\dfrac{(x+1)^{10}}{(2x+1)^5}}\right)\right]$.

Solution.

Note: From the previous example, we see that using the properties of logarithms prior to differentiation may simplify our derivatives overall.

3.8.3. Logarithmic Differentiation

Let's try to use the natural logarithm to manipulate products and powers effectively for a convenient calculation of the derivative.

Algorithm **3.63.** **Logarithmic Differentiation**

1. Take natural logarithms of both sides of an equation $y = f(x)$ and use the Laws of Logarithms to conveniently reform the right side.
2. Differentiate implicitly with respect to x.
3. Solve the resulting equation for y'.
4. Replace y with $f(x)$.

Example **3.64.** Let $y = a^x$. Use logarithmic differentiation to prove that $y' = a^x \ln a$.

Solution.

Example **3.65.** Use logarithmic differentiation to find the derivative of

(a) $y = (x+1)^x$.

(b) $y = (\sin x)^{\ln x}$.

Solution.

Theorem 3.66. **The Number e as a Limit**: *The number e can be calculated as the limit*

$$e = \lim_{x\to 0}(1+x)^{1/x}. \tag{3.39}$$

Proof. If $f(x) = \ln x$, then $f'(x) = 1/x$ and $f'(1) = 1$. By the definition of derivative,

$$\begin{aligned} f'(1) &= \lim_{h\to 0}\frac{f(1+h)-f(1)}{h} = \lim_{x\to 0}\frac{f(1+x)-f(1)}{x} \\ &= \lim_{x\to 0}\frac{1}{x}\ln(1+x) = \lim_{x\to 0}\ln(1+x)^{1/x} \\ &= \ln\Big[\lim_{x\to 0}(1+x)^{1/x}\Big] = 1, \end{aligned} \tag{3.40}$$

from which (3.39) follows. □

Remark 3.67. The formula in (3.39) can be rewritten in various forms. A popular form reads

$$e = \lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n, \tag{3.41}$$

where the right-side is an increasing sequence, approaching e, as n grows.

Exercises 3.8

1. Suppose that the differentiable function $y = f(x)$ has an inverse and that the graph of f passes through the point $(1, 5)$ and has a slope of $1/2$ there. Can you find each of the following? If yes, find it.

   (a) $\dfrac{df^{-1}}{dx}(1/2)$.

   (b) $\dfrac{df^{-1}}{dx}(1)$.

   (c) $\dfrac{df^{-1}}{dx}(5)$.

2. Let $f(x) = x^2 - 4x - 5$, $x > 2$. Find $\dfrac{df^{-1}}{dx}(0)$.
   Hint: $0 = f(5)$

3. Suppose that the function f ans its derivative have the following values at $x = 0, 1, 2, 3, 4$.

x	0	1	2	3	4
$f(x)$	-4	3	-1	2	1
$f'(x)$	3	2	$5/4$	$2/3$	$1/5$

   Assuming the inverse function f^{-1} is differentiable, find the slope of $f^{-1}(x)$ at

   (a) $x = 1$ (b) $x = 2$ (c) $x = 3$

   Ans: (c) $1/2$

4. **Logarithmic Differentiation**. Use logarithmic differentiation to find the derivative of y.

   (a) $y = x(x + 1)(x + 2)$

   (b) $y = \sqrt{x(x + 1)}$

   (c) $y = \dfrac{\theta \sin\theta}{\sqrt{\sec\theta}}$

   (d) $y = (\tan\theta)\sqrt{2\theta + 1}$

5. Use logarithmic differentiation or the method used in Example 3.60 to find dy/dx.

   (a) $y = x^{\sqrt{x}}$

   (b) $y = (\sin x)^x$

   (c) $y^x = x^3 y$

   (d) $x = y^{xy}$

   Ans: (d) $\dfrac{1 - xy\ln y}{x^2(1 + \ln y)}$

3.9. Inverse Trigonometric Functions

Recall from Section 1.6.3

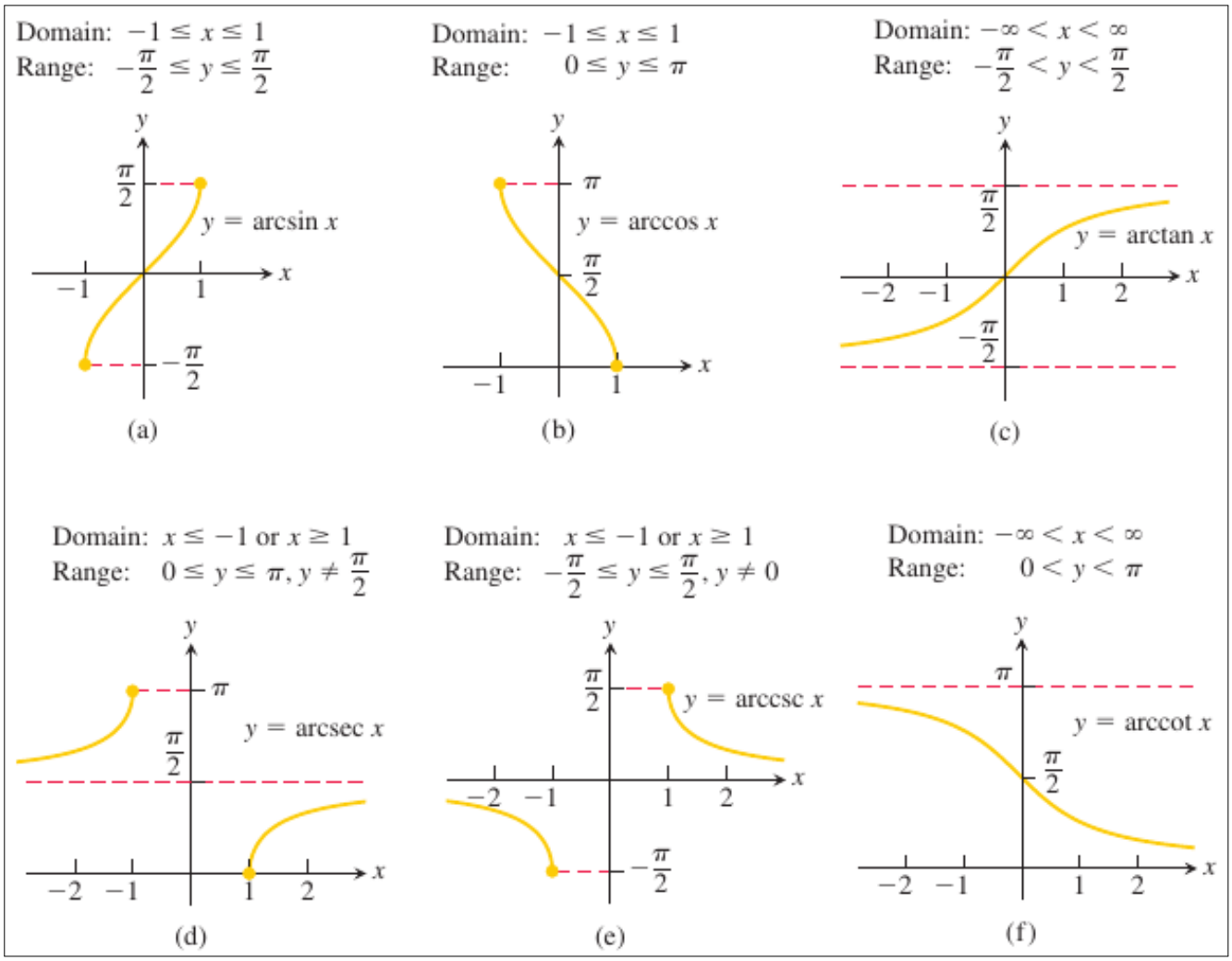


Figure 1.26 (p.53): Graphs of the six basic inverse trigonometric functions.

The trigonometric functions are not one-to-one. If we restrict their domains, we can make them one-to-one and they will have inverses. For example,

Definition 1.70 (p.53). (The Arcsine and Arccosine Functions)

$y =$ **arcsin** x is the number in $[-\pi/2, \pi/2]$ for which $\sin y = x$
$y =$ **arccos** x is the number in $[0, \pi]$ for which $\cos y = x$

Example 3.68. Find the exact value for the expression:

(a) $\cos^{-1}(1/2)$

(b) $\arctan(1)$

(c) $\arcsin\left(-\dfrac{\sqrt{3}}{2}\right)$

Solution.

Derivatives of the Inverse Trigonometric Functions

The derivatives can be found by using the definition of inverses.

- $\boldsymbol{y = \arcsin x}$: Then, $x = \sin y$. Implicit differentiation gives

$$1 = (\cos y)y' \quad \Longrightarrow \quad y' = \frac{1}{\cos y} = \frac{1}{\sqrt{1-\sin^2 y}} = \frac{1}{\sqrt{1-x^2}}$$

- $\boldsymbol{y = \arctan x}$: Then, $x = \tan y$. Implicit differentiation gives

$$1 = (\sec^2 y)y' \quad \Longrightarrow \quad y' = \frac{1}{\sec^2 y} = \frac{1}{1+\tan^2 y} = \frac{1}{1+x^2}$$

- The derivatives can be found similarly for other inverse trigonometric functions.

Summary **3.69.** Here we collect the derivatives of the inverse trigonometric functions.

$$\begin{aligned}
\frac{d}{dx}(\arcsin x) &= \frac{1}{\sqrt{1-x^2}} \quad (|x|<1) & \frac{d}{dx}(\arccos x) &= -\frac{1}{\sqrt{1-x^2}} \quad (|x|<1)\\
\frac{d}{dx}(\arctan x) &= \frac{1}{1+x^2} & \frac{d}{dx}(\mathbf{arccot}\, x) &= -\frac{1}{1+x^2}\\
\frac{d}{dx}(\mathbf{arcsec}\, x) &= \frac{1}{|x|\sqrt{x^2-1}} \quad (|x|>1) & \frac{d}{dx}(\mathbf{arccsc}\, x) &= -\frac{1}{|x|\sqrt{x^2-1}} \quad (|x|>1)
\end{aligned} \tag{3.42}$$

Note: We can also combine these derivatives with ALL the rules we've previously covered.

Example **3.70.** Find the derivative of the functions.

(a) $y = \arcsin(\sqrt{2}t)$

(b) $y = \sec^{-1}(5s)$

(c) $y = \ln(\mathbf{arccot}\,(9x^2))$

(d) $y = x\arcsin x + \sqrt{1-x^2}$

Exercises 3.9

1. Find exact angles:

   (a) $\arcsin(-1/2)$
   (b) $\arccos(1/2)$
   (c) $\tan^{-1}\sqrt{3}$
   (d) $\csc^{-1}\sqrt{2}$

   Ans: (d) $\pi/4$

2. Find the values.

   (a) $\sin\left(\cos^{-1}\left(\dfrac{1}{\sqrt{2}}\right)\right)$
   (b) $\tan\left(\arcsin\left(-\dfrac{1}{2}\right)\right)$

   Ans: (b) $-1/\sqrt{3}$

3. Find the derivative of y.

   (a) $y = \arccos(1/x)$
   (b) $y = \ln(\tan^{-1} x)$
   (c) $y = \tan^{-1}(\ln x)$
   (d) $y = \cos(x - \arccos x)$

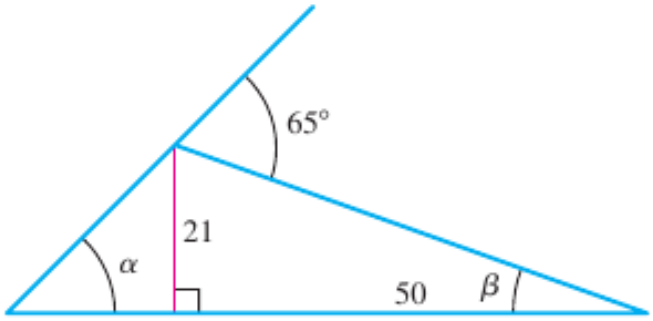
   Ans: (a) $1/(x^2\sqrt{1-1/x^2})$. (d) $-(1+1/\sqrt{1-x^2})\sin(x-\arccos x)$.

4. Use implicit differentiation to find dy/dx and then an equation of the tangent line to the curve at $P(0, 1/2)$.

$$\arcsin(x+y) + \arccos(x-y) = \frac{5\pi}{6}$$

   Ans: $y = 1/2$

5. Find the angle α (in degrees). ***Hint***: $\alpha + \beta = 65°$.



3.10. Related Rates

Note: In this section, we look at questions that arise when two or more related quantities are changing (w.r.t. a common variable, e.g., time). The problem of determining how the rate of change of one of them affects the rate of change of the others is called a **related rates problem**.

Example 3.71. Suppose we are pumping air into a spherical balloon.

- Both the volume and radius of the balloon are increasing over **time**.
- If V is the volume and r is the radius of the balloon at an instant of time, then
$$V = \frac{4}{3}\pi r^3. \tag{3.43}$$
- Using the Chain Rule, we differentiate both sides with respect to t to find an equation relating the rates of change of V and r,
$$\frac{dV}{dt} = \frac{dV}{dr} \cdot \frac{dr}{dt} = 4\pi r^2 \frac{dr}{dt}. \tag{3.44}$$
- The rates dV/dt and dr/dt are related by (3.44). If we know one, then the other can be computed by the relation.

Strategy 3.72. **Related Rates Problem Strategy**:

1. Let t denote time, and **name the variables and constants**.
   - We will assume that all variables are differentiable functions of t.
   - For most cases, to **draw a picture** would be helpful.
2. Write **an equation that relates the variables**.
   - You may have to combine two or more equations to get a single equation that relates the variable whose rate you want to the variables whose rates you know.
3. Differentiate with respect to t to obtain a **related rates equation**.
4. Use **known values (data)** to find **the unknown rate**.

Example **3.73.** A spherical balloon is inflated with helium at a rate of $100\pi\,\text{ft}^3/\text{min}$. How fast is the balloon's radius increasing at the instant the radius is 5 ft?

Solution.

Ans: $1\,\text{ft/min}$

Example **3.74.** The length of a rectangle is decreasing at a rate of 2 cm/sec and its width is increasing at a rate of 2 cm/sec. When the length is 12 cm and the width is 5 cm, how fast is the area of the rectangle changing?

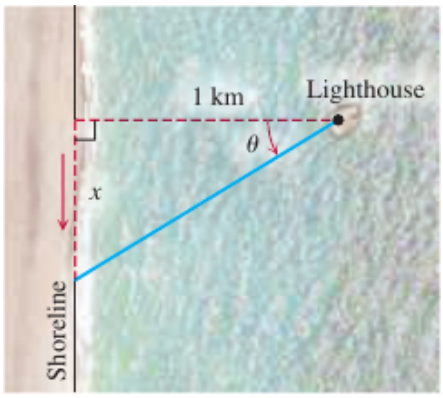
Solution.

Example 3.75. Sand falls from a conveyor belt at a rate of 10 m^3/min onto the top of a conical pile. The height of the pile is always three-eights of the base diameter. How fast is the height of the pile increasing when the pile is 4 m high?

Solution. Begin with $V = \frac{1}{3}\pi r^2 h$ and $h = \frac{3}{8} \cdot 2r$.

Example 3.76. A lighthouse sits 1 km offshore, and its beam of light rotates counterclockwise at the constant rate of 3 full circles per minute. At what rate is the image of the beam moving down the shoreline when the image is 1 km from the spot on the shoreline nearest the lighthouse?

Solution. Begin with $x = \tan\theta$.



Ans: 12 km/min

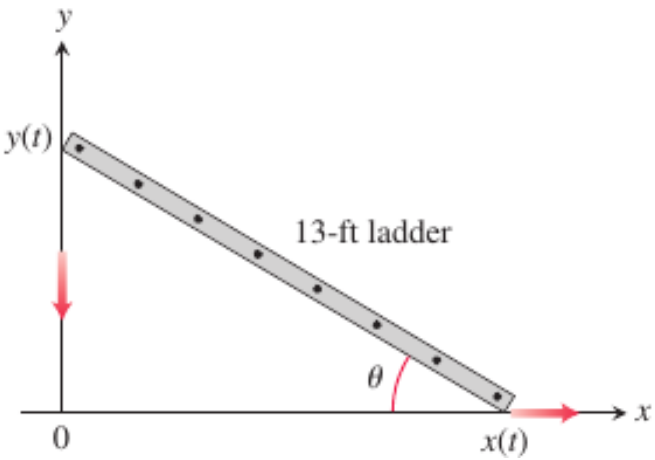
Example 3.77. A sliding ladder. A 13 ft ladder is leaning against a house when its base starts to slide away. By the time the base is 12 ft from the house, the base is moving at the rate of 5 ft/sec.

(a) How fast is the top of the ladder sliding down the wall then?

(b) At what rate is the area of the triangle formed by the ladder, wall, and ground changing then?

(c) At what rate is the angle θ between the ladder and the ground changing then?

Solution. Begin with
(a) $h^2 + b^2 = 13^2 = 169$,
(b) $A = \frac{1}{2}hb$, and
(c) $\tan\theta = \frac{h}{b}$.



Ans: (b) -59.5 ft^2/sec. (c) -1 rad/sec.

Exercises 3.10

1. If $x^2 + y^2 = 25$ and $dx/dt = -2$, then what is dy/dt when $x = 3$ and $y = -4$?

2. (a) Assume that $y = 5x$ and $dx/dt = 2$. Find dy/dt.
   (b) Assume that $2x + 3y = 12$ and $dy/dt = -2$. Find dx/dt.

   Ans: (b) 3.

3. Let x and y be differentiable functions of t and let $s = \sqrt{x^2 + y^2}$ be the distance between the points $(x, 0)$ and $(0, y)$ in the xy-plane.

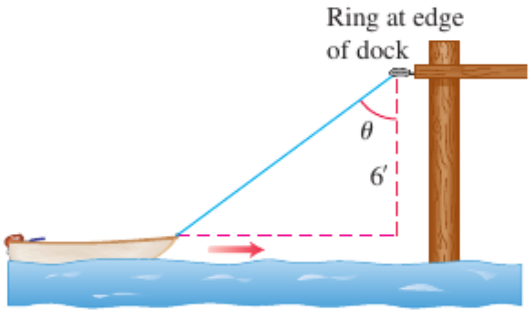
   (a) How is ds/dt related to dx/dt if y is constant?
   (b) How is ds/dt related to dx/dt and dy/dt if neither x nor y is constant?
   (c) How is dx/dt related to dy/dt if s is constant?

   Ans: (b) $\dfrac{ds}{dt} = \dfrac{x}{\sqrt{x^2+y^2}}\dfrac{dx}{dt} + \dfrac{y}{\sqrt{x^2+y^2}}\dfrac{dy}{dt}$.

> **Note**: Let $s = xy$, for example. Then, we know $\dfrac{ds}{dt} = \dfrac{dx}{dt}y + x\dfrac{dy}{dt}$. Such Product Rule can be rephrased as the following: $\dfrac{ds}{dt} = \dfrac{ds}{dx}\dfrac{dx}{dt} + \dfrac{ds}{dy}\dfrac{dy}{dt} = y\dfrac{dx}{dt} + x\dfrac{dy}{dt}$.

4. **Hauling in a dinghy**. A dinghy is pulled toward a dock by a rope from the bow through a ring on the dock 6 ft above the bow. The rope is hauled in at the rate of 2 ft/sec.

   (a) How fast is the boat approaching the dock when 10 ft of rope are out?
   (b) At what rate is the angle θ changing at this instant (see the figure)?



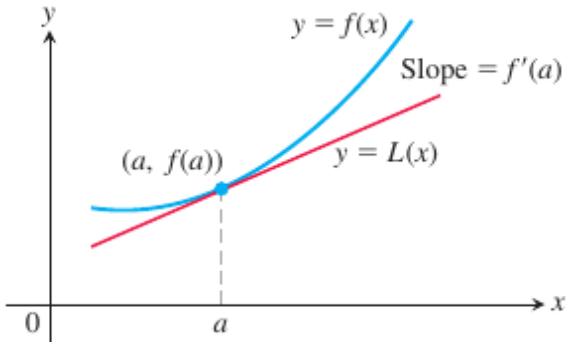
Hint: Begin with (a) $\tan\theta = x/6$ and (b) $r\cos\theta = 6$.

3.11. Linearization and Differentials

3.11.1. Linearization

In general, the **tangent line** to $y = f(x)$ at a point $x = a$, where f is differentiable, passes through the point $(a, f(a))$, so its **point-slope equation** is

$$y = f(a) + f'(a)(x - a).$$



Definition 3.78. If f is differentiable at $x = a$, then the approximating function

$$L(x) := f(a) + f'(a)(x - a) \tag{3.45}$$

is the **linearization** of f at a. The approximation

$$f(x) \approx L(x) \tag{3.46}$$

of f by L is the **linear approximation** (or, **tangent line approximation**) of f at a. The point $x = a$ is the **center** of the approximation.

Note: As long as this line remains close the graph of f, as we move off the point of tangency, $L(x)$ gives a good approximation to $f(x)$.

Example 3.79. Find the linearization $L(x)$ of $f(x) = x^3 - 3x + 2$ at $a = 0$.

Solution.

Example **3.80.** Find the linearization $L(x)$ of $f(x) = \sqrt[3]{x}$ at $a = -8$.

Solution.

Example **3.81.** Find the linearization $L(x)$ of $f(x) = 2\tan x$ at $a = \pi$.

Solution.

Example **3.82.** Use a linear approximation to estimate the given number: $(1.0002)^{50}$.

Solution.

3.11.2. Differentials

We often use the Leibniz notation dy/dx to represent the derivative of y with respect to x. Contrary to its appearance, it is not a ratio. We now introduce **two new variables dx and dy**, **of which their ratio (when exists) is equal to the derivative**.

Definition 3.83. Let $y = f(x)$ be a differentiable function. The **differential** dx is an independent variable. The **differential** dy is a dependent variable, defined as

$$dy = f'(x)dx. \tag{3.47}$$

Remark 3.84. What is dy?
Often the variable dx is chosen to be Δx, the change in x. Then the differential dy is the change in the linearization of f at $x = a$, ΔL.

Example 3.85. Let $f(x) = x^5 + 37x$.

(a) Find dy.

(b) Find the value of dy when $x = 1$ and $dx = 0.2$.

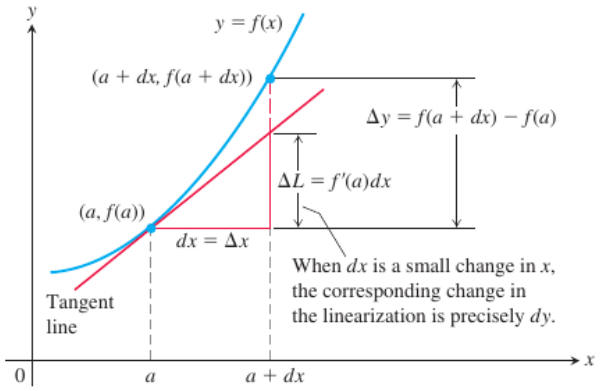


Figure 3.6: Geometrically, the differential $dy = \Delta L$, when $x = a$ and $dx = \Delta x$.

Example 3.86. Find the differential dy of the following functions.

(a) $y = 4e^{x^2}$

(b) $y = \ln\left(\dfrac{x+1}{\sqrt{x-1}}\right)$

Solution.

Example 3.87. Find the differential dy and evaluate it for the given values of x and dx.

(a) $f(x) = 2x^2 + 4x - 3$, $x = -2$, $dx = 0.1$

(b) $y = x^{-1}$, $x = 0.5$, $dx = 0.1$

Solution.

Exercises 3.11

1. **Common linear approximations at $x = 0$.** Find the linearizations of the following functions at $x = 0$.

   (a) $\sin x$
   (b) $\cos x$
   (c) $\tan x$
   (d) e^x

   Ans: (b) $L(x) = 1$. (c) $L(x) = x$.

2. (i) Find $f(a)$. (ii) Find a linearization at a suitably chosen integer near a at which the given function and its derivative are easy to evaluate. (iii) Use the linearization to estimate f at a.

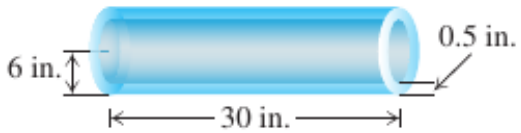
   (a) $f(x) = x^2 + 2x, \quad a = 0.1$
   (b) $f(x) = 2x^2 + 3x - 3, \quad a = -0.9$

   Ans: (b) (i) -4.08. (ii) $L(x) = -x - 5$. (iii) -4.1.

3. Each function $f(x)$ changes value when x changes from x_0 to $x_0 + dx$.
   Find $\Delta f = f(x_0 + dx) - f(x_0)$ and $df = f'(x_0)dx$.

   (a) $f(x) = 2x^2 + 4^x - 3, \quad x_0 = -1, \quad dx = 0.1$
   (b) $f(x) = x^3 - x, \quad x_0 = 1, \quad dx = 0.05$

4. **Estimating volume**. Estimate the volume of material in a cylindrical shell with length 30 in., radius 6 in., and shell thickness 0.5 in.



Hint: The whole volume $V = \pi r^2 \cdot 30 = 30\pi r^2$. Thus $dV = 60\pi r \cdot dr$. Now, what are r and dr?

Definition 3.88. A quadratic approximation $Q(x) = b_0 + b_1(x - a) + b_2(x - a)^2$ to $f(x)$ at $x = a$ is defined to satisfy the properties:

(i) $Q(a) = f(a)$
(ii) $Q'(a) = f'(a)$
(ii) $Q''(a) = f''(a)$

Thus, $b_0 = f(a)$, $b_1 = f'(a)$, and $b_2 = \dfrac{f''(a)}{2}$.

5. CAS Let $f(x) = \sqrt{1 + x}$.

   (a) Find the linear and quadratic approximations of f at $x = 0$.
   (b) Graph f and its linear and quadratic approximations over the interval $[-1, 2]$.

CHAPTER 4

Applications of Derivatives

In this chapter, ...

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4.1. Extreme Values of Functions on Closed Intervals

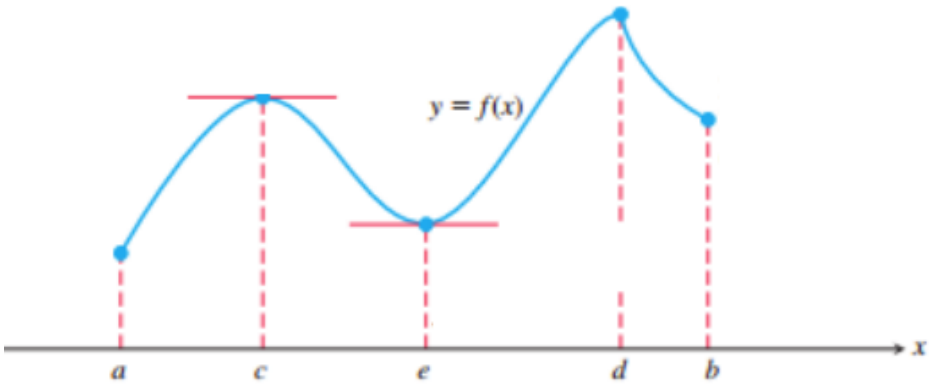
Definition 4.1. Let f be function with domain D. Then f has

- **absolute maximum** value on D at a point c
  if $f(c) \geq f(x)$ for all x in D
- **absolute minimum** value on D at a point c
  if $f(c) \leq f(x)$ for all x in D

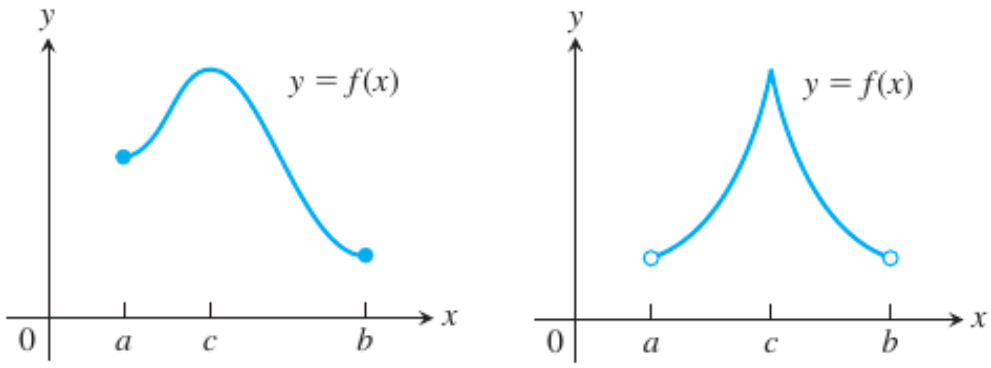
The number $f(c)$ is a

- **local maximum** value at a point c
  if $f(c) \geq f(x)$ when x is in an open interval containing c
- **local minimum** value at a point c
  if $f(c) \leq f(x)$ when x is in an open interval containing c

Example 4.2. For each point, a–e, identify whether it corresponds to a local/absolute maximum or minimum.



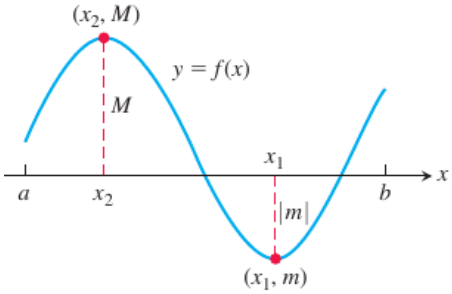
Example 4.3. Determine from the graph whether the function has any absolute extreme values on $[a, b]$.



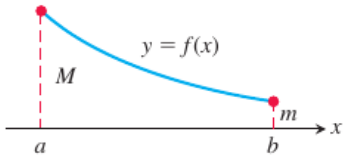
Theorem 4.4. **The Extreme Value Theorem.** *If f is **continuous** on a **closed interval** $[a, b]$, then f attains both an absolute maximum value M and an absolute minimum value m in $[a, b]$, i.e.,*

- *there are numbers x_1 and x_2 in $[a, b]$ with $f(x_1) = m$, $f(x_2) = M$, and*

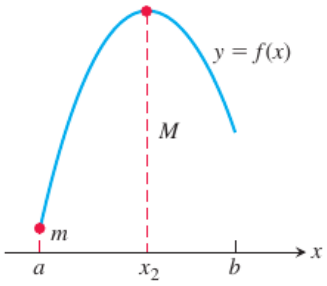
$$m \le f(x) \le M, \quad \textbf{for every } x \in [a, b]. \tag{4.1}$$



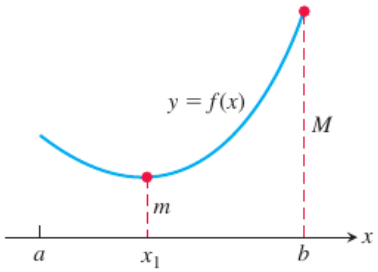
Maximum and minimum at interior points



Maximum and minimum at endpoints



Maximum at interior point, minimum at endpoint



Minimum at interior point, maximum at endpoint

Finding Extrema

How do we know where to look for a function's extrema? The following theorem and definition can help.

Theorem 4.5. **The First Derivative Theorem for Local Extreme Values.** *If f has a local maximum or minimum value at an interior point c of its domain, and if f' is defined at c, then*

$$f'(c) = 0.$$

Definition 4.6. **An interior point in the domain** of a function f where **f' is zero or undefined** is a **critical point** of f.

Example 4.7. Determine all critical points for each function.

(a) $f(x) = 6x^2 - x^3$ (b) $f(x) = 4x - \tan x$ (c) $f(x) = x - 3x^{\frac{2}{3}}$

Solution.

Strategy 4.8. Finding the Absolute Extrema of a Continuous Function f on a Finite Closed Interval

1. Find all **critical points** of f on the interval.
2. Evaluate f at all **critical points** and **endpoints**.
3. Take the largest and smallest of these values.

Example 4.9. Find the absolute maximum and minimum values of each function on the given interval.

(a) $f(x) = 3x^4 - 4x^3 - 12x^2 + 1, \quad 0 \le x \le 4$

(b) $f(x) = \csc x, \quad \frac{\pi}{3} \le x \le \frac{2\pi}{3}$

(c) $f(x) = \frac{1}{x} + \ln x, \quad \frac{1}{e} \le x \le 4$

Exercises 4.1

1. Sketch the graph of each function and determine whether the function has any absolute extreme values on its domain. Explain how your answer is consistent with the Extreme Value Theorem, p.171.

   (a) $f(x) = |x|, \quad -1 < x < 2$

   (b) $g(x) = \begin{cases} -x, & 0 \le x < 1 \\ x - 1, & 1 \le x \le 2 \end{cases}$

   Clue: (b) g is not continuous.

2. Find the absolute maximum and minimum values of each function on the given interval. Then graph the function. Identify the points on the graph where the absolute extrema occur, and include their coordinates.

   (a) $f(x) = 4 - x^3, \quad -2 \le x \le 1$

   (b) $g(x) = -\dfrac{1}{x}, \quad -2 \le x \le -1$

   (c) $h(t) = 2 - |t|, \quad -1 \le t \le 3$

   (d) $F(x) = \sec x, \quad -\dfrac{\pi}{3} \le x \le \dfrac{\pi}{6}$

   Ans: (d) absolute maximum: $(-\pi/3, 2)$, absolute minimum: $(0, 1)$.

3. Determine all critical points for each function.

   (a) $f(x) = x(x-3)^3$

   (b) $g(x) = \dfrac{x^2}{x-2}$

   Clue: (b) Is $x = 2$ is a critical point of g? See Definition 4.6.

   Ans: (b) $x = 0,\ 4$

4. CAS You will use a CAS to help find the absolute extrema of the given function over the specified closed interval. Perform the following steps.

   (1) Plot the function over the interval to see its general behavior there.
   (2) Find the interior points where $f' = 0$. (You may want to plot f' as well.)
   (3) Find the interior points where f' does not exist.
   (4) Evaluate the function at all points found in parts (2) and (3) and at the endpoints of the interval.
   (5) Find the function's absolute extreme values on the interval and identify where they occur.

   (a) $f(x) = \sqrt{x} + \cos x, \quad [0,\ 2\pi]$

   (b) $f(x) = \ln(2x + x\sin x), \quad [1,\ 15]$

4.2. The Mean Value Theorem

To arrive at the Mean Value Theorem, we first investigate a special case.

Theorem 4.10. ***Rolle's Theorem.*** *Suppose that $y = f(x)$ is **continuous** over the **closed interval** $[a, b]$, and **differentiable** at every point of its interior (a, b). If $f(a) = f(b)$, then there is at least one number $c \in (a, b)$ at which*

$$f'(c) = 0. \tag{4.2}$$

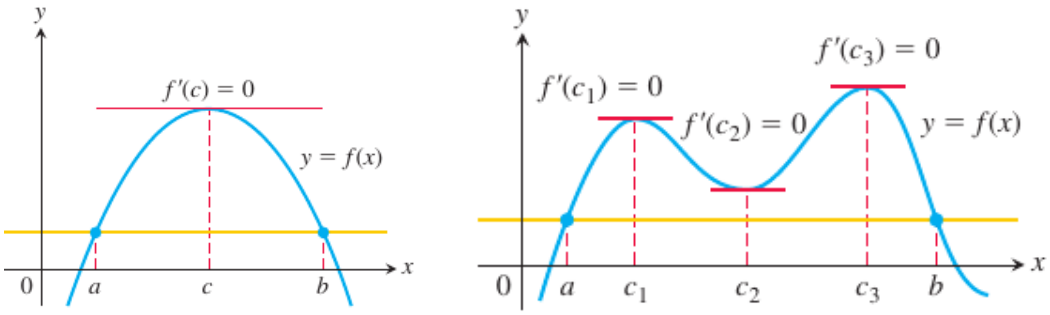


Figure 4.1: Rolle's Theorem

Note: The **Mean Value Theorem**, first stated by **Joseph-Louis Lagrange**, is a slanted version of Rolle's Theorem. The Mean Value Theorem guarantees that there is a point where the tangent line is parallel to the secant line that joins A and B.

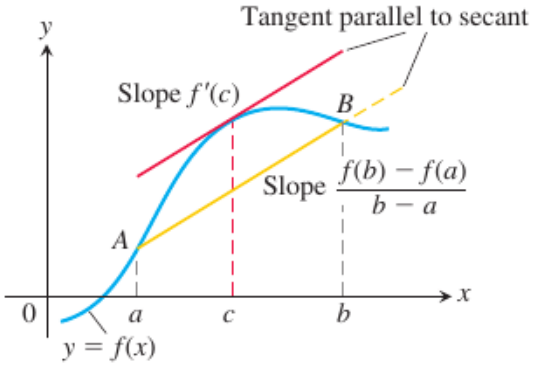


Figure 4.2: The Mean Value Theorem

Theorem 4.11. **The Mean Value Theorem (MVT).** *Suppose $y = f(x)$ is **continuous** over a **closed interval** $[a, b]$, and **differentiable** on the interval's interior (a, b). Then there is at least one point $c \in (a, b)$ at which*

$$f'(c) = \frac{f(b) - f(a)}{b - a}. \tag{4.3}$$

Example 4.12. Verify that the function satisfies the hypotheses of the MVT on the given interval. Find the value or values c that satisfy (4.3).

(a) $f(x) = x^2 + 2x - 1$ $[0, 1]$

(b) $f(x) = x + \dfrac{1}{x}$ $\left[\dfrac{1}{2}, 2\right]$

(c) $f(x) = \ln(x - 1)$ $[2, 4]$

Solution.

Example 4.13. Suppose that $f'(x) \leq 1$ for $1 \leq x \leq 4$. Show that $f(4) - f(1) \leq 3$.

Solution.

Example 4.14. It took 14 seconds for a mercury thermometer to rise from $-19°$C to $100°$C when it was taken from a freezer and placed in boiling water.

(a) Show that somewhere along the way, the temperature of the mercury was rising at a rate of $8.5°$ C/sec.

(b) Can we find a moment when the temperature of the mercury was rising at a rate of $9°$ C/sec?

Solution.

Corollary 4.15. Mathematical Consequences of the MVT

1. *If $f'(x) = 0$ at each point x of an open interval (a, b), then $f(x) = C$ for all $a \leq x \leq b$, where C is a constant.*
2. *If $f'(x) = g'(x)$ at each point x in an open interval (a, b), then there exists a constant C such that $f(x) = g(x) + C$ for all $a \leq x \leq b$. That is, $f - g$ is a constant function on (a, b).*

Example 4.16. Find the function $f(x)$ whose derivative is $\sin x$ and whose graph passes through the point $(0, 2)$.

Solution.

Ans: $f(x) = -\cos x + 3$.

Example 4.17. Find all possible functions with the given derivative.

(a) $y' = 2x - 1$ (b) $y' = \sin 2t + \sqrt{t}$

Solution.

Exercises 4.2

1. Find the value or values of c that satisfy the equation $f'(c) = \dfrac{f(b) - f(a)}{b - a}$.

   (a) $f(x) = x^3 - x^2, \quad [-1, 2]$ (b) $f(x) = \sqrt{x - 1}, \quad [1, 3]$

2. Suppose that $f(0) = 5$ and that $f'(x) = 2$ for all x. Must $f(x) = 2x + 5$ for all x? Give reasons for your answer.

3. Suppose that $f'(x) = 2x$ for all x. Find $f(2)$ if

   (a) $f(0) = 0$ (b) $f(1) = 0$ (c) $f(-2) = 3$

   Ans: (b) $f(2) = 3$.

4. Find the function with the given derivative whose graph passes through the point P.

   (a) $f'(x) = \dfrac{1}{x^2} + 2x, \quad P(-1, 1)$ (b) $g'(t) = \sec t \tan t - 1, \quad P(0, 0)$

   Ans: (b) $g(t) = \sec t - t - 1$

5. Show that for any numbers x and y, the sine inequality $|\sin x - \sin y| \leq |x - y|$ is true. ***Hint***: Use the MVT.

6. **Challenge** Suppose that f'' is continuous on $[a, b]$ and that f has three zeros in the interval. Show that f'' has at least one zero in (a, b). ***Hint***: You may assume that f has three *distinct* zeros. The problem is to apply Rolle's Theorem.

4.3. Monotonic Functions and the First Derivative Test

Corollary 4.18. **Another corollary to the Mean Value Theorem.** *Suppose that f is continuous on $[a, b]$ and differentiable on (a, b).*

*(a) If $f'(x) > 0$ at each point $x \in (a, b)$, then f is **increasing** on $[a, b]$.*

*(b) If $f'(x) < 0$ at each point $x \in (a, b)$, then f is **decreasing** on $[a, b]$.*

Remark 4.19. Let x_1 and x_2 be any two points in $[a, b]$ with $x_1 < x_2$.

- Then the Mean Value Theorem says that

$$f(x_2) - f(x_1) = f'(c)(x_2 - x_1), \tag{4.4}$$

  for some $c \in (x_1, x_2)$. Corollary 4.18 follows from (4.4).
- A function that is increasing or decreasing on an interval is said to be **monotonic** on the interval.

Example 4.20. Find the critical points of $f(x) = x^3 - 12x - 5$ and identify the open intervals on which f is increasing and on which f is decreasing.

Solution.

x		-2		2	
$f'(x)$		0		0	
$f(x)$					
Behavior of f					

First Derivative Test for Local Extrema

Suppose that c is a critical point of a continuous function f, and that f is differentiable at every point in some interval containing c, except possibly at c itself. Moving across this interval from left to right,

1. if f' changes from negative to positive at c,
   then f has a local minimum at c;
2. if f' changes from positive to negative at c,
   then f has a local maximum at c;
3. if f' does not change sign at c,
   then f has no local extremum at c.

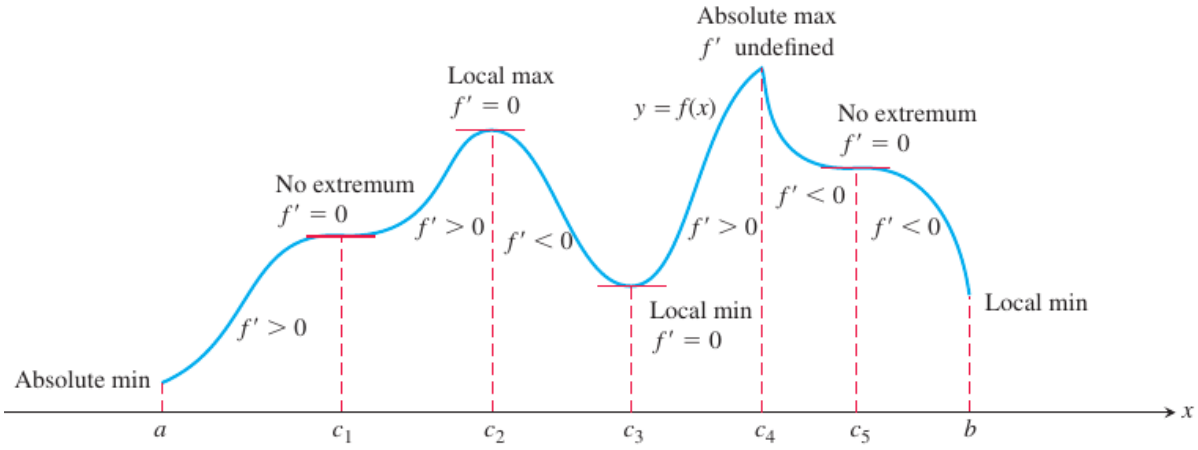


Figure 4.3: First Derivative Test for local extrema.

Example 4.21. Let $f(x) = (x^2 - 3)e^x$.

(a) Find the critical points of the function.

(b) Identify the open intervals on which f is increasing and on which f is decreasing.

(c) Find the function's local and absolute extreme values.

Exercises 4.3

1. (i) Find the open intervals on which the function is increasing and decreasing. (ii) Identify the function's local and absolute extreme values, if any, saying where they occur.

   (a) $f(x) = -x^3 + 3x^2$ (b) $g(x) = x^4 - 4x^3 + 4x^2$

2. (a) Prove that $f(x) = x - \ln x$ is increasing for $x > 1$.
   (b) Using part (a), show that $\ln x < x$ if $x > 1$.

   Hint: (a) $f'(x) = 1 - 1/x$, which is clearly positive for $x > 1$.

3. CAS (i) Identify the function's local extreme values in the given domain, and say where they occur. (ii) Which of the extreme values, if any, are absolute? (iii) Support your findings with a CAS.

   (a) $f(x) = x^3 - 3x^2, \quad -\infty < x \le 3$ (b) $g(x) = \dfrac{x-2}{x^2-1}, \quad 0 \le x < 1$

4.4. Concavity and Curve Sketching

Definition 4.22. The graph of a differentiable function $y = f(x)$ is

(a) **concave up (convex)** on an open interval I if f' is increasing on I;

(b) **concave down** on an open interval I if f' is decreasing on I.

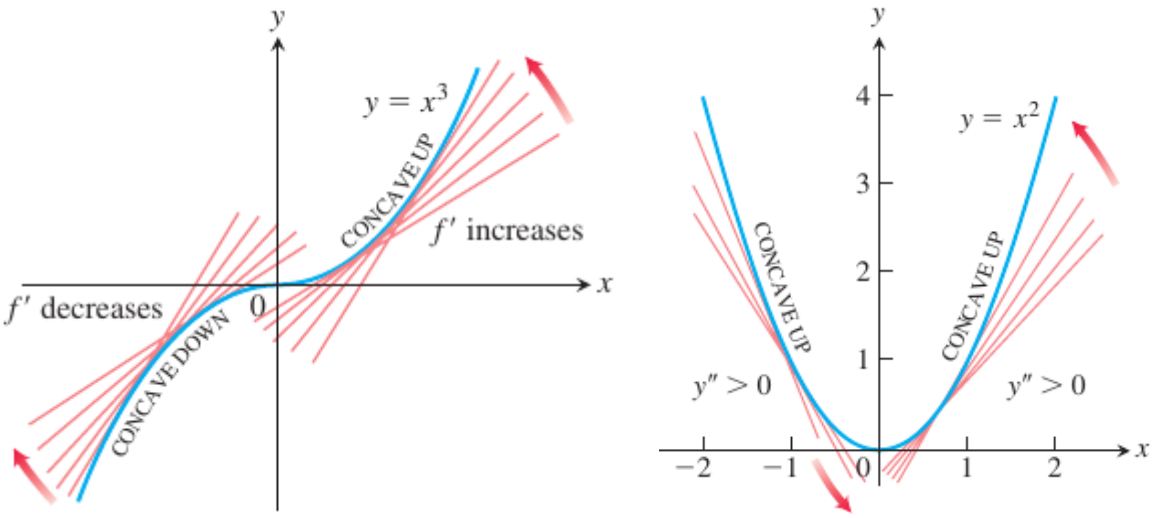


Figure 4.4: Concavity

The Second Derivative Test for Concavity

Let $y = f(x)$ be twice-differentiable on an interval I.

(a) If $f'' > 0$ on I, the graph of f over I is concave up.

(b) If $f'' < 0$ on I, the graph of f over I is concave down.

Definition 4.23. A point $(c, f(c))$ where the graph of a function has a tangent line and where the concavity changes is a **point of inflection**.

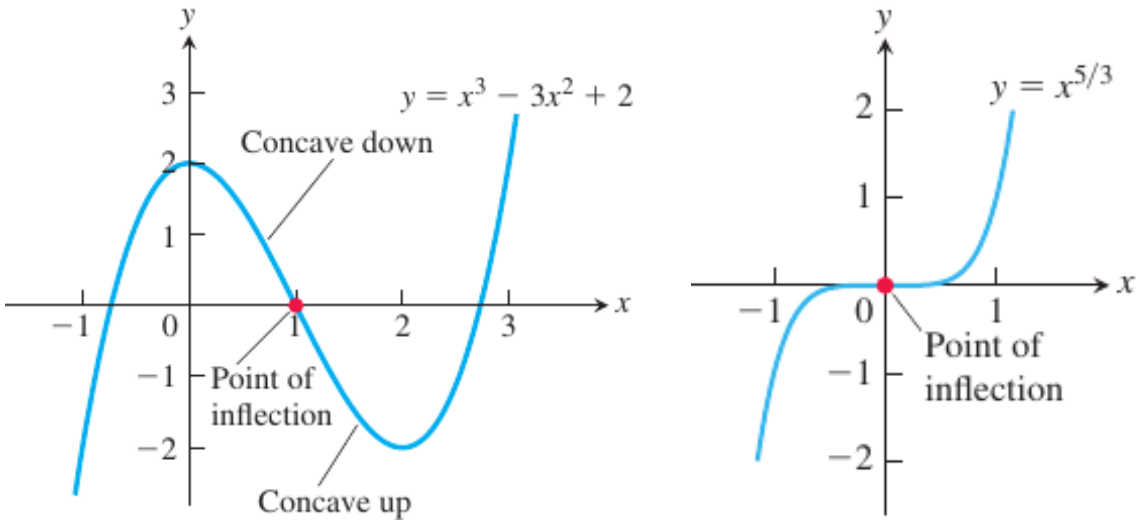
Example 4.24. For $f(x) = x^3$, as in Figure 4.4, $(0, 0)$ is an inflection point.

Remark 4.25. At a point of inflection $(c, f(c))$, either $f''(c) = 0$ or $f''(c)$ fails to exist.

Example 4.26. Determine the concavity and find the inflection points of

(a) $f(x) = x^3 - 3x^2 + 2$ (b) $g(x) = x^{5/3}$

Solution.



Theorem 4.27. **Second Derivative Test for Local Extrema**
Suppose f'' is continuous on an open interval that contains $x = c$.

(a) If $f'(c) = 0$ and $f''(c) < 0$, then f has a local maximum at $x = c$.

(b) If $f'(c) = 0$ and $f''(c) > 0$, then f has a local minimum at $x = c$.

(c) If $f'(c) = 0$ and $f''(c) = 0$, then the test fails. The function f may have a local maximum, a local minimum, or neither.

Example 4.28. Sketch a graph of the function $f(x) = x^4 - 4x^3 + 10$. Identify the coordinates of any local extreme points, inflection points, and concavity.

Solution.

x		0		2		3	
$f'(x)$		0				0	
$f''(x)$				0			
$f(x)$							
Behavior of f							

Strategy 4.29. Procedure for Graphing $y = f(x)$

1. Identify the domain of f and any symmetries the curve may have.
2. Find the derivatives y' and y''.
3. Find the critical points of f, if any, and identify the function's behavior at each one.
4. Find where the curve is increasing and where it is decreasing.
5. Find the points of inflection, if any occur, and determine the concavity of the curve.
6. Plot key points, such as the intercepts and the points found in Steps 3–5, and sketch the curve together with any asymptotes that exist.

Example **4.30.** Graph the following functions using appropriate methods from the graphing procedures, identify the coordinates of any local extreme points and inflection points.

(a) $y = x^3 - 3x + 3$

(b) $y = 5\sqrt{3}x + 10\cos x, \quad 0 \leq x \leq 2\pi$

Example **4.31.** Graph the following functions using appropriate methods from the graphing procedures, identify the coordinates of any local extreme points and inflection points.

(a) $y = \ln(3 - x^2)$

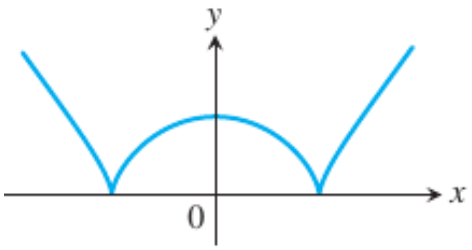
(b) $y = \dfrac{x}{x^2 - 1}$

Exercises 4.4

1. For the functions, identify (i) the inflection points, (ii) local maxima and minima, and (iii) the intervals on which the functions are concave up and concave down.

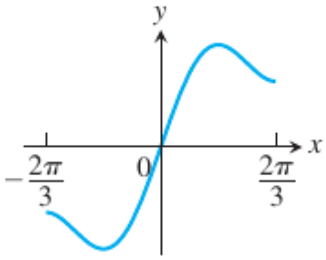
(a)

$y = \frac{3}{4}(x^2 - 1)^{2/3}$



(b)

$y = x + \sin 2x, -\frac{2\pi}{3} \le x \le \frac{2\pi}{3}$



Hint: (b) $y' = 1 + 2\cos 2x$, $y'' = -4\sin 2x$, and $\frac{-4\pi}{3} \le 2x \le \frac{4\pi}{3}$. Thus $y' = 0$ implies $2x = \pm\frac{2\pi}{3}$, $\pm\frac{4\pi}{3}$, and $y'' = 0$ implies $2x = 0, \pm\pi$. Thus, for example, the three inflection points are $(0, 0)$ and $\pm(\frac{\pi}{2}, \frac{\pi}{2})$.

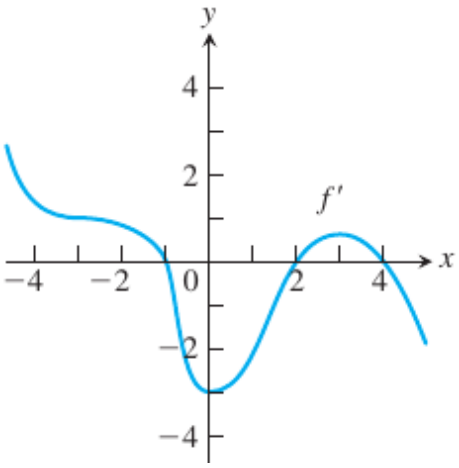
2. Identify the coordinates of any local and absolute extreme points and inflection points. Then, graph the function.

(a) $y = x^2(6 - x^2) - 4$

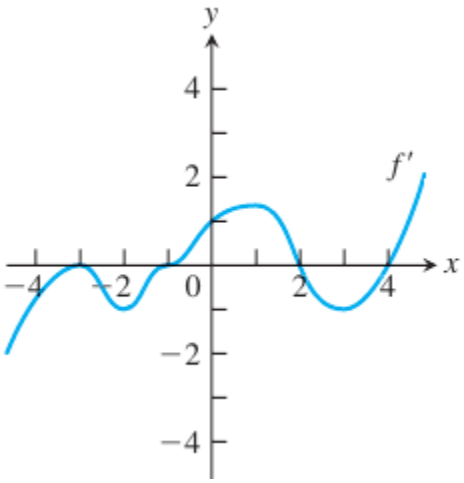
(b) $y = \cos x + \sqrt{3}\sin x, \quad 0 \le x \le 2\pi$

3. The graph of f' is given. Determine x-values corresponding to local minima, local maxima, and inflection points for the graph of f.

(a)



(b)



Hint: (b) Local minima: at $x = -1, 4$. Inflection points: at $x = -3, 1, 3$. Note that $f'' = 0$ at $x = -1$; however the corresponding point is not an inflection point.

4.5. Indeterminate Forms and L'Hôpital's Rule

4.5.1. L'Hôpital's Rule

Definition 4.32. Consider a limit of the form

$$\lim_{x\to a}\frac{f(x)}{g(x)}. \tag{4.5}$$

(a) If both $f(x)\to 0$ and $g(x)\to 0$ as $x\to a$, then this limit may or may not exist and is called an **indeterminate form of type** $0/0$.

(b) If both $f(x)\to \infty$ and $g(x)\to \infty$ as $x\to a$, then this limit may or may not exist and is called an **indeterminate form of type** ∞/∞.

Theorem 4.33. L'Hôpital's Rule
Asuume that f and g are differentiable and $g'(x)\neq 0$ on an open interval I containing a (except possibly at a). Suppose that

$$\lim_{x\to a} f(x)=0 \quad \textit{and} \quad \lim_{x\to a} g(x)=0 \tag{4.6}$$

or that

$$\lim_{x\to a} f(x)=\pm\infty \quad \textit{and} \quad \lim_{x\to a} g(x)=\pm\infty. \tag{4.7}$$

(In other words, we have an indeterminate form of type $\frac{0}{0}$ or $\frac{\infty}{\infty}$.) Then

$$\lim_{x\to a}\frac{f(x)}{g(x)} = \lim_{x\to a}\frac{f'(x)}{g'(x)}. \tag{4.8}$$

Remark 4.34.

- We can continue to differentiate f and g so long as we still get $\frac{0}{0}$ or $\frac{\infty}{\infty}$ form.
- L'Hôpital's Rule does not apply when either the numerator or the denominator has a finite nonzero limit.

Example 4.35. Use L'Hôpital's Rule to evaluate the limit. Compare this with the methods studied in Chapter 2.

(a) $\lim_{x \to -2} \dfrac{x+2}{x^2-4}$

The method from Chap.2 L'Hôpital's Rule

(b) $\lim_{x \to \infty} \dfrac{5x^2-3x}{7x^2+1}$

The method from Chap.2 L'Hôpital's Rule

Example 4.36. Use L'Hôpital's Rule to find the limit.

(a) $\lim_{t\to 0} \dfrac{\sin 5t}{2t}$

(b) $\lim_{\theta\to 0} \dfrac{\cos\theta - 1}{e^{\theta} - \theta - 1}$

(c) $\lim_{x\to 0} \dfrac{x - \sin x}{x^3}$

(d) $\lim_{x\to 0} \dfrac{x2^x}{2^x - 1}$

4.5.2. Other Indeterminate Forms

Indeterminate Products (Type $0 \cdot \infty$)

Example 4.37. Use L'Hôpital's Rule to find the limit.

(a) $\lim_{x\to\infty} x^3 e^{-x^2}$ (b) $\lim_{x\to 0^+} x^2 \ln x$

Solution.

Indeterminate Differences (Type $\infty - \infty$)

Example 4.38. Find the limit of functions on the $\infty - \infty$ form.

(a) $\lim_{x\to 0} \left[\frac{1}{\sin x} - \frac{1}{x}\right]$ (b) $\lim_{x\to\infty} [\ln 2x - \ln(x+1)]$

Solution.

Ans: (a) 0.

Indeterminate Powers: 1^∞, 0^0, ∞^0

If $\lim_{x\to a} \ln f(x) = L$, then

$$\lim_{x\to a} f(x) = \lim_{x\to a} e^{\ln f(x)} = e^L. \tag{4.9}$$

Here a may be either finite or infinite.

Example 4.39. Apply L'Hôpital's Rule to show that

$$\lim_{x\to 0^+} (1+x)^{1/x} = e. \tag{4.10}$$

Solution. Start with $\lim_{x\to 0^+} \ln f(x) = \lim_{x\to 0^+} \dfrac{\ln(1+x)}{x}$.

Example 4.40. Find $\lim_{x\to\infty} x^{1/x}$

Solution.

Ans: 1.

Exercises 4.5

1. Use L'Hôpital's Rule to evaluate the limit.

   (a) $\lim_{x\to 0} \dfrac{1-\cos x}{x^2}$ (b) $\lim_{x\to\infty} \dfrac{\ln(1+1/x)}{\sin(1/x)}$ (c) $\lim_{x\to 0^+} \sqrt{x}\ln x$

   Ans: (b) 1. (c) 0.

2. Use L'Hôpital's Rule to evaluate the limit.

   (a) $\lim_{x\to 0} \dfrac{\sin x^2}{x}$

   (b) $\lim_{x\to\infty} \dfrac{\ln(x+1)}{\log_2 x}$

   (c) $\lim_{x\to 0^+} [\ln x - \ln\sin x]$

   (d) $\lim_{h\to 0} \dfrac{e^h-(1+h)}{h^2}$

   Ans: (c) 0.

3. L'Hôpital's Rule may not help with the limits. Try it first – you will just keep on cycling. Find the limits some other way.

   (a) $\lim_{x\to(\pi/2)^+} \dfrac{\sec x}{\tan x}$ (b) $\lim_{x\to\infty} \dfrac{e^{x^2}}{xe^x}$ (c) $\lim_{x\to\infty} \dfrac{2^x-3^x}{3^x+4^x}$

   Hint: (a) Try to simplify the fraction. (b) Apply the natural logarithm.

   Ans: (b) ∞. (c) 0.

4. Challenge For what values of a and b is

$$\lim_{x\to 0}\left[\frac{\sin x}{x^2}+\frac{a}{x}+\frac{\sin bx}{x}\right] = 2?$$

5. CAS This problem explores the difference between the limits

$$\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^x \quad \text{and} \quad \lim_{x\to\infty}\left(1+\frac{1}{x^2}\right)^x$$

   (a) Use L'Hôpital's Rule to show that $\lim_{x\to\infty}\left(1+\dfrac{1}{x}\right)^x = e$.

   (b) Graph

$$f(x)=\left(1+\frac{1}{x}\right)^x \quad \text{and} \quad g(x)=\left(1+\frac{1}{x^2}\right)^x$$

   together for $x \geq 0$. How does the behavior of g compare with that of f? Estimate the value of $\lim_{x\to\infty} g(x)$.

   (c) Confirm your estimate of $\lim_{x\to\infty} g(x)$ by calculating it with L'Hôpital's Rule.

4.6. Applied Optimization

Strategy 4.41. Solving Applied Optimization Problems

1. *Read the problem*. Read the problem until you understand it. What is given? What is the unknown quantity to be optimized?
2. *Introduce variables*. List every relevant relation in the problem as an equation. In most problem it is helpful to draw a picture.
3. *Write an equation for the unknown quantity*. Express the quantity to be optimized as a function of **a single variable**. This may require considerable manipulation.
4. *Test the critical points and endpoints in the domain of the function found in the previous step*. Use what you know about the shape of the function's graph. Use the first and second derivatives to identify and classify the function's critical points.

Example 4.42. What is the smallest perimeter possible for a rectangle whose area is 16 in^2, and what are its dimensions?

Solution.

Example 4.43. A 2048 ft^3 open-top rectangular tank with a square base x ft on a side and y ft deep is to be built with its top flush with the ground to catch runoff water. What dimensions of the tank will minimize the weight?

Solution.

Ans: $x = 16$, $y = 8$

Example 4.44. You have been asked to design a one-liter can shaped like a right circular cylinder (Figure 4.5). What dimensions will use the least material?

Solution. Surface area: $A = 2\pi r^2 + 2\pi rh$, with $\pi r^2 h = 1000$.

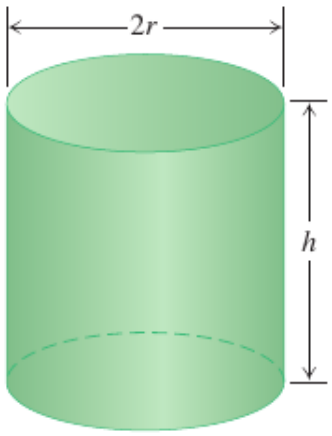


Figure 4.5: The one-liter can.

Ans: $r = \sqrt[3]{500/\pi}$ and $h = 2r$.

Example **4.45.** A rectangular plot of farmland will be bounded on one side by a river and on the other three sides by a single-strand electric fence. With 800 m of wire at your disposal, what is the largest area you can enclose, and what are its dimensions?

Solution.

Example **4.46.** A 216 m^2 rectangular pea patch is to be enclosed by a fence and divided into two equal parts by another fence parallel to one of the sides. What dimensions for the outer rectangle will require the smallest total length of fence? How much fence will be needed?

Solution.

Example **4.47.** Find a positive number for which the sum of it and its reciprocal is the smallest (least) possible.

Solution.

Exercises 4.6

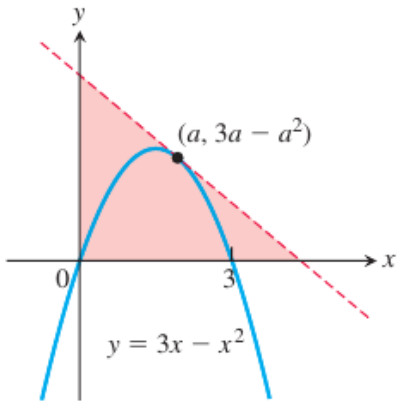
1. A rectangle has its base on the x-axis and its upper two vertices on the parabola $y = 12 - x^2$. What is the largest area the rectangle can have, and what are its dimensions?
Hint: Using symmetry, let $[-x, x]$ be the base, where $x > 0$. Then the area $A = 2xy = 2x(12-x^2)$.

2. Two sides of a triangle have lengths a and b, and the angle between them is θ. What value of θ will maximize the triangle's area? What kind of triangle is it?
Hint: $A = \frac{1}{2}ab\sin\theta$.

Ans: $\frac{\pi}{2}$.

3. Find the point on the line $\dfrac{x}{a} + \dfrac{y}{b} = 1$ that is closest to the origin. ***Clue***: Let (x, y) be a point on the line. Then the square-distance from the point to the origin is $d^2 = x^2 + y^2$. Note that the equation of the line can be rewritten as $y = b(1 - x/a) = b - \dfrac{b}{a}x$. Thus $d^2 = x^2 + (b - \frac{b}{a}x)^2$, which can be minimized.

4. Among all triangles in the first quadrant formed by the x-axis, the y-axis, and tangent lines to the graph of $y = 3x - x^2$, what is the smallest possible area?
Clue: First of all, $a \in (3/2, 2)$. The tangent line reads $y = (3 - 2a)(x - a) + 3a - a^2$, of which the x-intercept is $x = \dfrac{a^2}{2a - 3}$. Now, find the y-intercept. Then the area of the triangle is $A = \dfrac{1}{2}xy$, as a function of a.



Ans: $A = 8$ when $a = 2$.

5. What value of a makes $f(x) = x^2 + \dfrac{a}{x}$ have
   (a) a local minimum at $x = 2$?
   (b) a point of inflection at $x = 1$?

Ans: (a) 16. (b) -1.

4.7. Newton's Method

In this section we study a numerical method called **Newton's method** or the **Newton-Raphson method**, which is a technique to approximate the solutions to an equation

$$f(x) = 0. \tag{4.11}$$

- Newton's method estimates the solutions using tangent lines of the graph of $y = f(x)$ near the points where f is zero.
- A value of x where f is zero is called a **root** of the function f and a **solution** of the equation $f(x) = 0$.

Procedure for Newton's Method

We can derive a formula for generating the successive approximations in the following way.

- Given an **initial approximation** x_0, the point-slope equation for the tangent to the curve at $(x_0, (x_0))$ is

$$y = f'(x_0)(x - x_0) + f(x_0). \tag{4.12}$$

- We can find where it crosses the x-axis by setting $y = 0$:

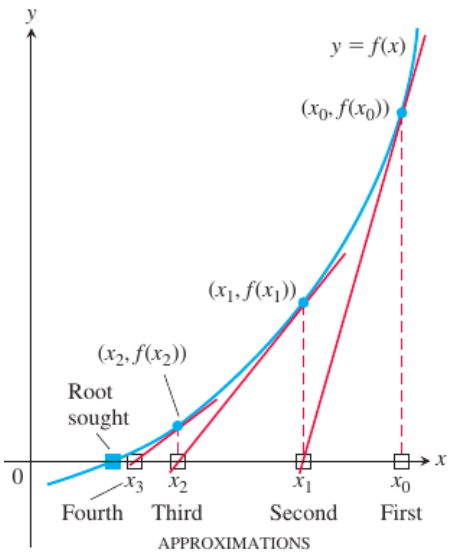
$$f'(x_0)(x - x_0) = -f(x_0),$$

which implies

$$x = x_0 - \frac{f(x_0)}{f'(x_0)}, \tag{4.13}$$

when $f'(x_0) \neq 0$.

- This value of x is the **next approximation** x_1.
- Repeat the steps to find new approximations.



Algorithm 4.48. Newton's Method

1. Guess a first approximation to a solution of the equation $f(x) = 0$. A graph of $y = f(x)$ may help.
2. Use the first approximation to get a second, the second to get a third, and so on, using the formula

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}, \quad f'(x_n) \neq 0. \tag{4.14}$$

Example 4.49. Approximate the positive root of the equation

$$f(x) = x^2 - 2 = 0.$$

Solution. Since $f'(x) = 2x$,

$$\begin{aligned} x_{n+1} &= x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{x_n^2 - 2}{2x_n} = x_n - \frac{1}{2}\left(x_n - \frac{2}{x_n}\right) \\ &= \frac{1}{2}\left(x_n + \frac{2}{x_n}\right) \end{aligned}$$

newton_method.m

```
% Solving f(x) = x^2 - 2 =0

x=1;
for n = 1:6
   x = (x + 2/x)/2;
   fprintf('x_%d = %.15f\n',n,x)
end
```

Result

```
x_1 = 1.500000000000000
x_2 = 1.416666666666667
x_3 = 1.414215686274510
x_4 = 1.414213562374690
x_5 = 1.414213562373095
x_6 = 1.414213562373095
```

Details of the correction:

newton_method2.m

```
% Solving f(x) = x^2 - 2 =0

x=1;
for n = 1:6
   h = (x-2/x)/2;
   x = x -h;
   fprintf('x_%d = %.15f; h = %.15f\n',n,x,h)
end
```

Result

```
x_1 = 1.500000000000000; h = -0.500000000000000
x_2 = 1.416666666666667; h =  0.083333333333333
x_3 = 1.414215686274510; h =  0.002450980392157
x_4 = 1.414213562374690; h =  0.000002123899820
x_5 = 1.414213562373095; h =  0.000000000001595
x_6 = 1.414213562373095; h =  0.000000000000000
```

Example 4.50. Find the x-coordinate of the point where the curve $y = x^3 - x$ crosses the horizontal line $y = 1$.

Solution. What we should solve is $x^3 - x = 1$. Let

$$f(x) = x^3 - x - 1.$$

Then $f(x) = -1$ and $f(2) = 5 \Rightarrow$ There is a solution in $(1, 2)$, by the IVT. Since $f'(x) = 3x^2 - 1$, Newton's method reads

$$x_{n+1} = x_n - \frac{x_n^3 - x_n - 1}{3x_n^2 - 1}.$$

newton_method3.m

```
% Solving f(x) = x^3 - x -1 =0

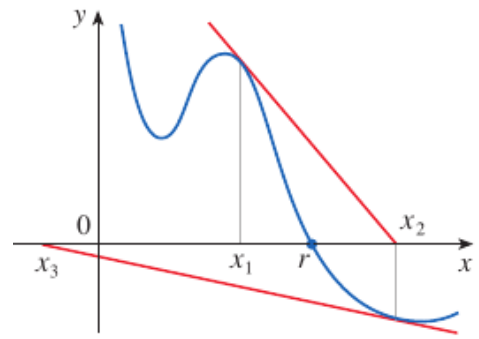
x=1.5;
for n = 1:6
   fx= x^3-x-1;
   h = fx/(3*x^2-1);
   x = x -h;
   fprintf('x_%d = %.15f; h = %.15f; f(xn)= %.15f\n',n,x,h,fx)
end
```

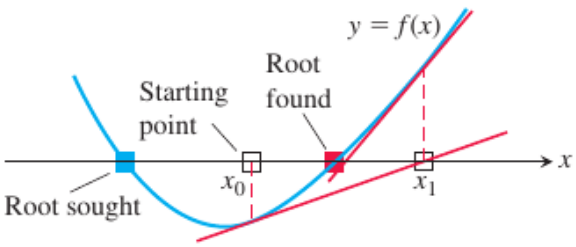
Result

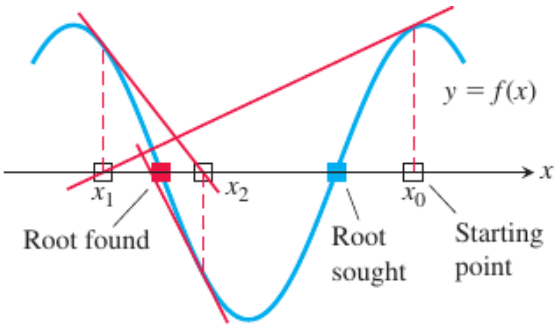
```
x_1 = 1.347826086956522; h = 0.152173913043478; f(xn)= 0.875000000000000
x_2 = 1.325200398950907; h = 0.022625688005615; f(xn)= 0.100682173091148
x_3 = 1.324718173999054; h = 0.000482224951853; f(xn)= 0.002058361916663
x_4 = 1.324717957244790; h = 0.000000216754264; f(xn)= 0.000000924377760
x_5 = 1.324717957244746; h = 0.000000000000044; f(xn)= 0.000000000000187
x_6 = 1.324717957244746; h = 0.000000000000000; f(xn)= 0.000000000000000
```

Convergence of the Approximations

- Newton's method does not always converge,
  particularly *when the initial approximation is not accurate enough.*
- If Newton's method does converge, it converges to a root.
- When Newton's method converges to a root, it may not be the root you have in mind.







Exercises 4.7

1. Using your calculator (or pencil-and-paper), run two iterations of Newton's method to find x_2 for given f and x_0.

   (a) $f(x) = x^4 - 2$, $x_0 = 1$

   (b) $f(x) = xe^x - 1$, $x_0 = 0.5$

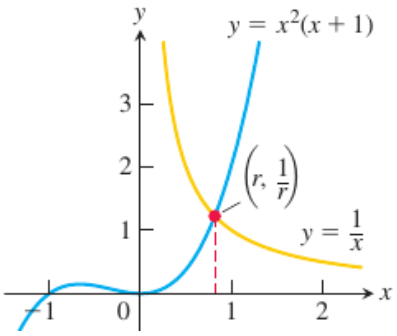
   Ans: (b) $x_2 = 0.68393972$

2. CAS Implement Newton's method to solve $f(x) = \arctan(x) = 0$,

   (a) with $x_0 = \pi/2$.

   (b) with $x_0 = \pi/4$.

3. CAS The graphs of $y = x^2(x+1)$ and $y = 1/x$ $(x > 0)$ intersect at one point $x = r$. Use Newton's method to estimate the value of r to eight decimal places.



4.8. Antiderivatives

Many problems require that we recover a function from its derivative, or from its rate of change. For instance, the laws of physics tell us the acceleration of an object falling from an initial height, and we can use this to compute its velocity and its height at any time. More generally, starting with a function f, we want to find a function F whose derivative is f.

4.8.1. Finding Antiderivatives

Definition 4.51. A function F is an **antiderivative** of f on an interval I if $F'(x) = f(x)$ for all x in I.

Example 4.52. Find an antiderivative for each of the following functions.

(a) $f(x) = 2x$ (b) $g(x) = \cos x$ (c) $h(x) = \sec^2 x + \dfrac{1}{2\sqrt{x}}$

Solution.

Note: If F is an antiderivative of f, so is $F + C$, for all constants C.

Theorem 4.53. *Let F and G be antiderivatives of f on an interval I. Then*

$$F(x) = G(x) + C, \quad \textit{for some } C. \tag{4.15}$$

Example 4.54. Find an antiderivative of $f(x) = 3x^2$ that satisfies $F(1) = -1$.

Solution.

Table 4.1: Antiderivative formulas, k a nonzero constant.

Function	General antiderivative
x^n	$\dfrac{1}{n+1}x^{n+1} + C, \quad n \neq -1$
$\sin kx$	$-\dfrac{1}{k}\cos kx + C$
$\cos kx$	$\dfrac{1}{k}\sin kx + C$
$\sec^2 kx$	$\dfrac{1}{k}\tan kx + C$
$\csc^2 kx$	$-\dfrac{1}{k}\cot kx + C$
$\sec kx \tan kx$	$\dfrac{1}{k}\sec kx + C$
$\csc kx \cot kx$	$-\dfrac{1}{k}\csc kx + C$

Example 4.55. Find the general antiderivative of each of the following functions.

(a) $f(x) = \dfrac{1}{\sqrt{x}}$ (b) $g(x) = \cos\dfrac{x}{2}$ (c) $h(x) = \dfrac{3}{\sqrt{x}} + \sin 2x$

Solution.

4.8.2. Initial Value Problems and Differential Equations

Note: Antiderivatives play important roles in mathematics and its applications. Methods and techniques for finding them are a major part of calculus, and we take up that study in Chapter 8.

- Finding an antiderivative for a function $f(x)$ is the same problem as finding a function $y(x)$ that satisfies the equation

$$\frac{dy}{dx} = f(x), \tag{4.16}$$

  which is called a **differential equation**.
- The function $y(x)$ is found by taking the antiderivative of $f(x)$.
- We can fix the arbitrary constant C, arising in the **antidifferentiation process**, by specifying an **initial value**

$$y(x_0) = y_0. \tag{4.17}$$

- The combination of a differential equation and an initial condition, (4.16) and (4.17), is called an **initial value problem**. Such problems play important roles in all branches of science.

Example **4.56.** Solve the initial value problems.

(a) $\dfrac{df}{dx} = \cos x + \sin x, \quad f(\pi) = 1$

(b) $\dfrac{d^2y}{dx^2} = 2 - 6x, \quad y'(0) = 4, y(0) = 1.$

Solution.

4.8.3. Indefinite Integrals

Definition **4.57.** The collection of all antiderivatives of f is called the **indefinite integral** of f with respect to x, and is denoted by

$$\int f(x)\,dx. \tag{4.18}$$

The symbol $\int$ is an **integral sign**. The function f is the **integrand** of the integral, and x is the **variable of integration**.

Example **4.58.** Evaluate $\int (x^2 - 2x + 5)dx$.

Solution.

Exercises 4.8

1. Find an antiderivative for each function. Check your answers by differentiation.

   (a) $x^3 - \dfrac{1}{x^3}$ (b) $\sqrt{x} + \dfrac{1}{\sqrt{x}} - \dfrac{1}{3}x^{-2/3}$ (c) $\dfrac{1}{1+4x^2}$

   Ans: (c) $\dfrac{1}{2}\arctan 2x$

2. Find the most general antiderivative or indefinite integral. You may need to try a solution and then adjust your guess. Check your answers by differentiation.

   (a) $\displaystyle\int 3\cos 5\theta\, d\theta$

   (b) $\displaystyle\int (3e^{-2x} + 2^x)dx$

   (c) $\displaystyle\int (4\sec x\tan x - 2\sec^2 x)dx$

   (d) $\displaystyle\int (2 + \tan^2 x)dx$

   Ans: (d) $x + \tan x + C$.

3. Which is right? Give a brief reason for your answer.

   (a) $\displaystyle\int x\sin x\, dx = \frac{x^2}{2}\sin x + C$

   (b) $\displaystyle\int x\sin x\, dx = -x\cos x + C$

   (c) $\displaystyle\int x\sin x\, dx = -x\cos x + \sin x + C$

4. Solve the initial value problems.

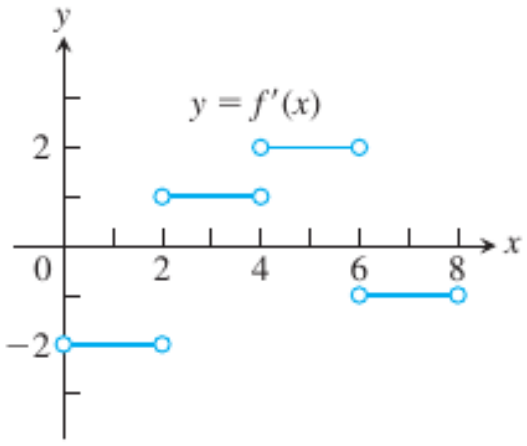
   (a) $\dfrac{ds}{dt} = 1 + \cos t, \quad s(0) = 4$

   (b) $y'' = -\cos x + 8\sin 2x$, $y'(0) = -2$, $y(0) = 3$

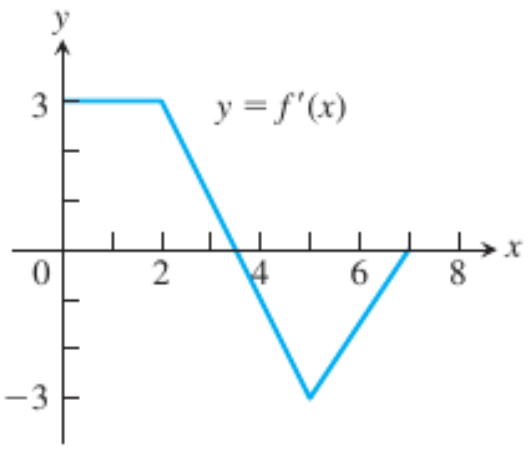
   Ans: (b) $y = \cos x - 2\sin 2x + 2x + 2$

5. The graph of f' is given. Assume that $f(0) = 1$ and sketch a possible continuous graph of f.

   (a)



   (b)



CHAPTER **P**

Projects

Finally we add projects.

Contents of Projects

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P.2. Newton's Method for Finding Roots and Global Extrema 215

P.1. Area Estimation of A Region

Problem P.1. It is common in reality that a **region** is saved by *a sequence of points*: For some $n > 0$,

$$(x_0, y_0),\ (x_1, y_1),\ \cdots,\ (x_n, y_n), \qquad (x_n, y_n) = (x_0, y_0). \tag{P.1.1}$$

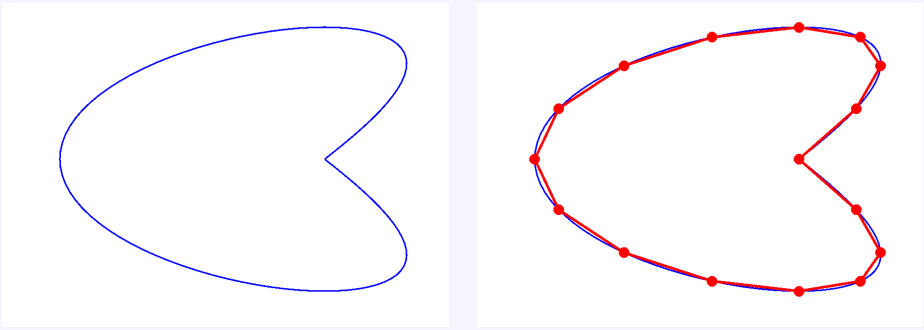


Figure P.1: A region and its approximation.

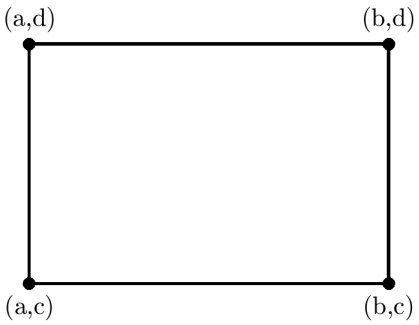
Here the question is:

If a sequence of points (P.1.1) represents a region, how can we compute its area accurately?

Derivation of Computational Formulas

Example P.2. Let's begin with a very simple example.

(a) **Find the area of the rectangle** $[a, b] \times [c, d]$.



Solution. We know the area = $(b - a) \cdot (d - c)$. It can be rewritten as

$$b \cdot (d - c) - a \cdot (d - c) = b \cdot (d - c) + a \cdot (c - d)$$

from which we may guess that

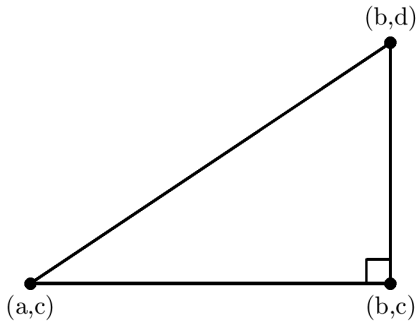
$$\text{Area} = \sum_i x_i^* \cdot \Delta y_i, \tag{P.1.2}$$

where the sum is carried out over line segments L_i and x_i^* denotes the mid value of x on L_i.

(b) **Find the area of the triangle**.

Solution. We know the area = $\frac{1}{2}(b-a)\cdot(d-c)$.
Now, let's try to find the area using the formula (P.1.2):

$$\text{Area} = \sum_i x_i^* \cdot \Delta y_i.$$



Let L_1, L_2, L_3 be the bottom side, vertical side, and the hypotenuse, respectively.
Then

$$\begin{aligned}\text{Area} &= \frac{a+b}{2}\cdot(c-c) + \frac{b+b}{2}\cdot(d-c) + \frac{b+a}{2}\cdot(c-d)\\ &= 0 \quad + b\cdot(d-c) \quad + \frac{b+a}{2}\cdot(c-d)\\ &= \left(b - \frac{b+a}{2}\right)\cdot(d-c) = \frac{1}{2}(b-a)\cdot(d-c).\end{aligned}$$

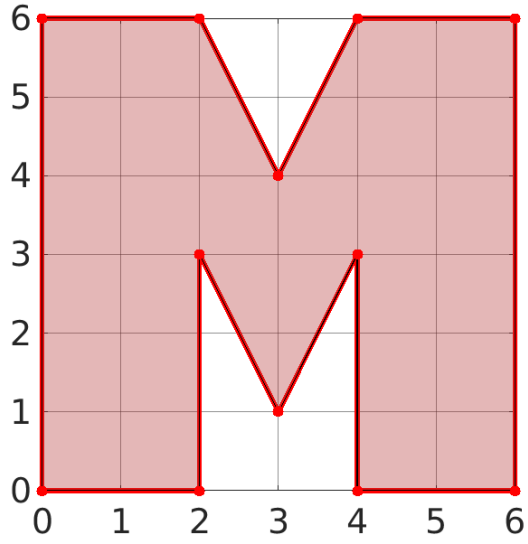
Okay. The formula is correct!

Note: Horizontal line segments makes **no** contribution to the area.

(c) **Let's verify the formula once more.**

The area of the M-shaped is 30.
Let's collect only nonzero values:

$$\begin{aligned}&2\cdot 3 - 2.5\cdot 2 + 3.5\cdot 2 - 4\cdot 3\\ &\qquad +6\cdot 6\\ &\qquad -3.5\cdot 2 + 2.5\cdot 2\\ &= 6 - 5 + 7 - 12\\ &\qquad +36\\ &\qquad -7 + 5\\ &= 30\end{aligned}$$



Again, the formula is correct!!

Summary **P.3.** The above work can be summarized as follows.

- Let a region R be represented as a sequence of points

$$(x_0, y_0),\ (x_1, y_1),\ \cdots,\ (x_n, y_n), \qquad (x_n, y_n) = (x_0, y_0). \tag{P.1.3}$$

- Let L_i be the i-th line segment connecting (x_{i-1}, y_{i-1}) and (x_i, y_i), $n = 1, 2, \cdots, n$. Then the area of R can be computed using the formula

$$\text{Area}(R) = \sum_{i=1}^{n} x_i^* \cdot \Delta y_i, \tag{P.1.4}$$

where

$$x_i^* = \frac{x_{i-1} + x_i}{2}, \quad \Delta y_i = y_i - y_{i-1}.$$

What You Will Do

1. **A given dataset**: First, download a dataset saved in `heart-data.txt`: https://skim.math.msstate.edu/LectureNotes/heart-data.txt
   - Draw a figure for it.
   - Use the formula (P.1.4) to find the area.
2. **Repeat Part 1 for another dataset**: For a dataset, you may choose one of the following.
   - Search the internet
   - Generate a dataset, e.g. points representing a circle
     - The last point must be the same as the first one: $(x_n, y_n) = (x_0, y_0)$
     - The larger n you choose, the more accurately the data would represent the region.
     - Analyze accuracy improvement of the area as n grows.

P.2. Newton's Method for Finding Roots and Global Extrema

In this project, you will implement Newton's method to find roots and global extrema of functions. Consider two functions defined on closed intervals.

$$\begin{aligned} f(x) &= 3x^4 - 4x^3 - 36x^2 - x + 20, \quad [-3, 4] \\ g(x) &= \frac{\sin(x^2 - \frac{\pi}{15})}{1 + x}, \quad [0, \pi] \end{aligned} \tag{P.2.1}$$

Problem P.4. Implement Newton's method. For each of the functions:

1. Find all roots.
2. Find the global minimum and the global maximum.

What You Will Do

1. Implement Newton's method, Algorithm 4.48, p. 201.
2. **Roots**. For each of f and g:
   (a) Plot the graph.
   (b) Guess an initial approximation for each root.
   (c) Apply Newton's method to approximate the root, in 10 decimal accuracy.
   (d) Add points to the plot, to see if the computation is correct.
3. **Global Extrema**. For each of f and g:
   (a) Plot and recognize where global extrema are located.
   (b) Find the derivative of the function.
   (c) With an initial approximation, apply Newton's method to approximate the root of the derivative, in 10 decimal accuracy.
   (d) Evaluate the function at the resulting point.
   (e) Add points to the plot, to see if the computation is correct.

Report your work (code) with results (numbers and plots).

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